

Hands-on Lab: Customer Support Automation with SAP Service Cloud and SAP Joule Agents

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<https://vasscompany.com>

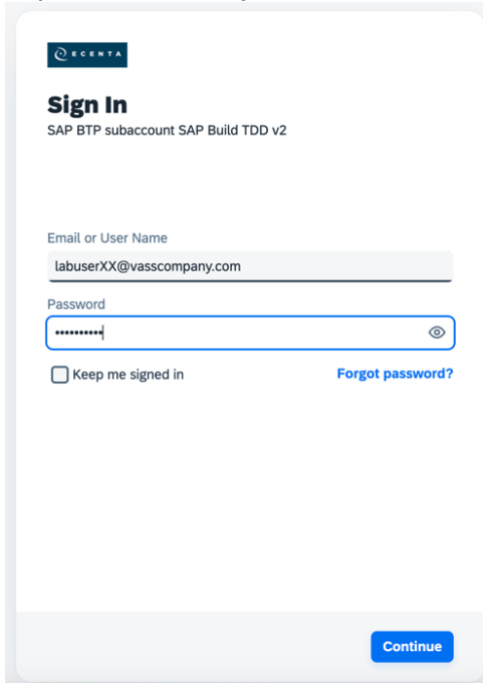
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Environment Information

1. Please use GOOGLE CHROME as your browser
2. SAP Build App URL: [SAP Build Apps](#)
3. Lab User Information
 - o User: **labuser<##>@vasscompany.com**
 - o Password: Newpa55!
4. Replace <##> with your Lab User Number assigned by the instructors



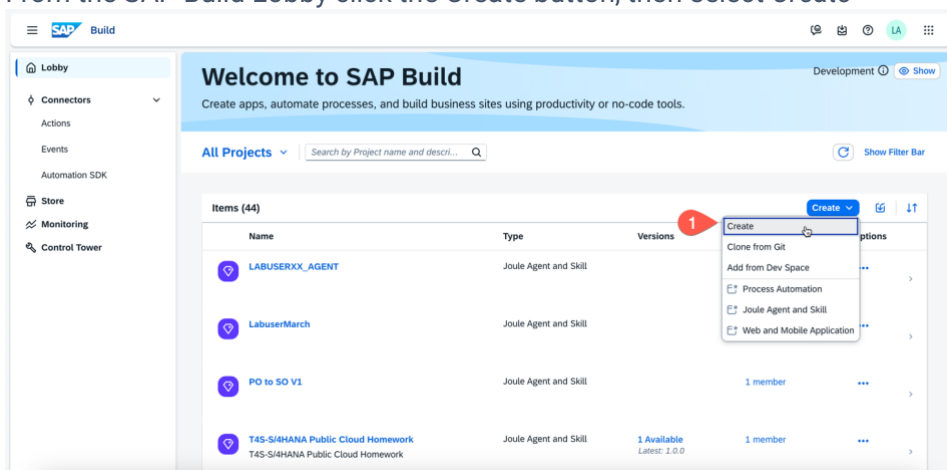
The screenshot shows a web browser window displaying the SAP Sign In page. At the top left is the SAP logo. The main heading is "Sign In" in bold, followed by the subtext "SAP BTP subaccount SAP Build TDD v2". Below this, there are two input fields: "Email or User Name" and "Password". The "Email or User Name" field contains the text "labuserXX@vasscompany.com". The "Password" field contains a series of asterisks. To the right of the password field is an eye icon for toggling visibility. Below the password field, there is a checkbox labeled "Keep me signed in" and a link labeled "Forgot password?". At the bottom right of the form is a blue button labeled "Continue".

Lab Goal

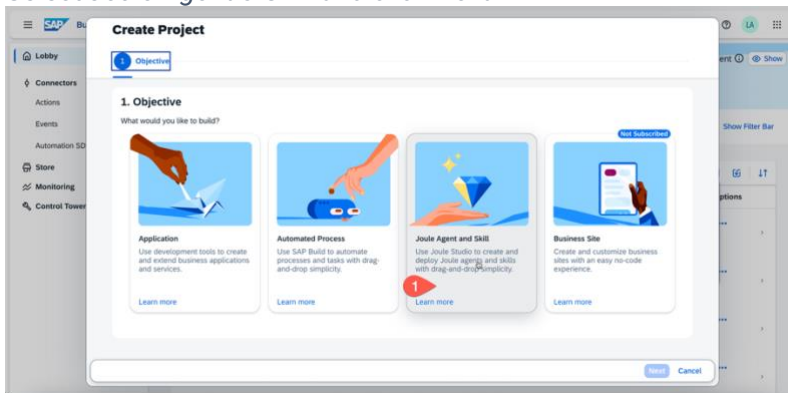
We will be building a Joule Agent that can review Cases within SAP Service Cloud V2, validate part numbers using a parts manual and then check for inventory within S/4HANA. For this lab, we have already set up the 'Actions' which call the API endpoints in SAP Service Cloud V2 and S/4HANA. Screenshots will be provided in the Lab Overview presentation. This is a very simple, self-contained scenario to demo the low-code environment where developers or analysts can create Joule Agents. This is not a production-ready scenario as all environments are distinctly different and require different inputs. This lab should provide a good, first, hands-on experience to encourage the lab user to explore more.

Create a Joule Project

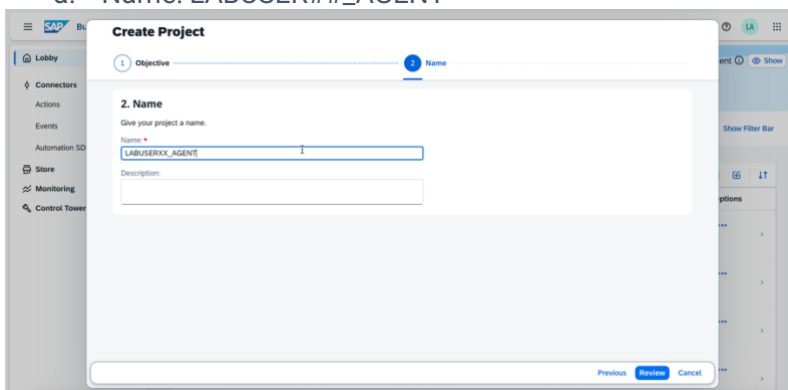
1. From the SAP Build Lobby click the Create button, then select *Create*



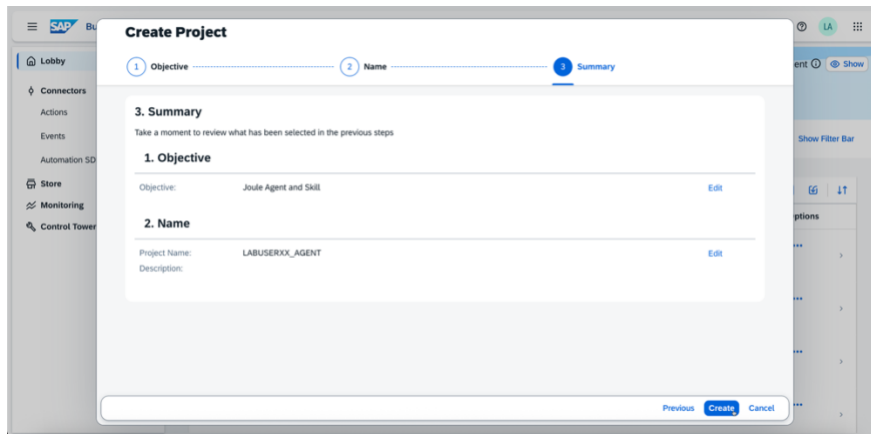
2. Select *Joule Agent & Skill* and click *Next*



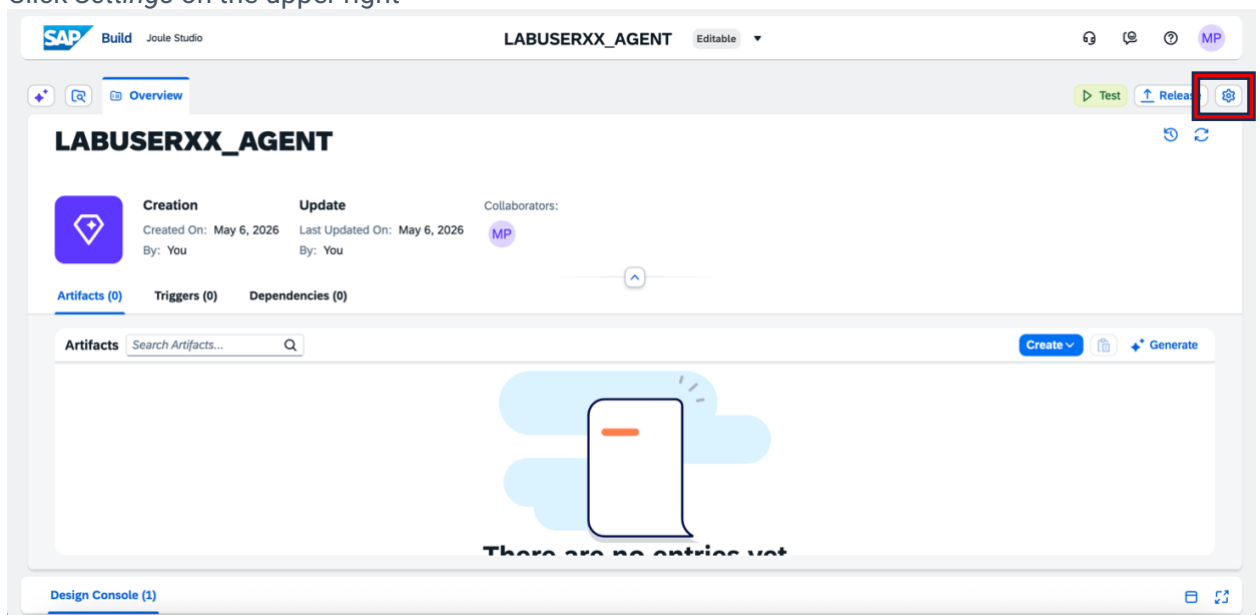
4. Enter a name for your Joule Skill in the following format
5. a. Name: LABUSER##_AGENT



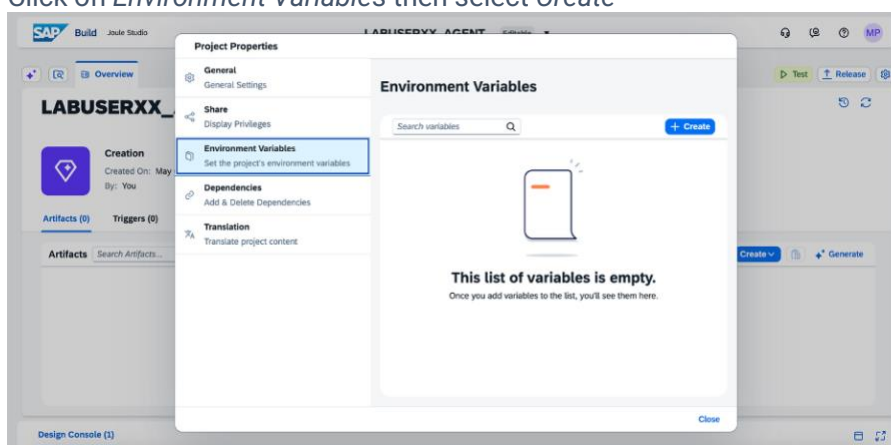
6. Click Review. Then click *Create*
- 7.



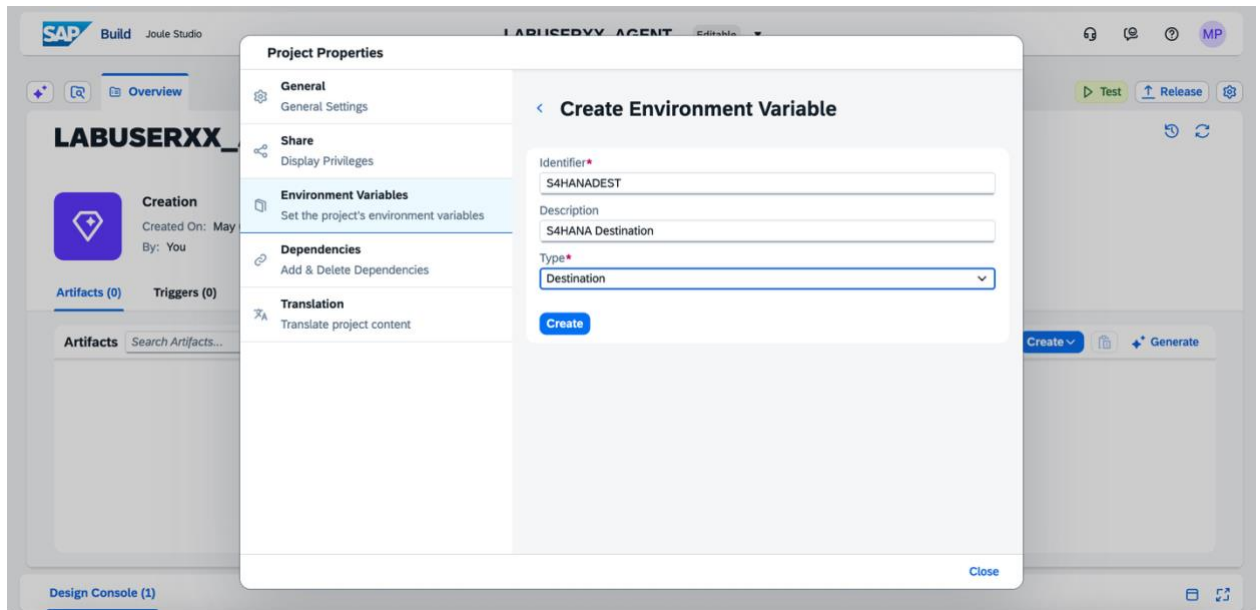
- 8.
9. A new browser tab will open (suppress the pop-up blocker)
10. Accept the Disclaimer if shown.
11. Click **Settings** on the upper right



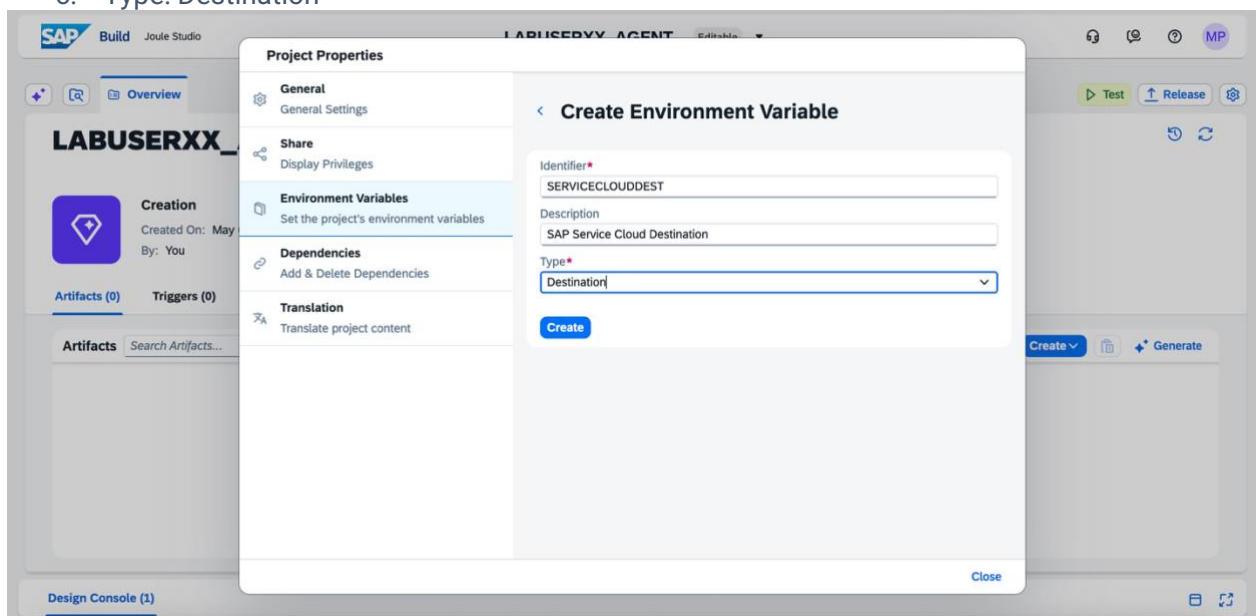
- 12.
13. Click on **Environment Variables** then select **Create**



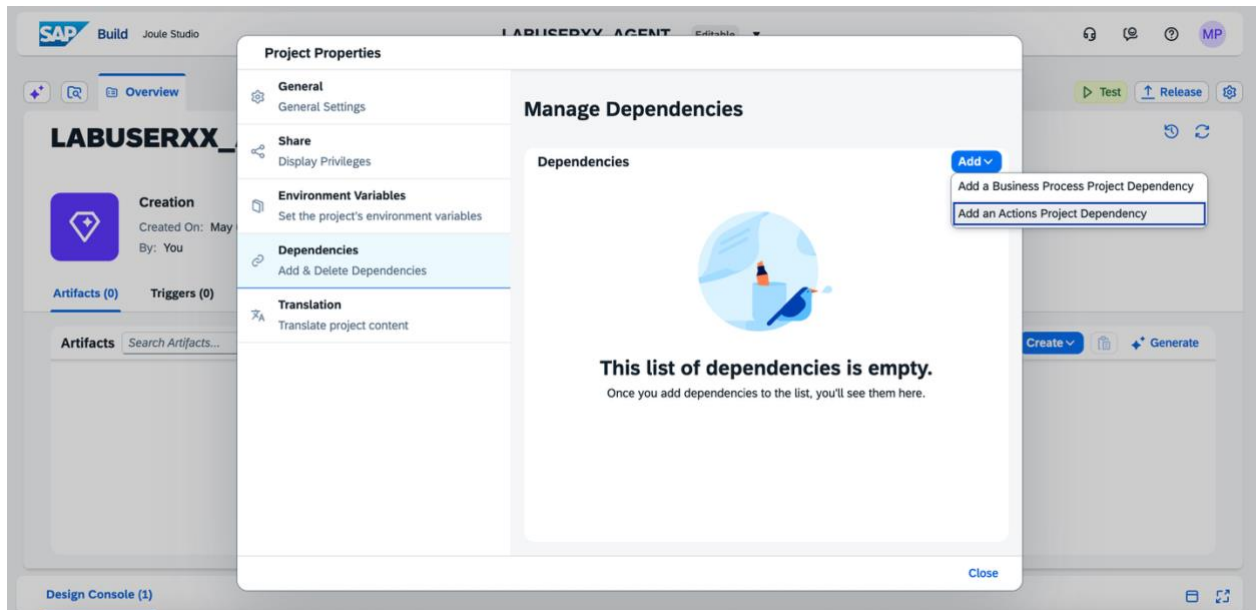
- 14.
15. Create the Destination with the data below and click Create again
 - a. Identifier: S4HANADEST
 - b. Description: S/4HANA Destination
 - c. Type: Destination



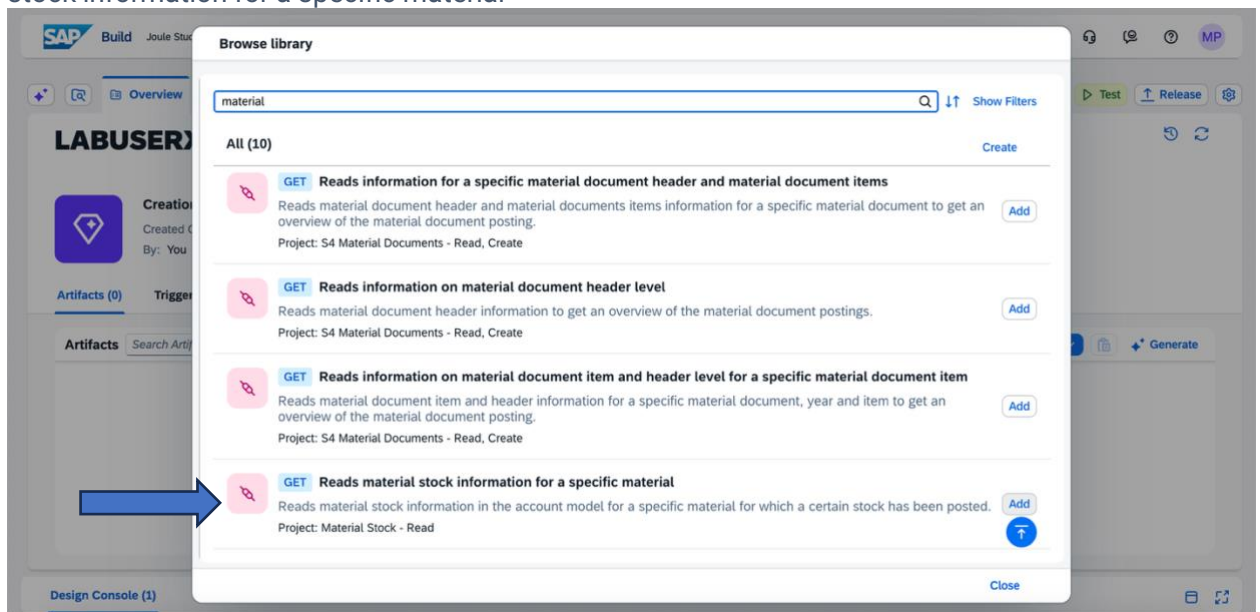
- 16.
17. Repeat for SAP Service Cloud with the data below
- a. Identifier: SERVICECLOUDDEST
 - b. Description: SAP Service Cloud Destination
 - c. Type: Destination



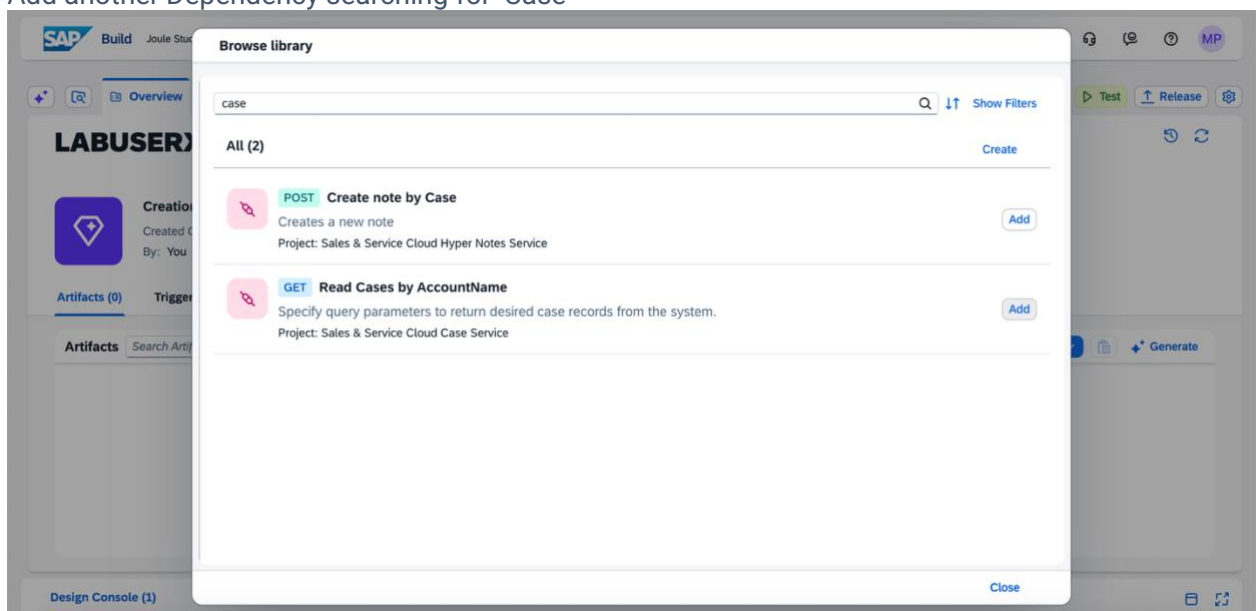
- 18.
19. Confirm you have two environment variables for Destinations
20. Next, Click on Dependencies, Click Add, Select 'Add an Actions Project Dependency'



21. When prompted to Search, enter 'Material' and select the Add button next to 'Reads material stock information for a specific material'

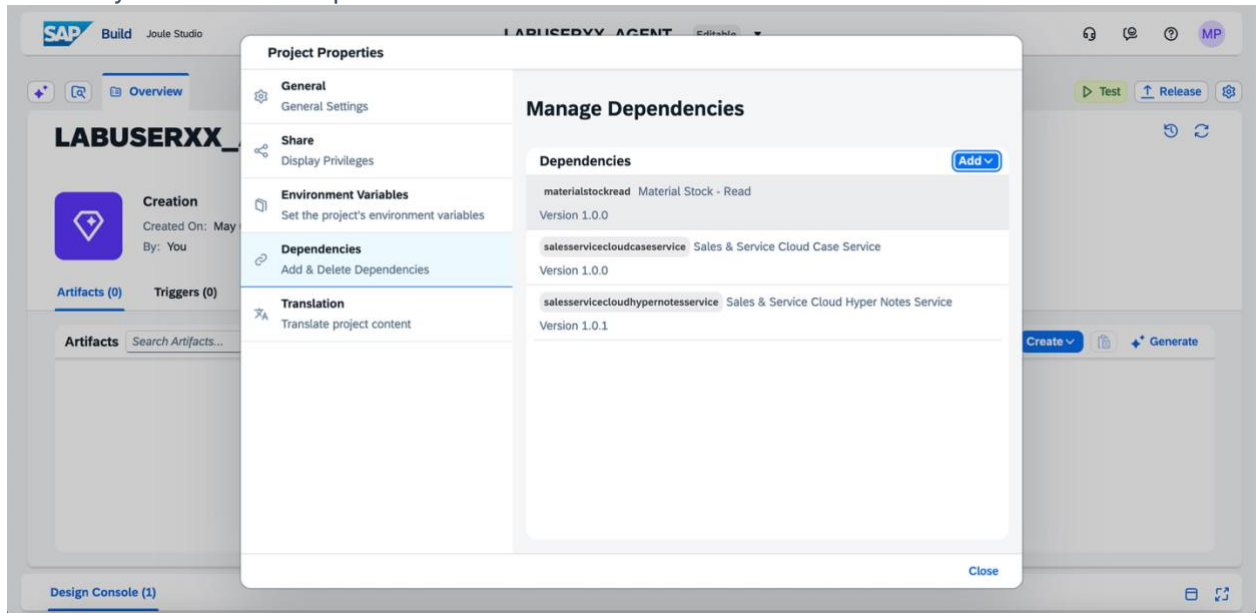


23. Add another Dependency searching for 'Case'



25.

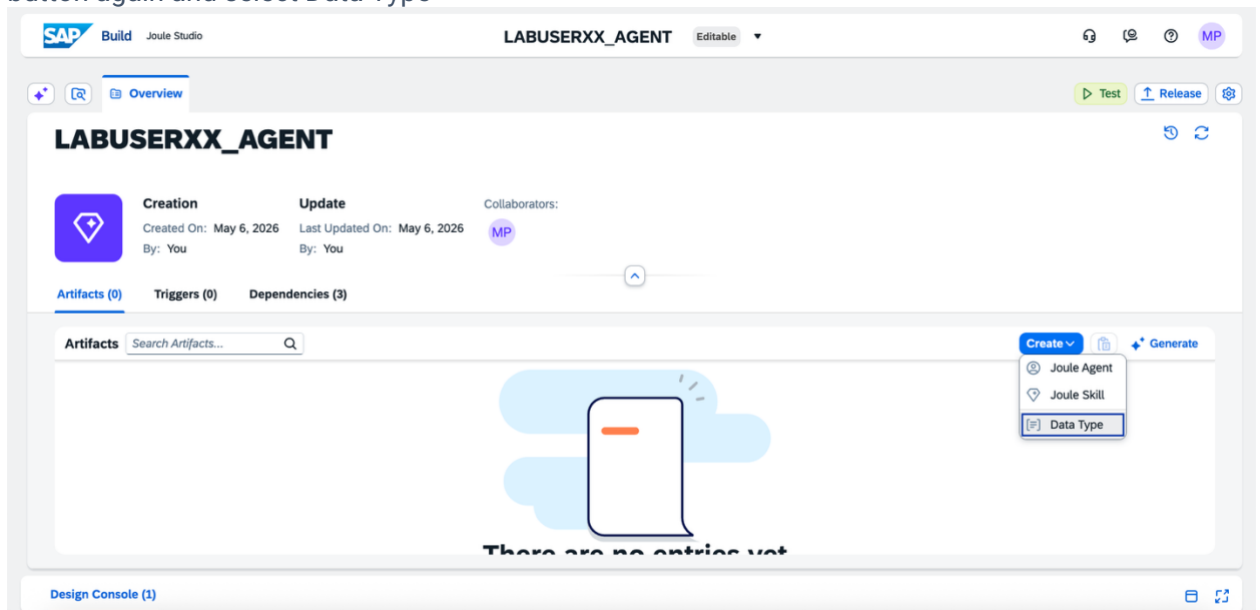
26. Add both Actions as dependencies
 - a. Create note by Case
 - b. Read Cases by AccountName
27. Confirm you have three dependencies added and click Close



28.

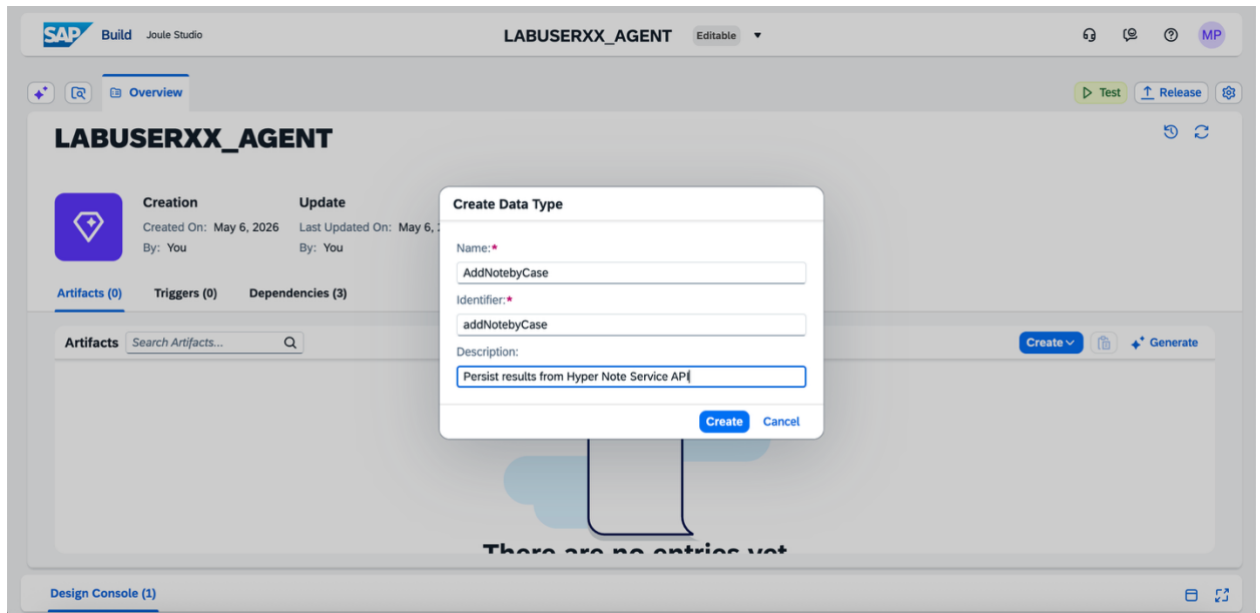
Create Data Types

29. Next, we're going to create a Data Type to persist the results of the APIs. Click the Create button again and select Data Type

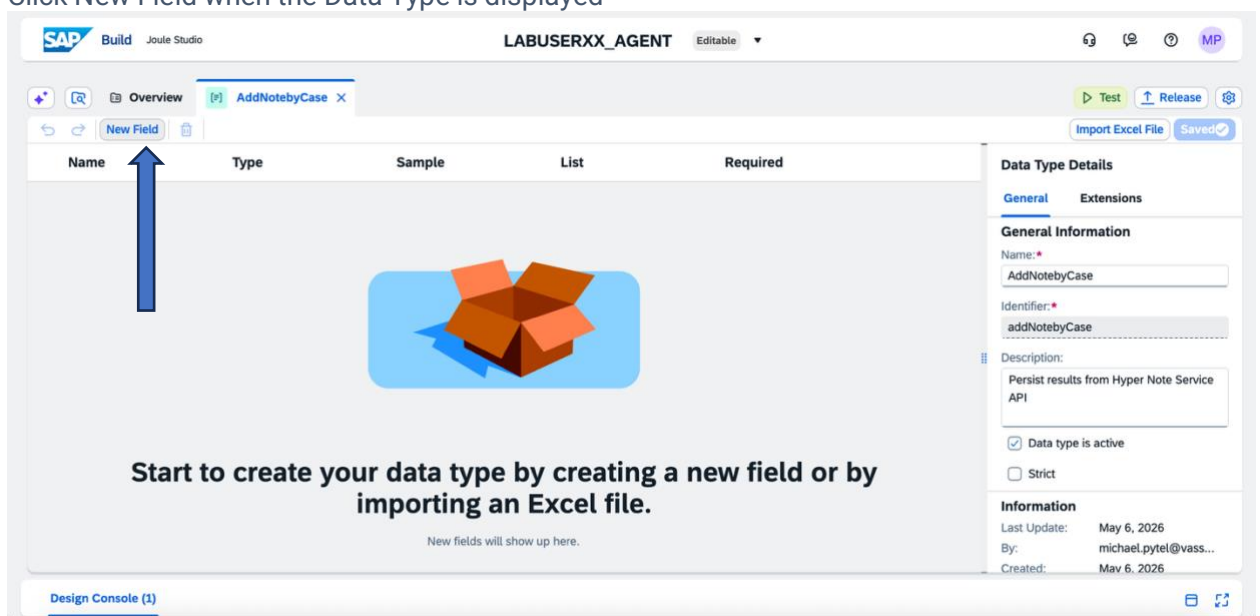


30.

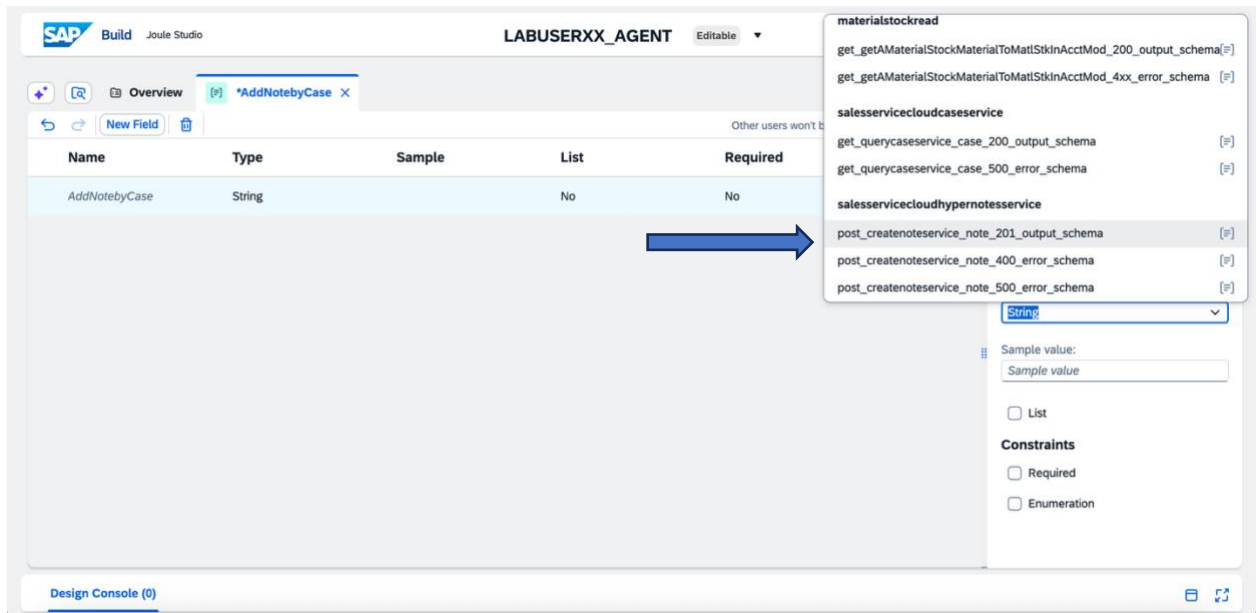
31. Enter the following details and click Create
 - a. Name: AddNotebyCase
 - b. Description: Persist results from Hyper Note Service API



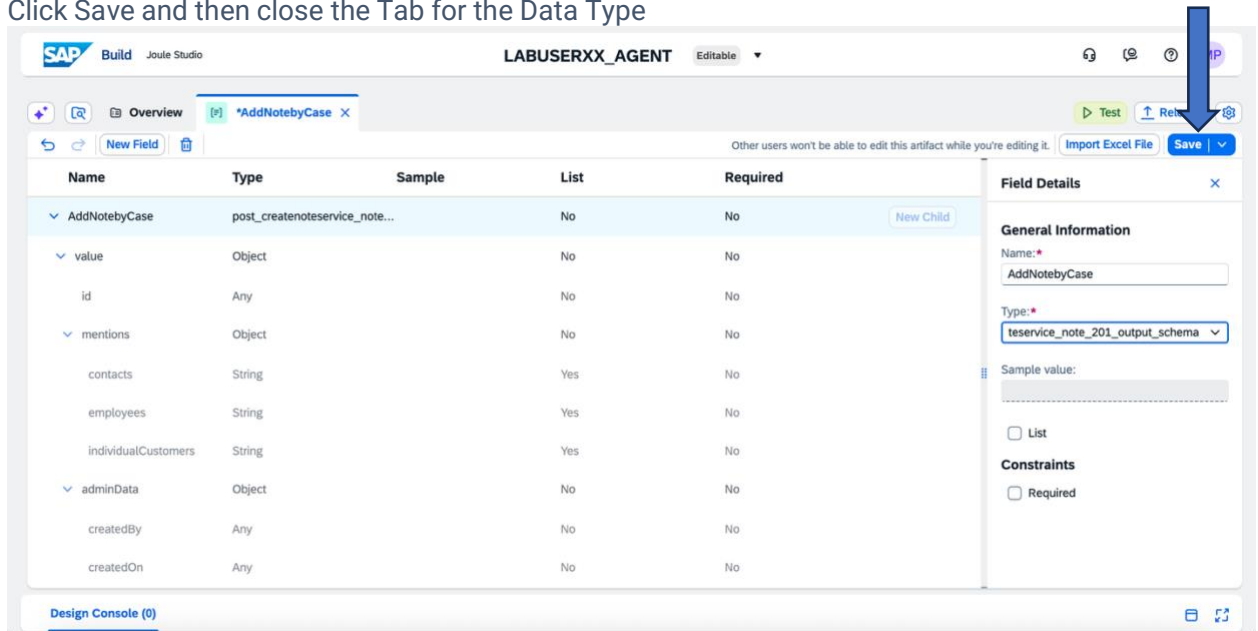
32.
 33. Click New Field when the Data Type is displayed



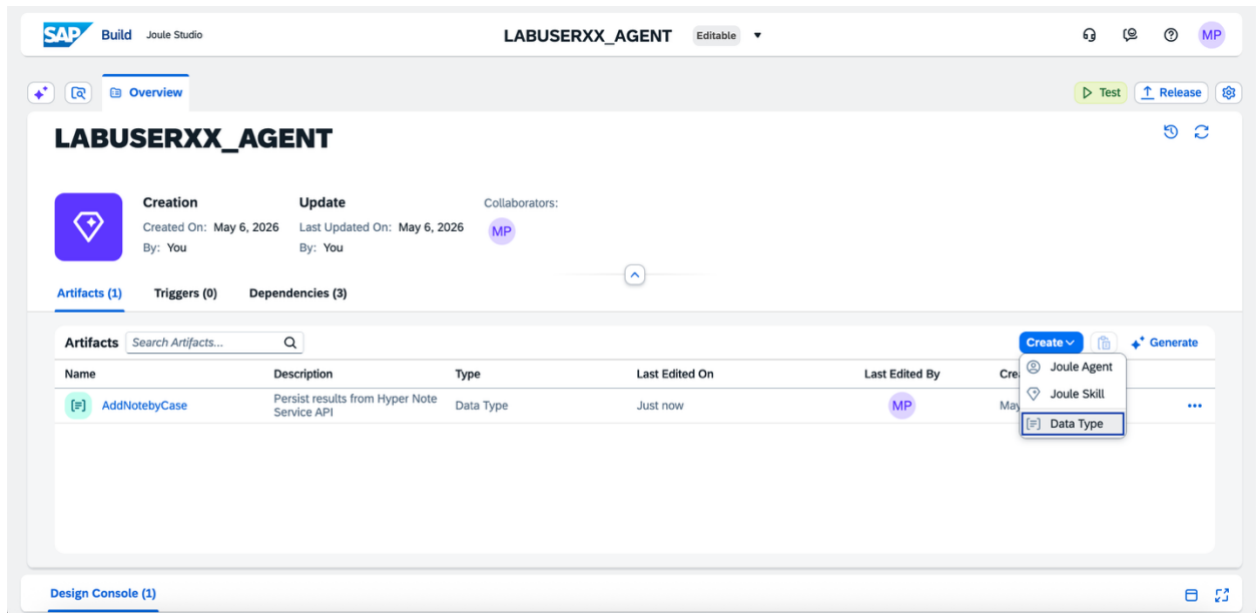
34.
 35. Enter the name 'AddNotebyCase'
 36. Select the Drop Down for Type and select the schema 'post_createnoteservice_note_201_output_schema'



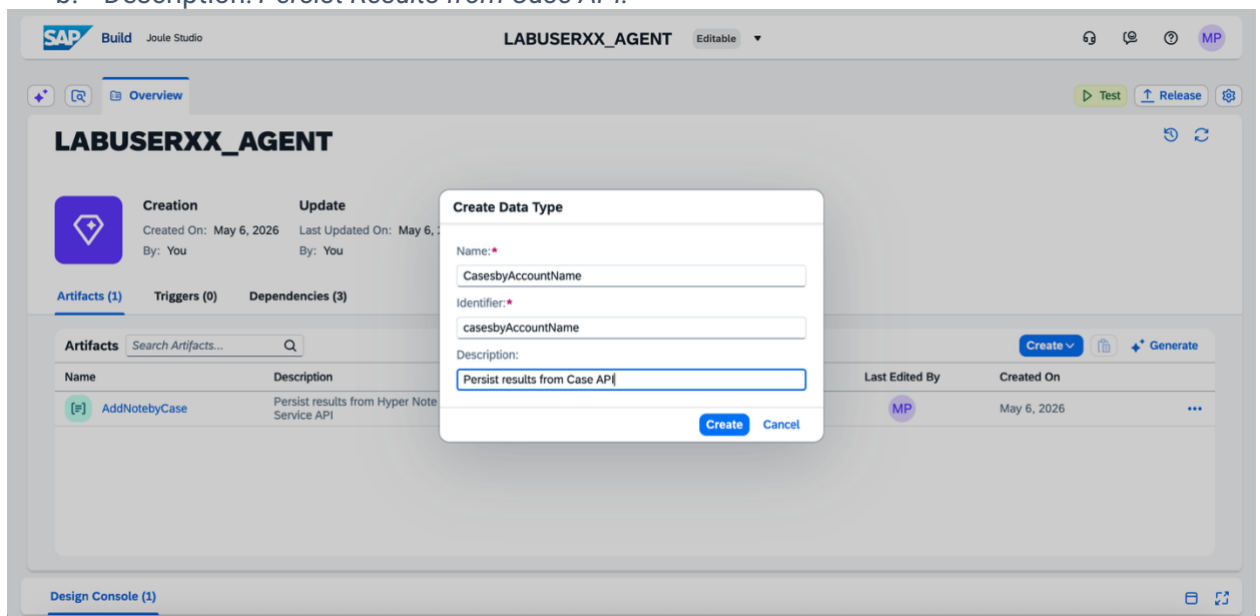
37. ...
 38. Click Tab Button and the resulting API schema should be displayed.
 39. Click Save and then close the Tab for the Data Type



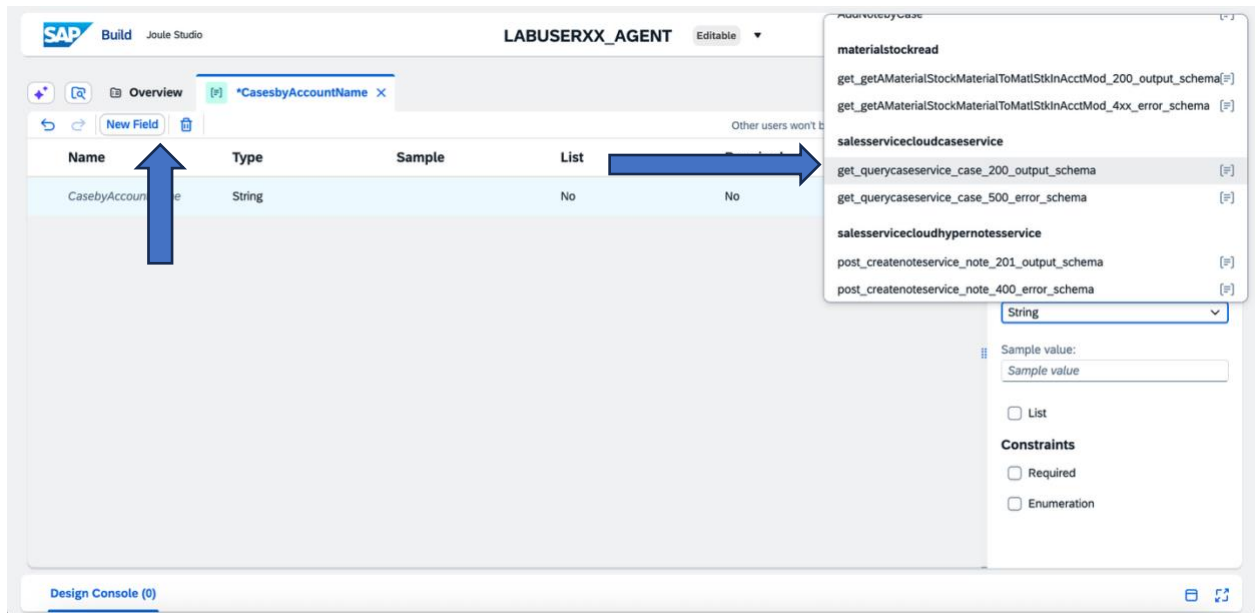
40. ...
 41. On the Overview screen, click Create > Data Type
 42. We will be adding a Data Type for each API



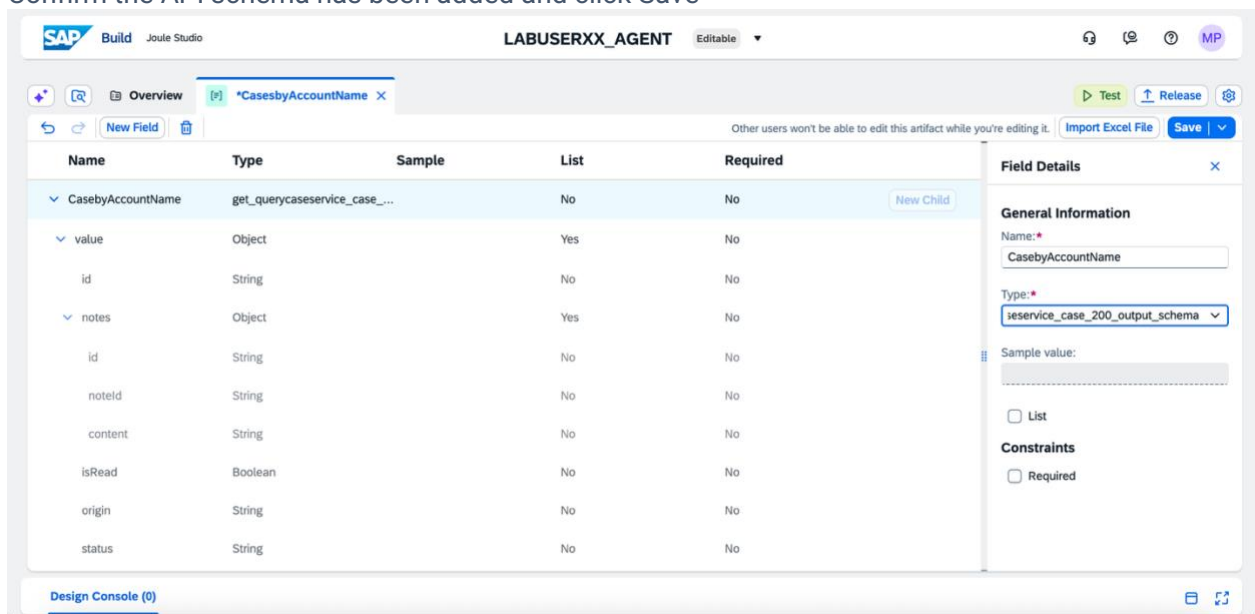
43.
 44. Enter the following details on the Data Type
- Name: CasesbyAccountName
 - Description: *Persist Results from Case API*.



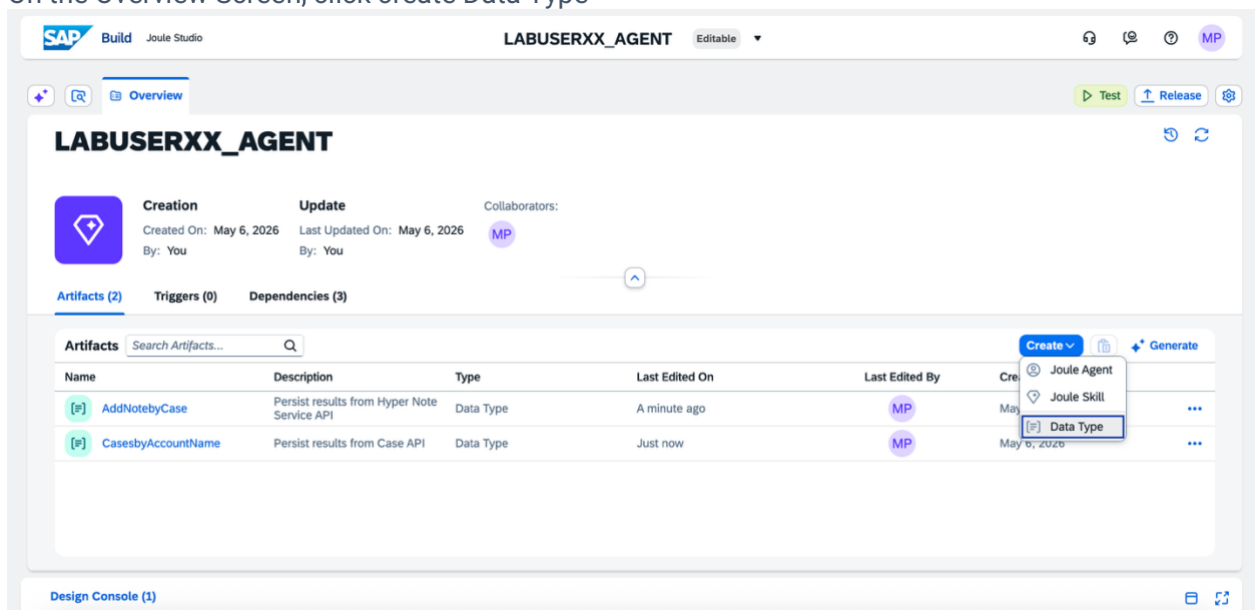
45.
 46. Click New Field, enter the name 'CasebyAccountName' and click the schema field
 47. Select the schema 'get_querycaseservice_case_200_output_schema'



48.
 49. Confirm the API schema has been added and click Save

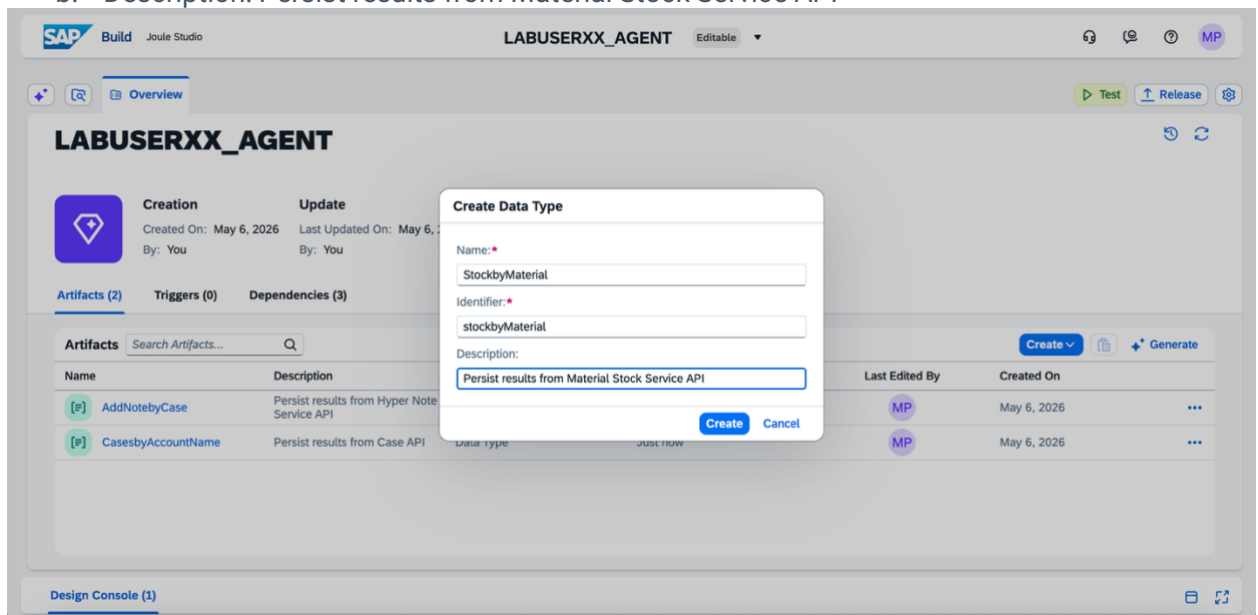


50.
 51. On the Overview Screen, click create Data Type

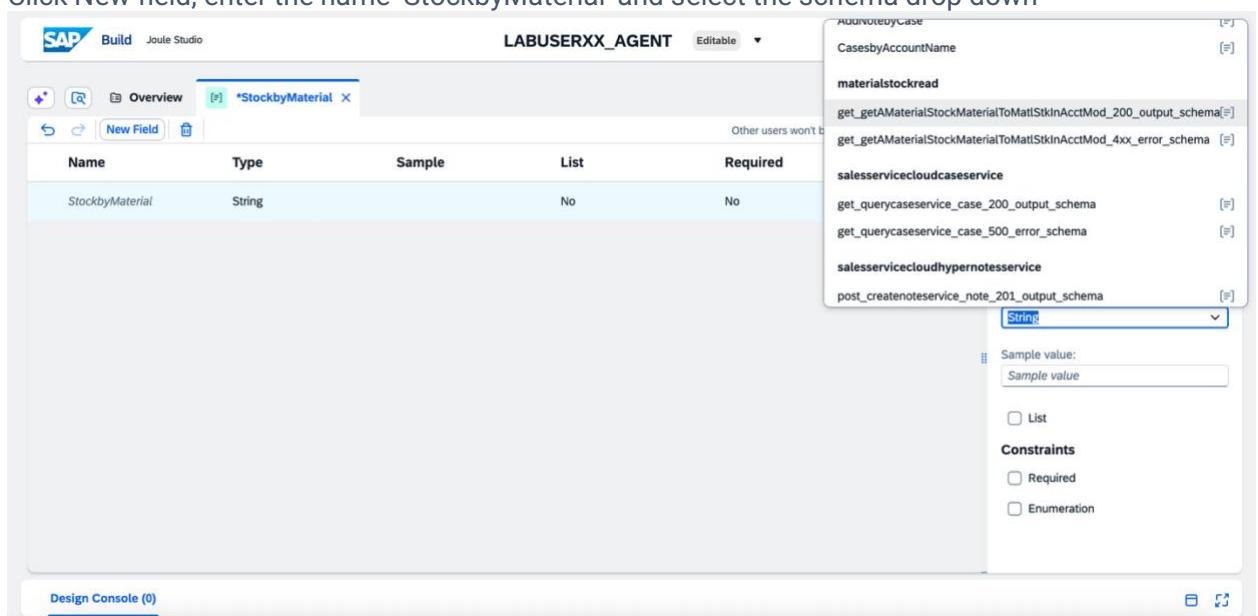


52.
 53. Enter the following details

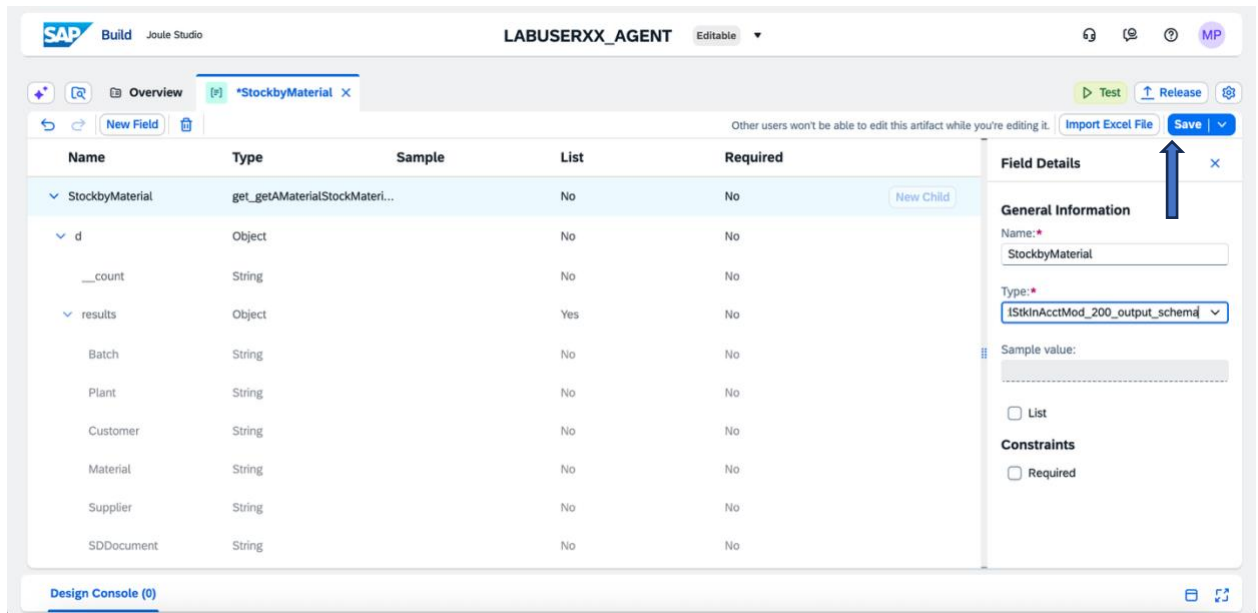
- a. Name: StockbyMaterial
- b. Description: Persist results from Material Stock Service API



- 54.
55. Click Create
56. Click New field, enter the name 'StockbyMaterial' and select the schema drop down

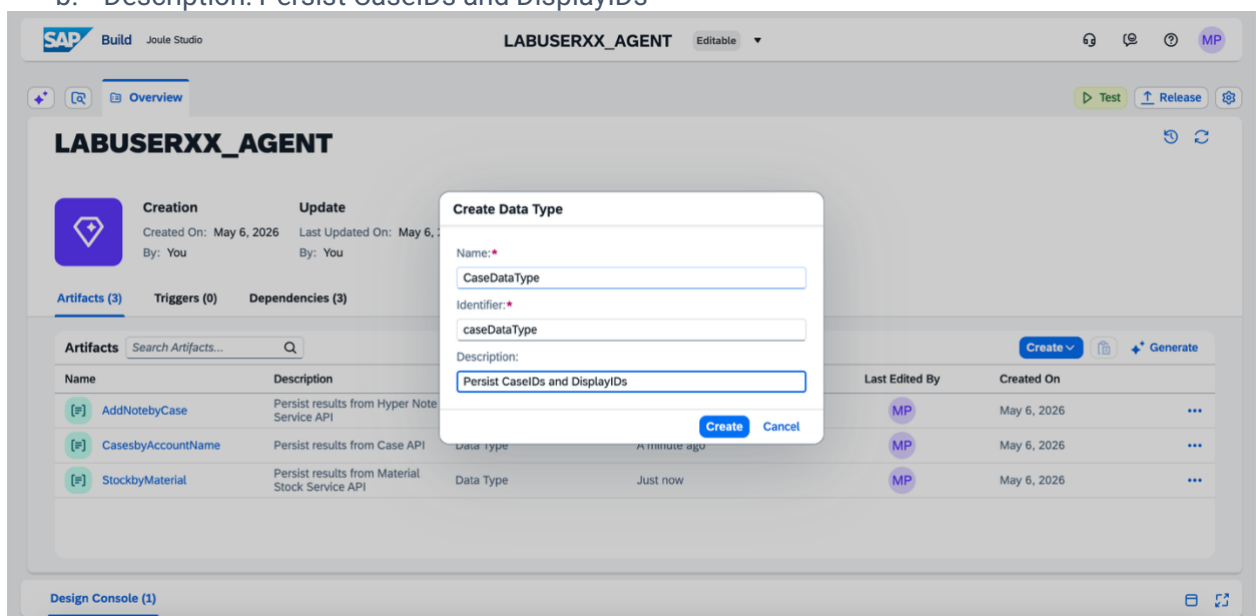


- 57.
58. Select the schema 'get_getAMaterialStockMaterialToMatlStknAcctMod_200_output_schema'
59. Verify the schema is populated and click Save

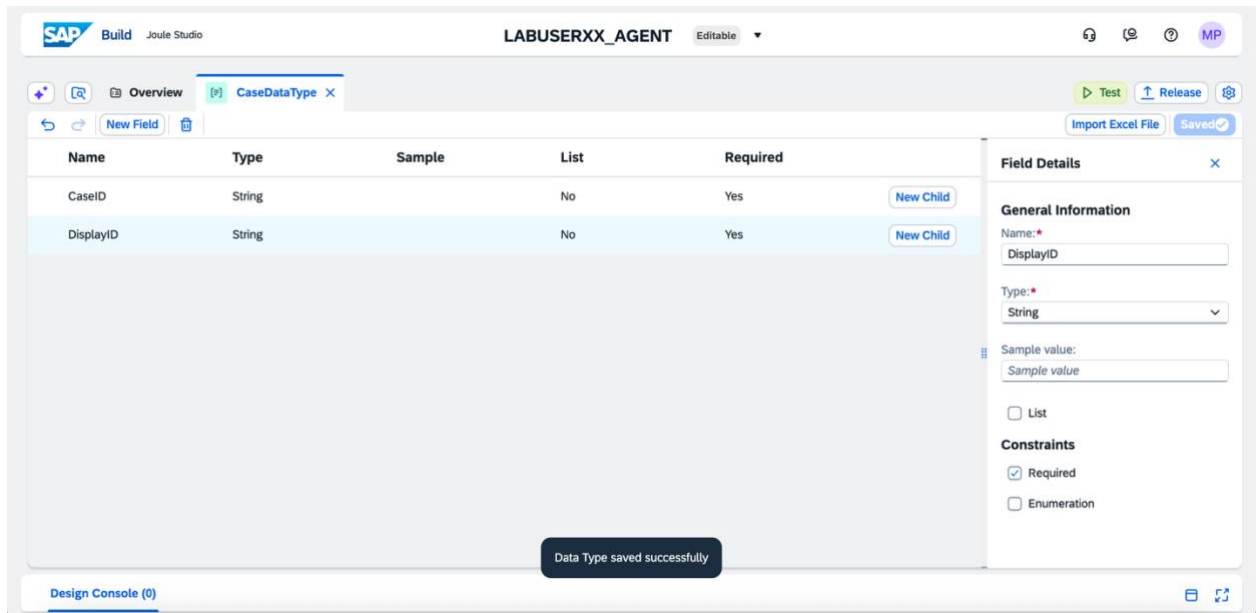


60. ...
61. For the final Data Type, click Create again from the Overview page.

62. Enter the details
- Name: CaseDataType
 - Description: Persist CaselDs and DisplayIDs



63. ...
64. Create two fields in the Data Type as shown below
- CaselD (Required)
 - DisplayID (Required)

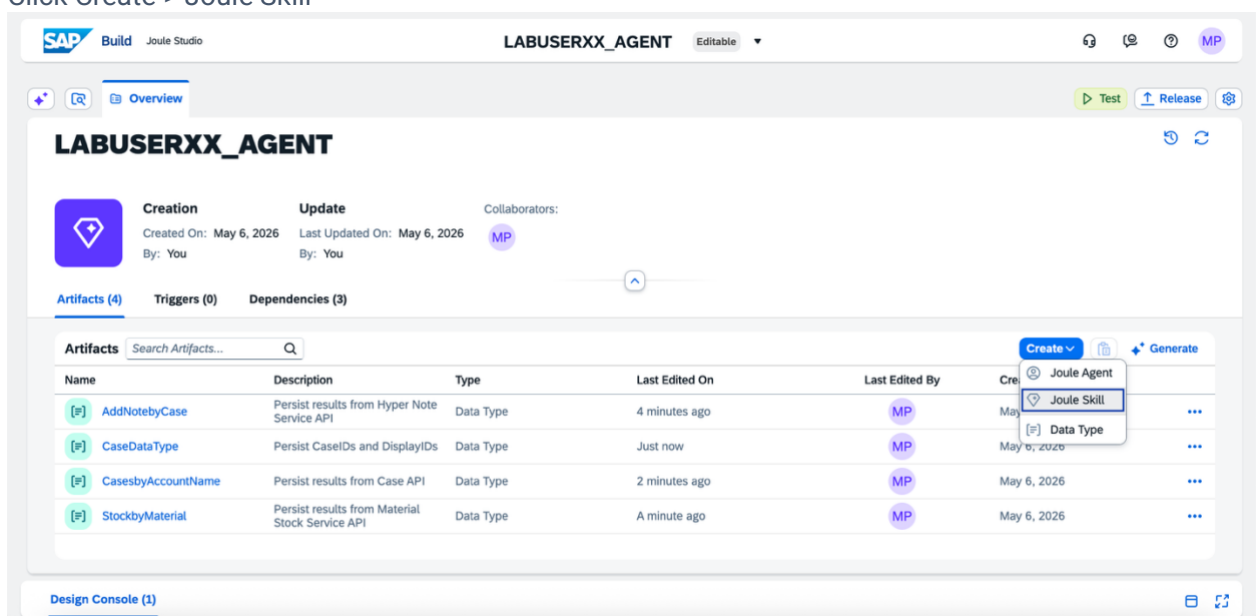


65.

66. Click Save and return to the Overview page.

Create Joule Skills

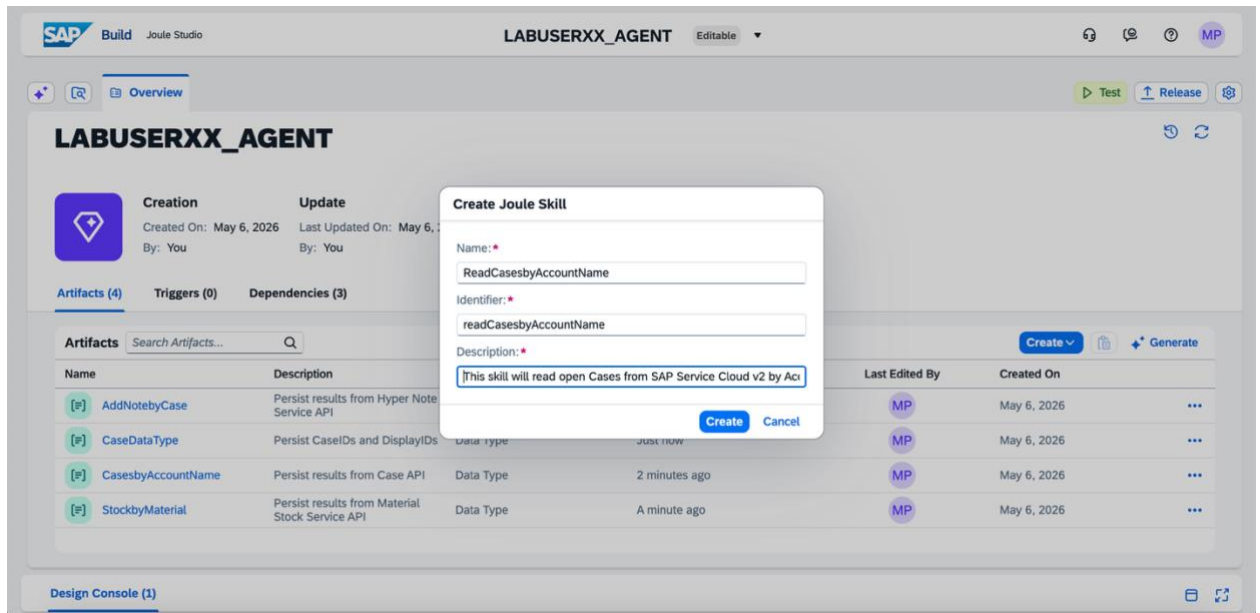
67. Click Create > Joule Skill



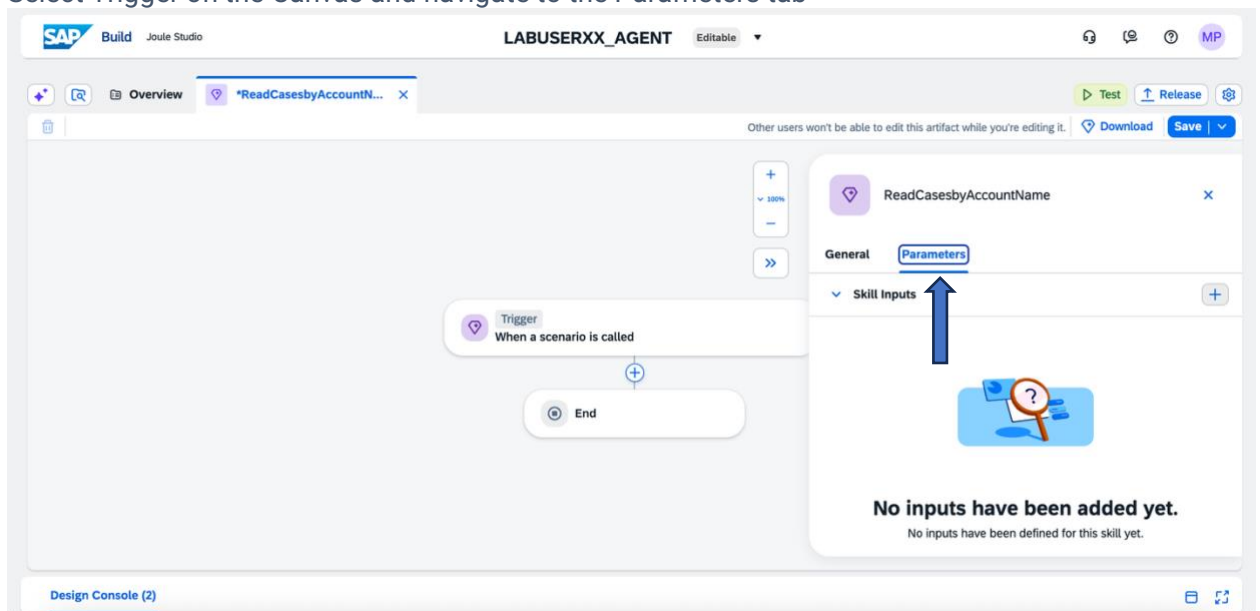
68.

69. Enter the following details

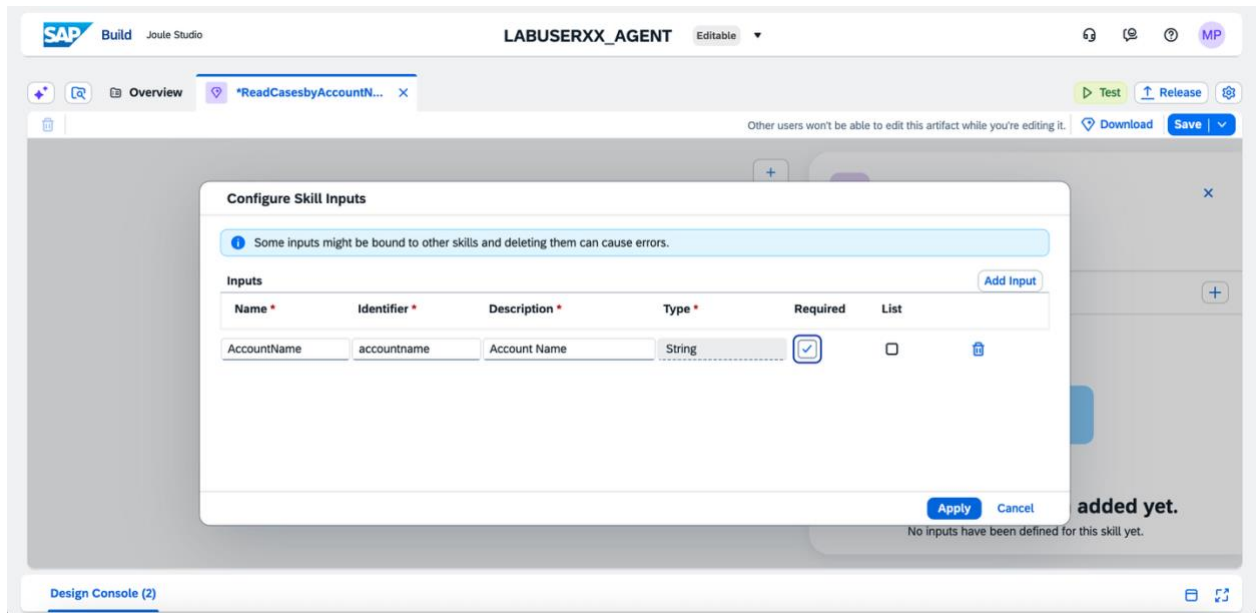
- Name: ReadCasesbyAccountName
- Description: This skill will read open Cases from SAP Service Cloud v2 by Account Name. Case details will include the status, contact details, the reporter, the case description, and related notes from the case.



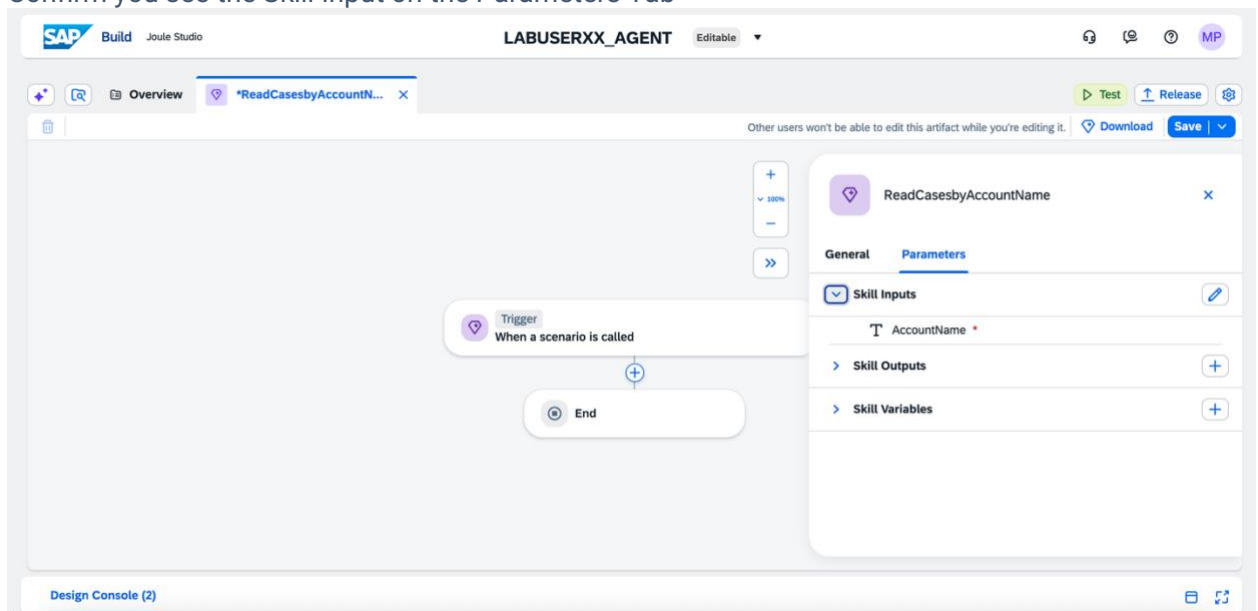
- 70.
71. Click Create
72. Select Trigger on the Canvas and navigate to the Parameters tab



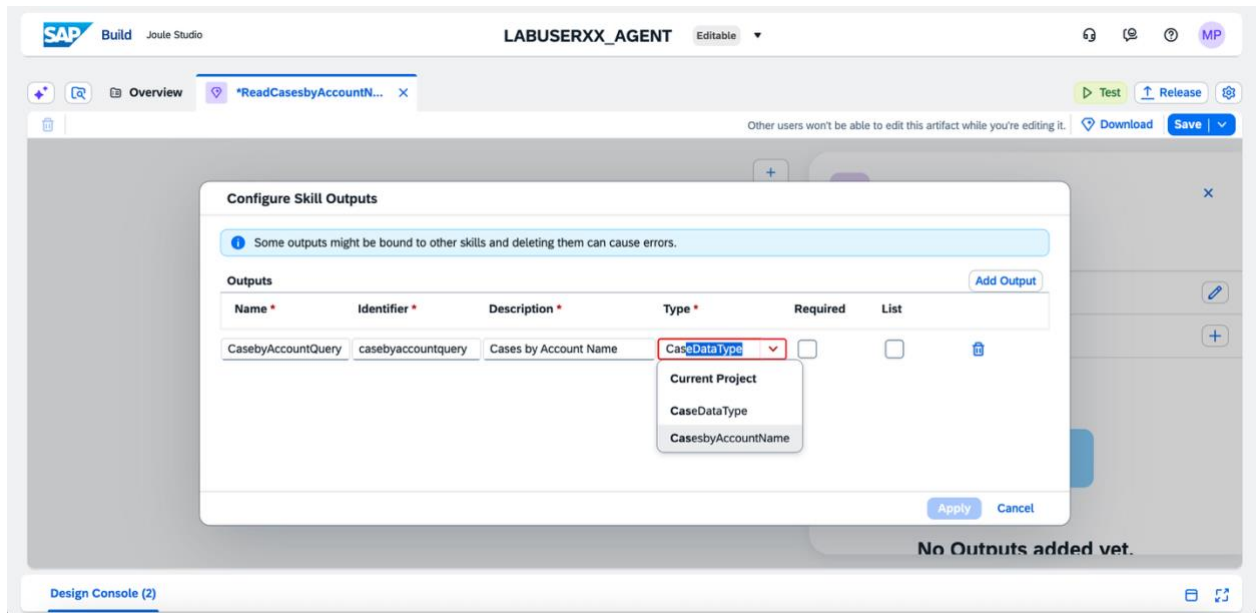
- 73.
74. Click the + button to add a Skill Input Parameter
75. Enter the details below



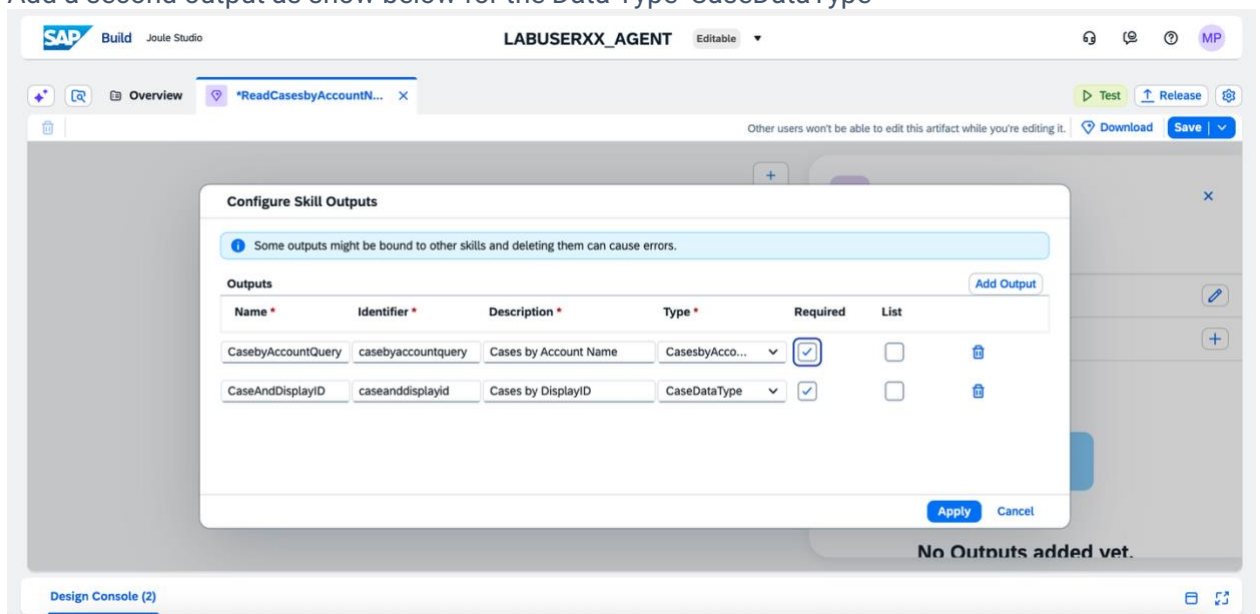
76. Click Apply
77. Click Apply
78. Confirm you see the Skill Input on the Parameters Tab



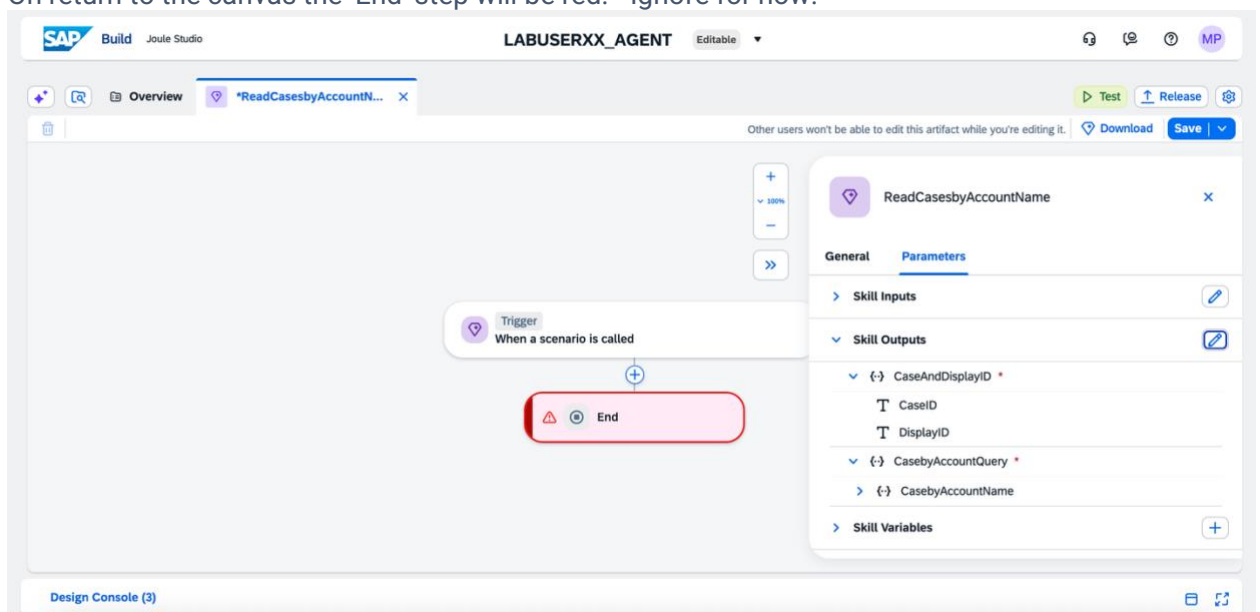
79. Expand Skill Outputs and click the + button to add an output
80. Expand Skill Outputs and click the + button to add an output
81. Enter the name 'CaseAccountQuery' and for the Type drop down select the DataType you created above 'CasesbyAccountName'



82. ...
83. Add a second output as show below for the Data Type 'CaseDataType'

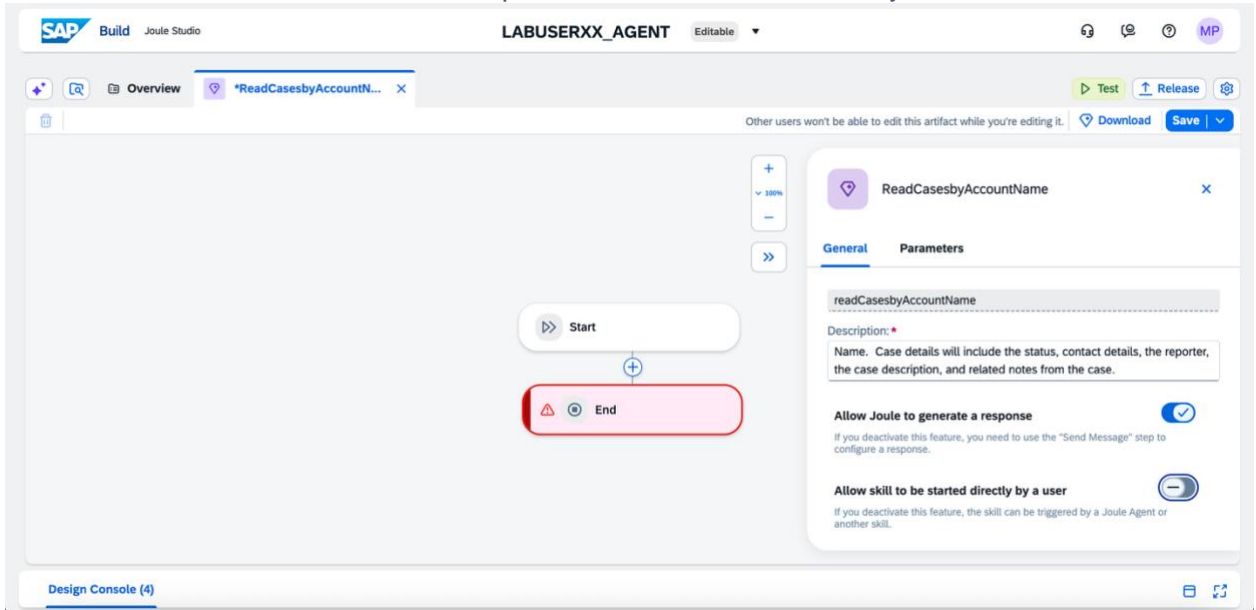


84. ...
85. Click Apply.
86. On return to the canvas the 'End' step will be red. Ignore for now.



87. ...

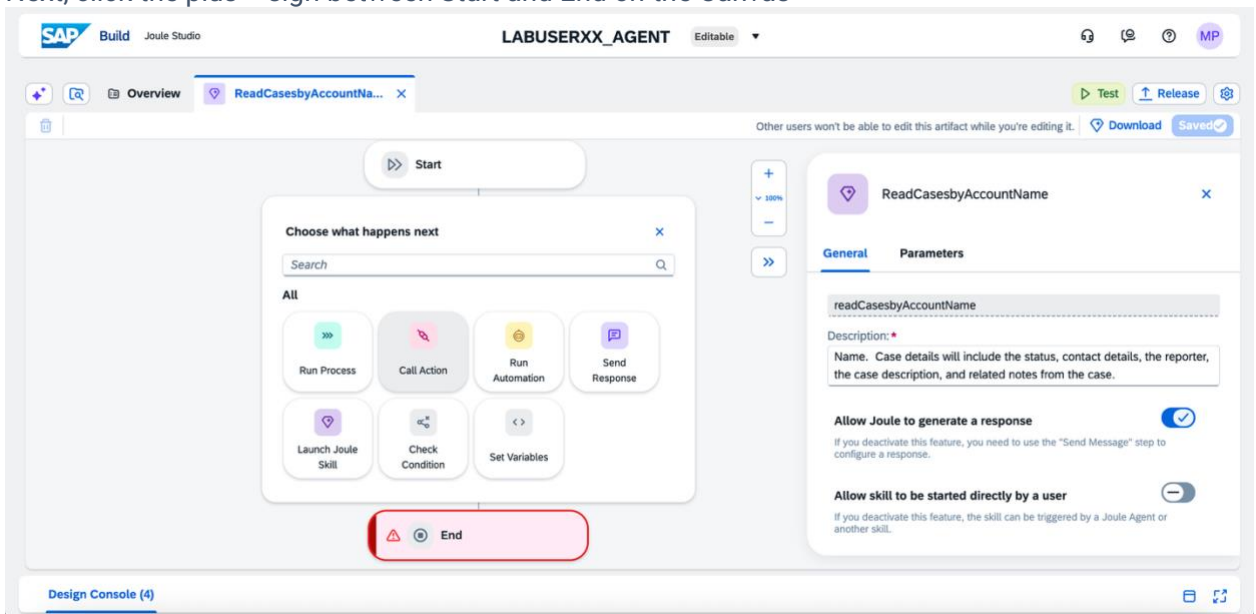
88. Return the General tab and disable the option 'Allow skill to be started by a user'



89.

90. Click Save in the upper right.

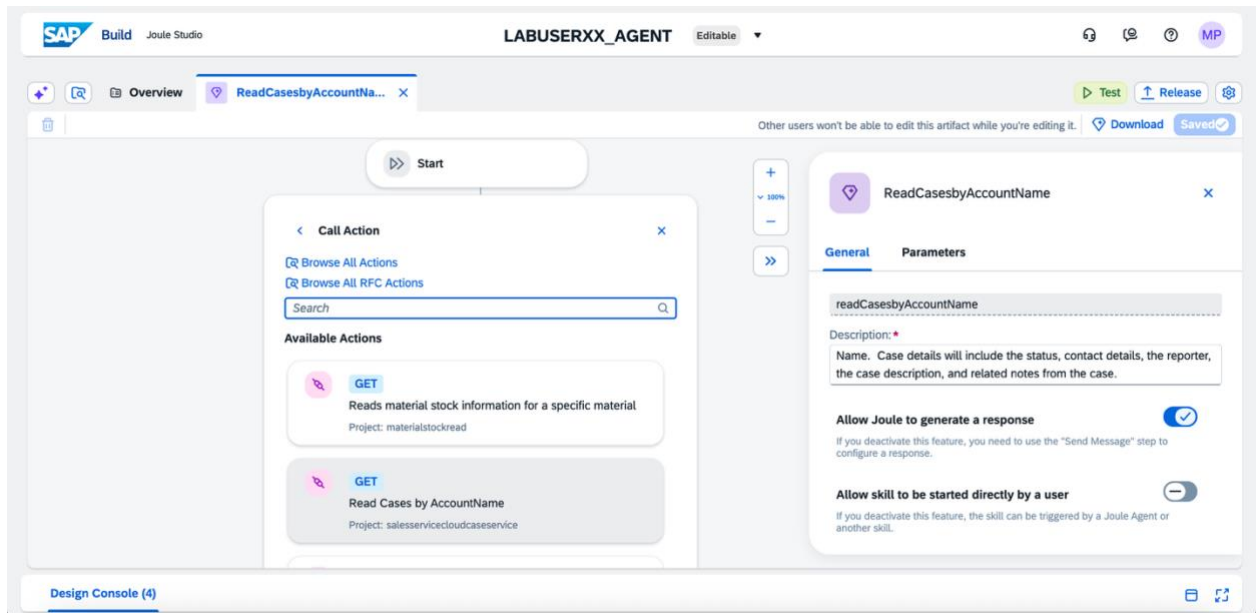
91. Next, click the plus + sign between Start and End on the Canvas



92.

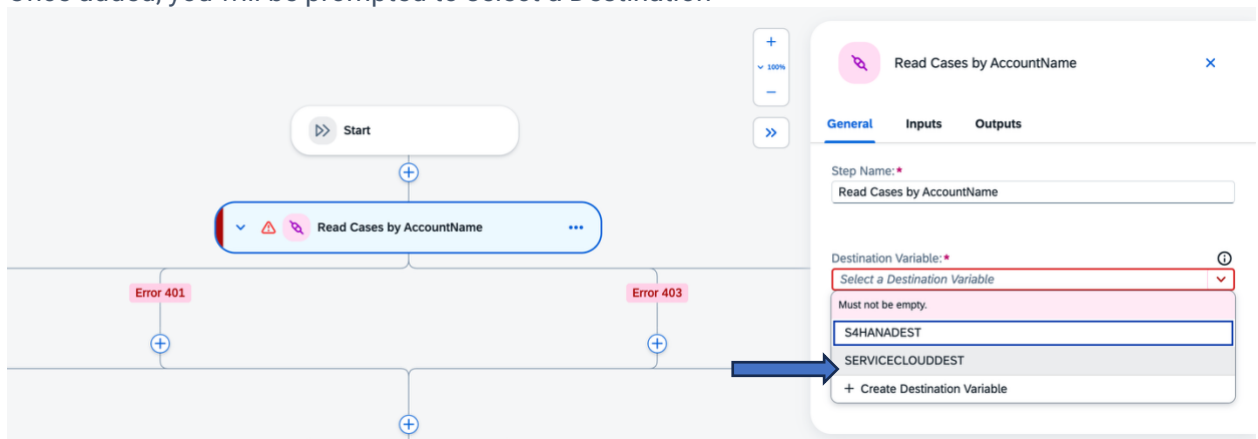
93. Click 'Call Action'

94. Select 'Read Cases by Account Name'



95.

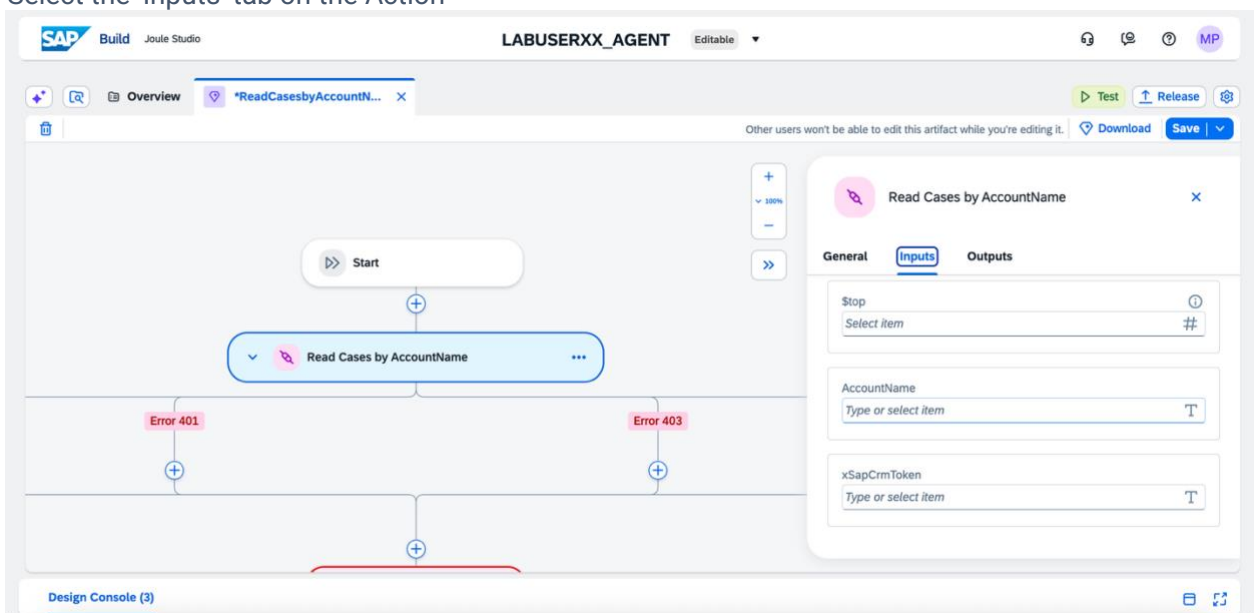
96. Once added, you will be prompted to select a Destination



97.

98. Select SERVICECLOUDDEST as the Destination Variable

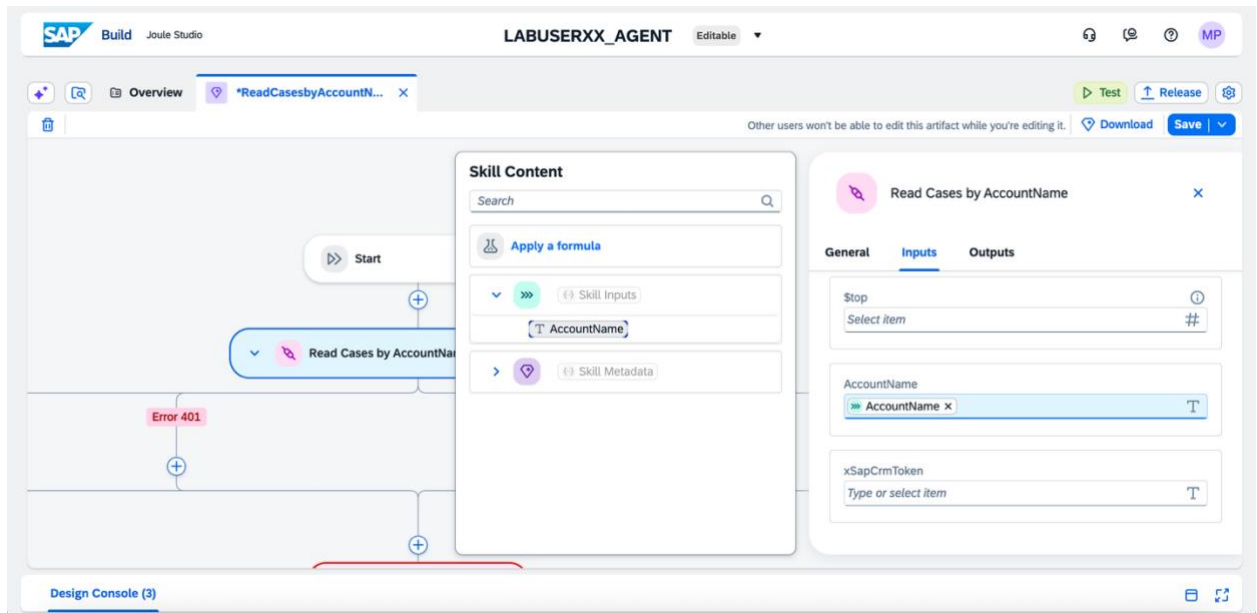
99. Select the 'Inputs' tab on the Action



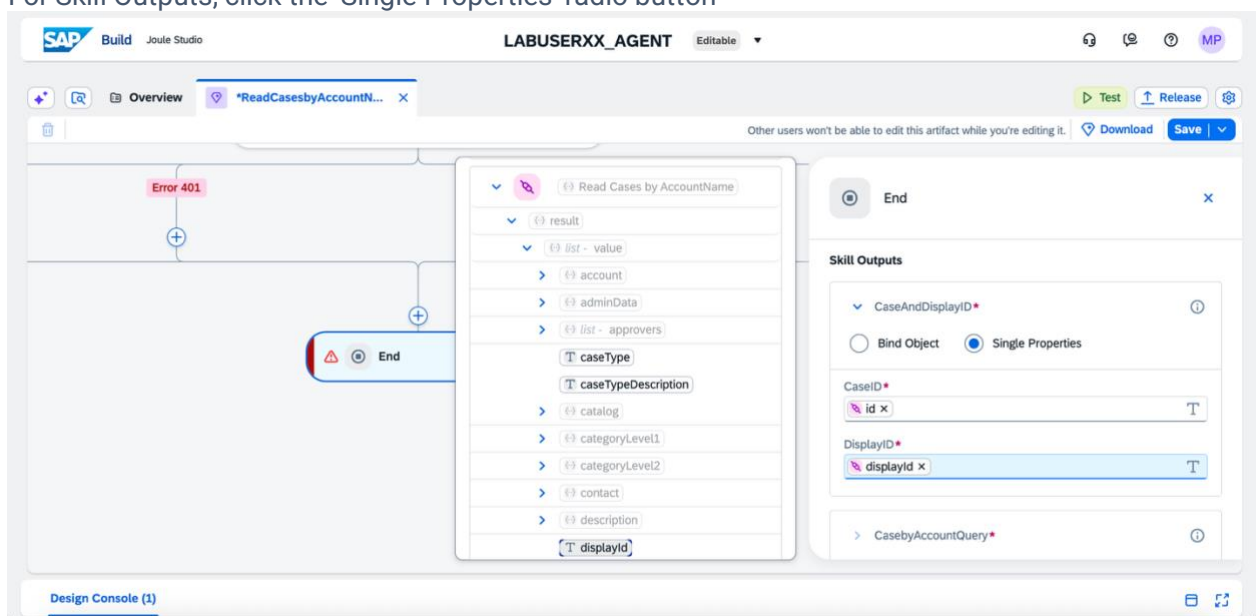
100.

101. Scroll down to the AccountName and select the field.

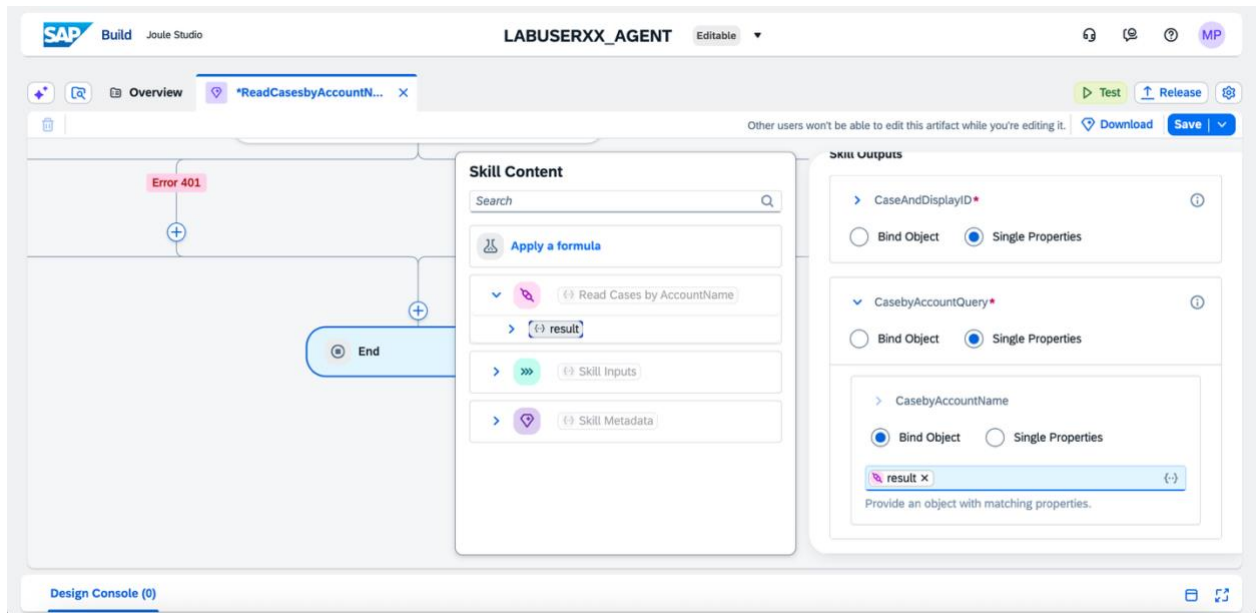
102. Select 'AccountName' from Skill Inputs



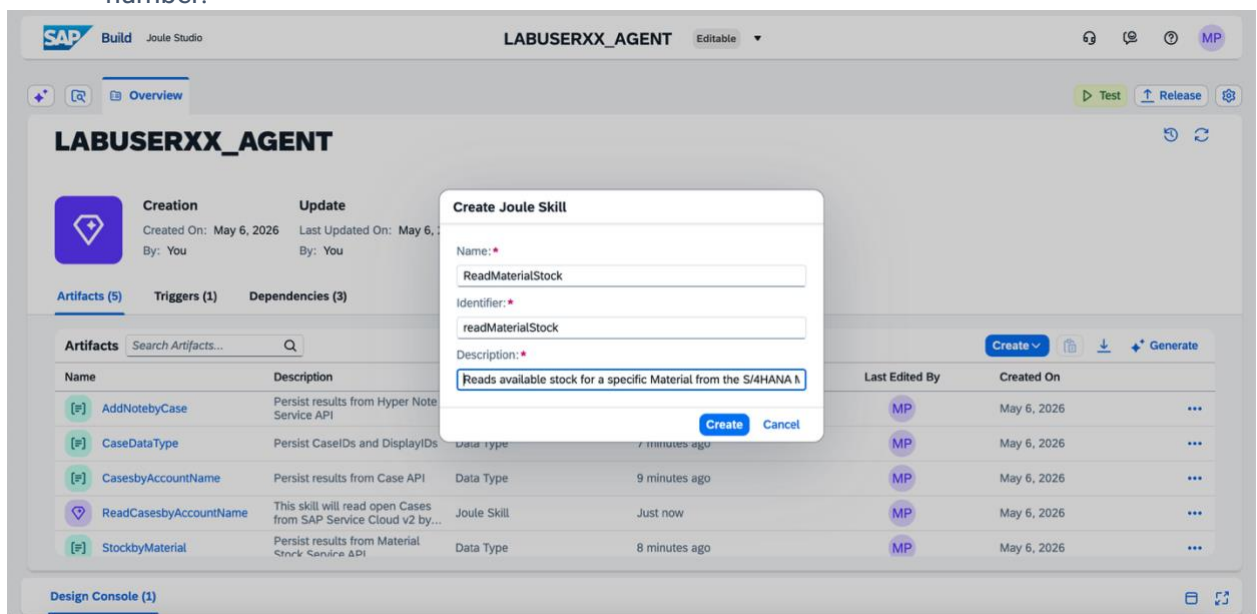
- 103.
104. Click Save
105. Next, select the End step (currently red)
106. For Skill Outputs, click the 'Single Properties' radio button



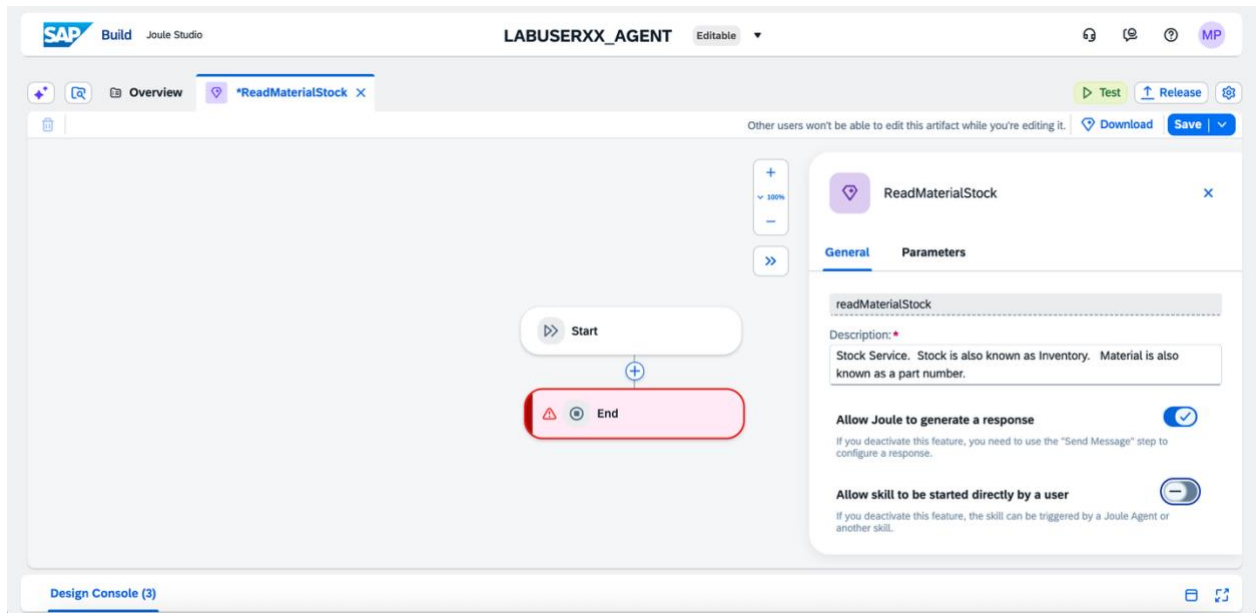
- 107.
108. Select the field for CaseID and map the API Response field 'id'
109. Select the field for DisplayID and map the API Response field 'displayId'
110. Open CasebyAccountQuery and select the radio button for 'Single Properties'



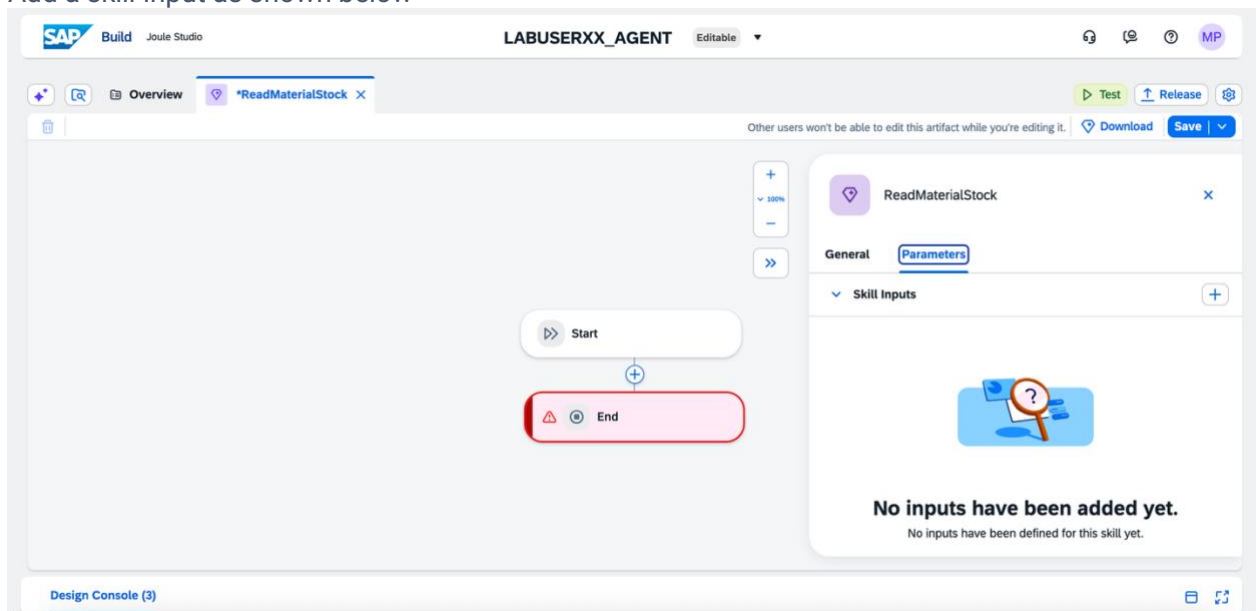
- 111.
112. Under CasebyAccountName, select the Bind Object radio button
113. Map the API 'result' to the CasebyAccountName. Click Save and close the tab for the Joule Skill.
114. Next, we're going to create a Joule Skill for ReadMaterialStock
115. From the Overview page, click Create > Joule Skill
 - a. Name: ReadMaterialStock
 - b. Description: Reads available stock for a specific Material from the S/4HANA Material Stock Service. Stock is also known as Inventory. Material is also known as a part number.



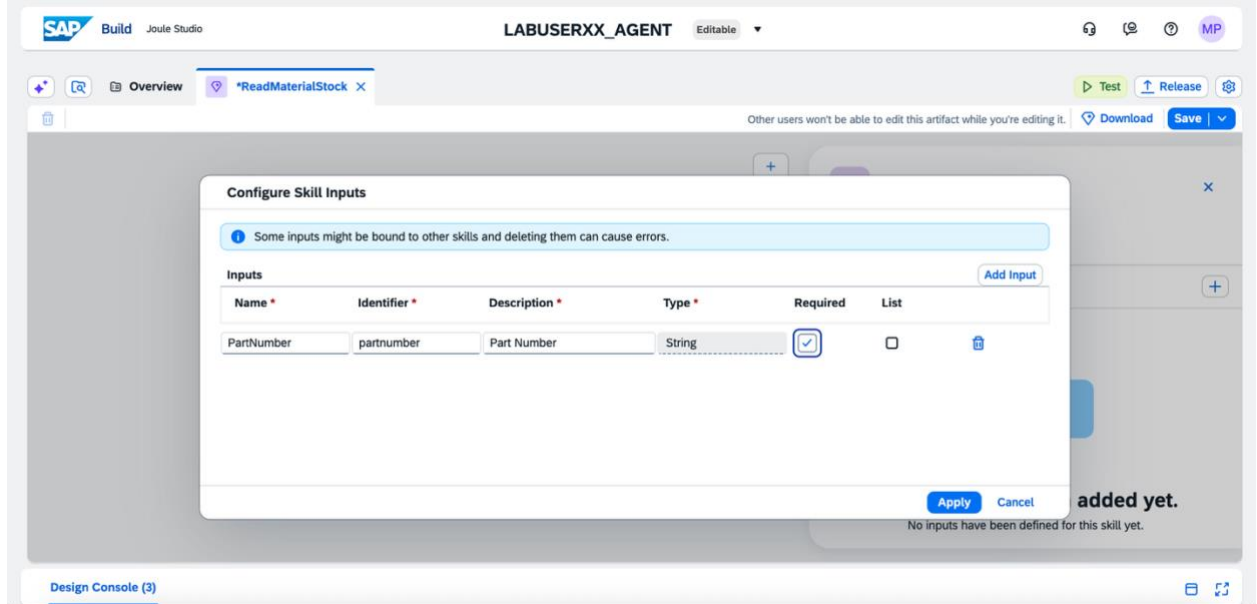
- 116.
117. Click Create
118. Immediately navigate to the General Tab of the Joule Skills and disable the option to allow the user to call the Joule Skill directly



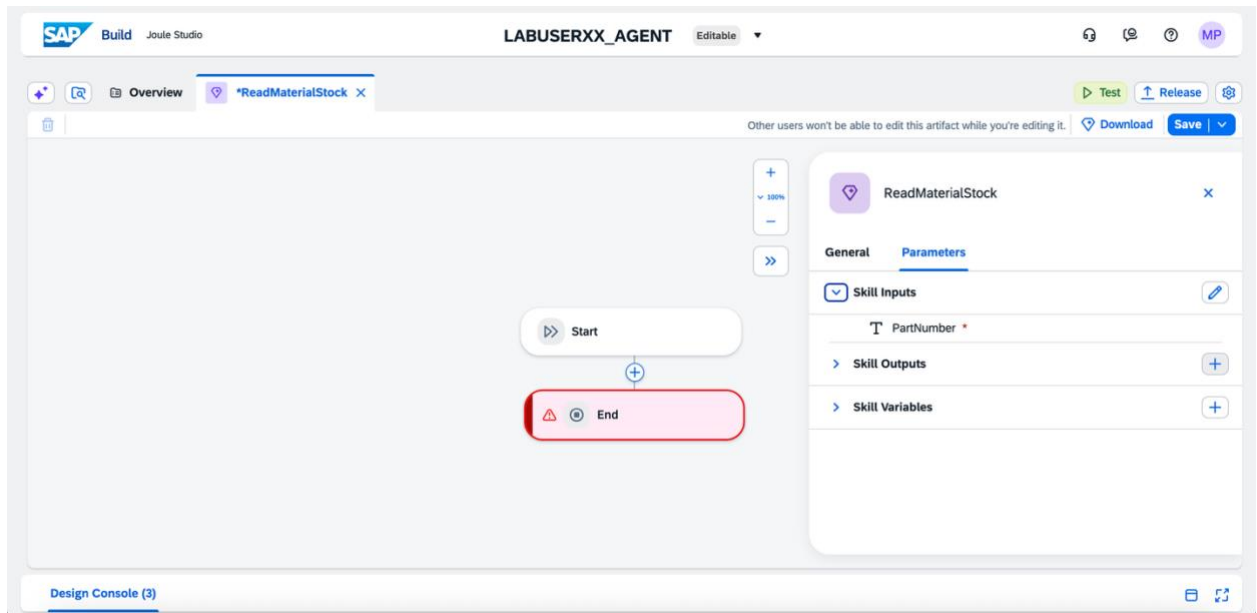
- 119.
120. Click the Parameters Tab
121. Add a skill Input as shown below



122.



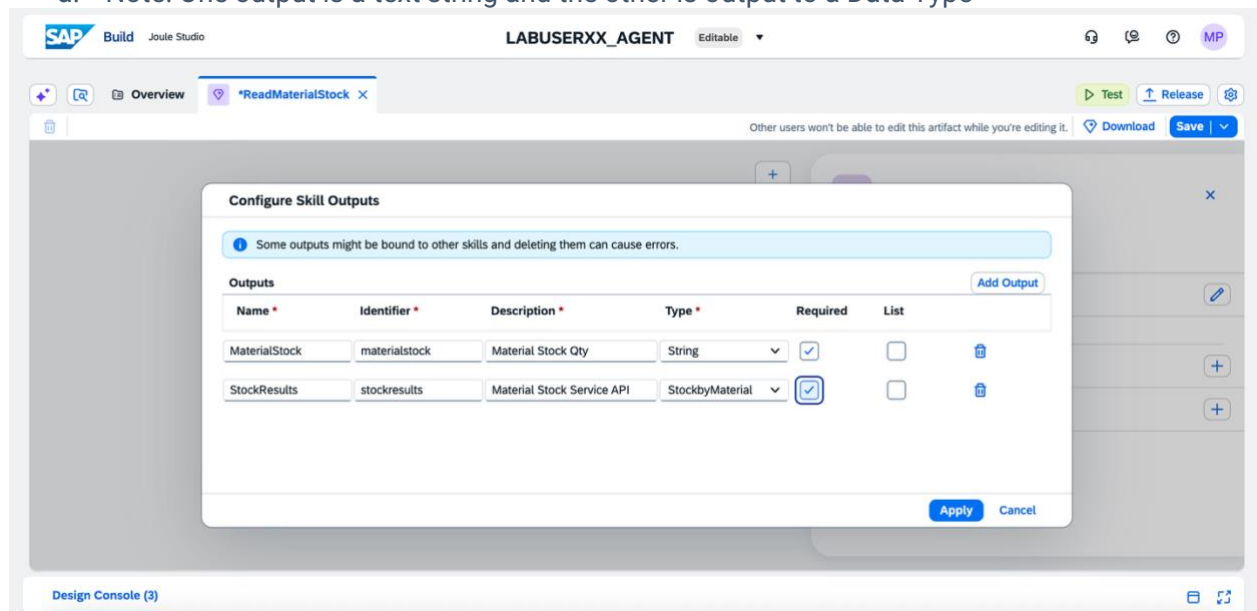
- 123.
124. Click Apply and confirm your Skill Input



125.

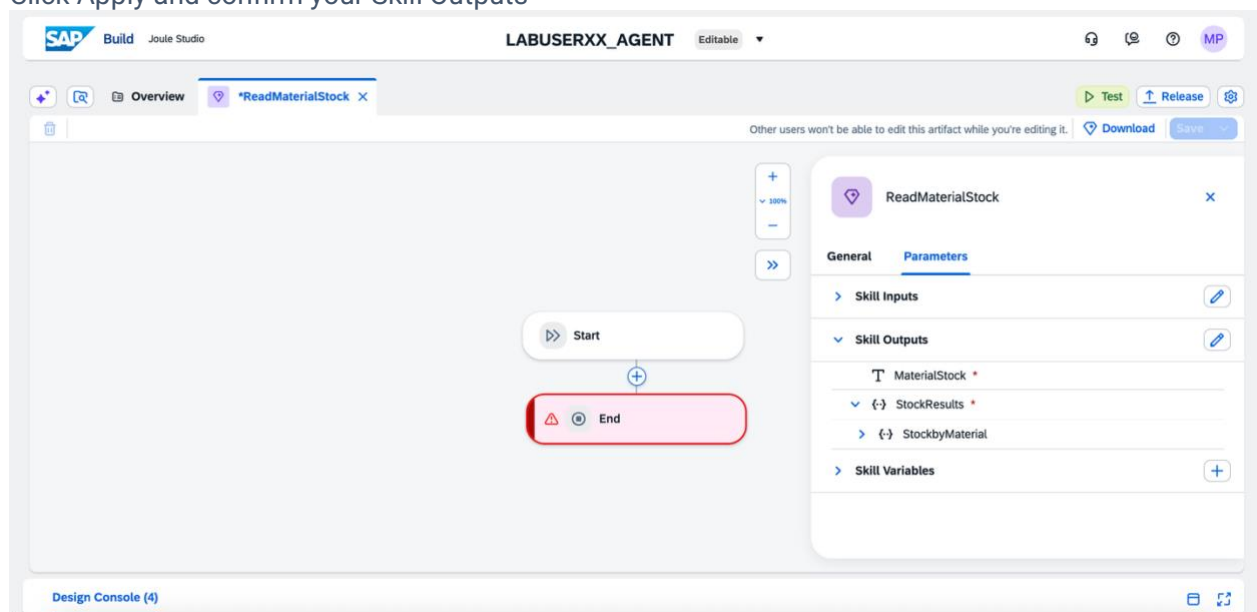
126. Next, add a Skill Output as shown below.

a. Note: One output is a text string and the other is output to a Data Type



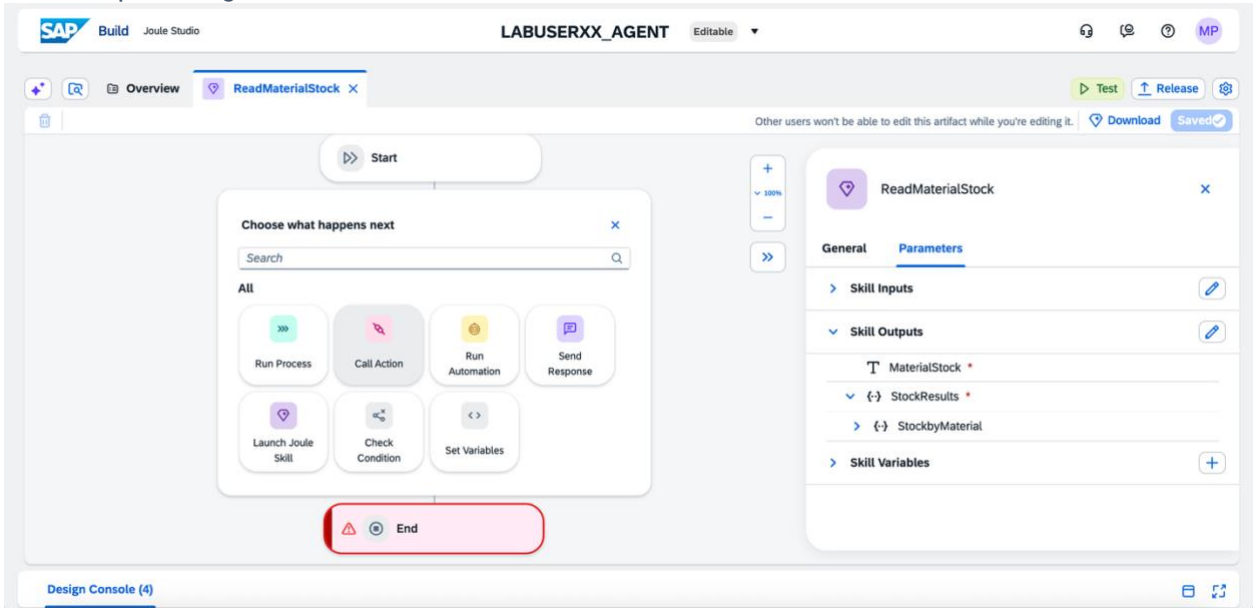
127.

128. Click Apply and confirm your Skill Outputs

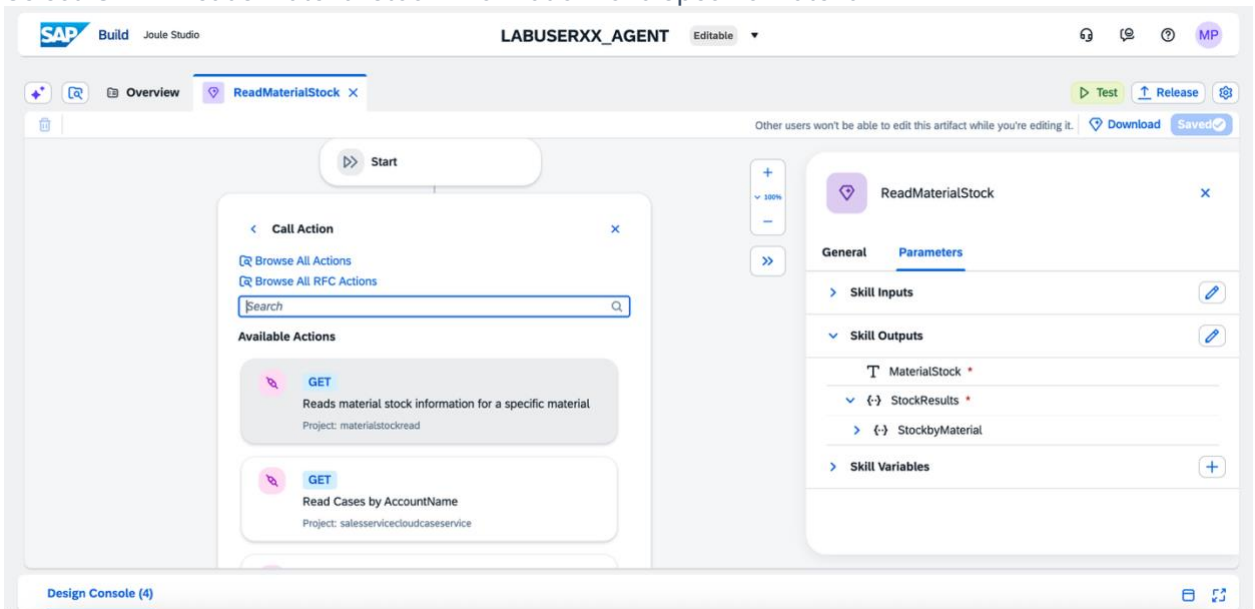


129.

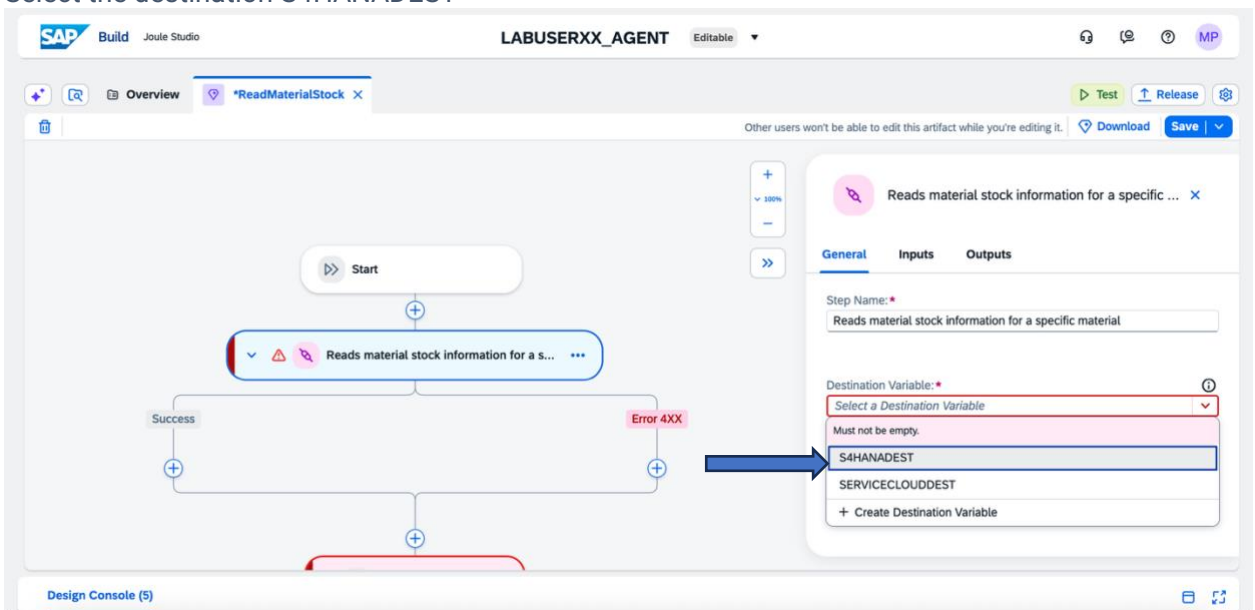
130. Click the plus + sign between Start and End to add a Call Action



131. Select 'GET – Reads Material stock information for a specific material'

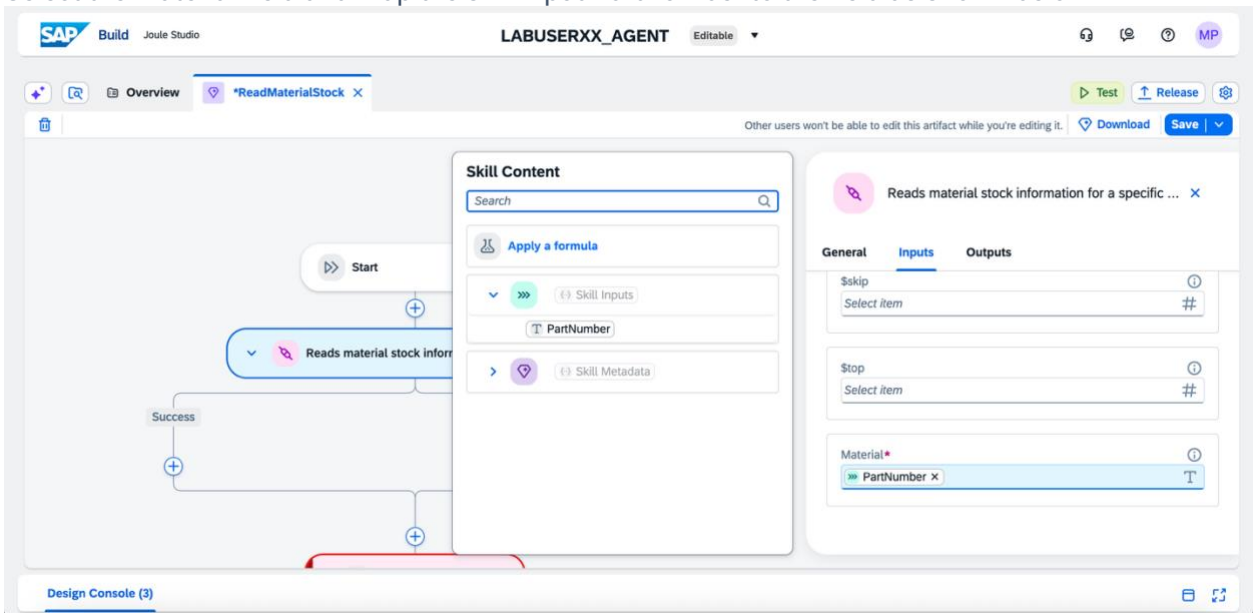


133. Select the destination S4HANADEST

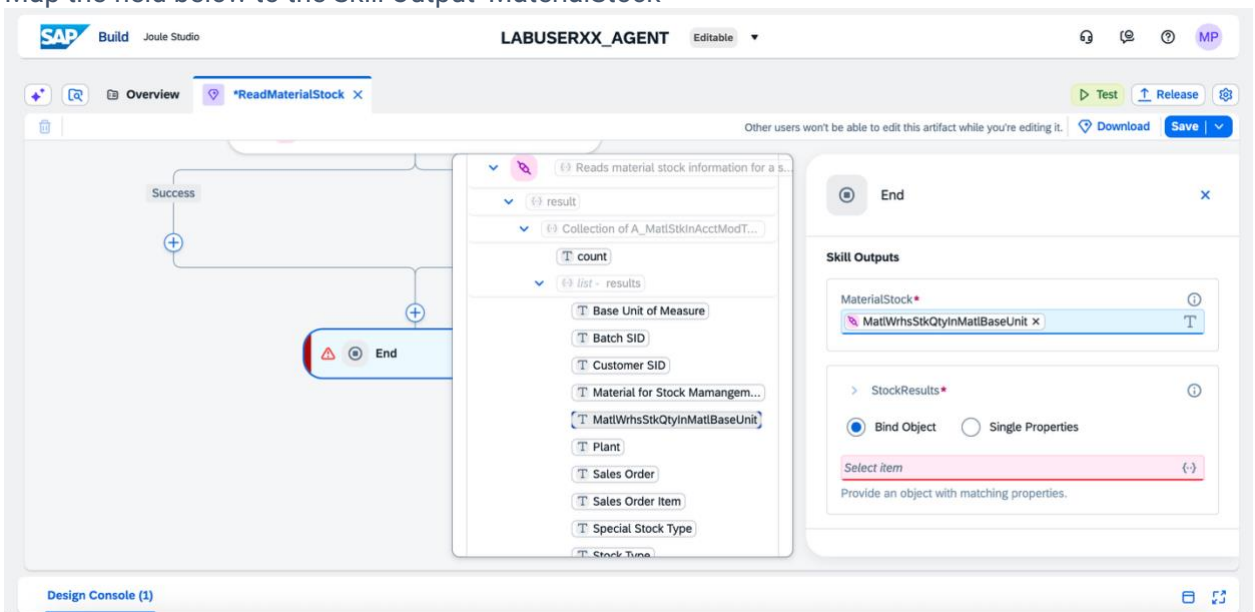


135.

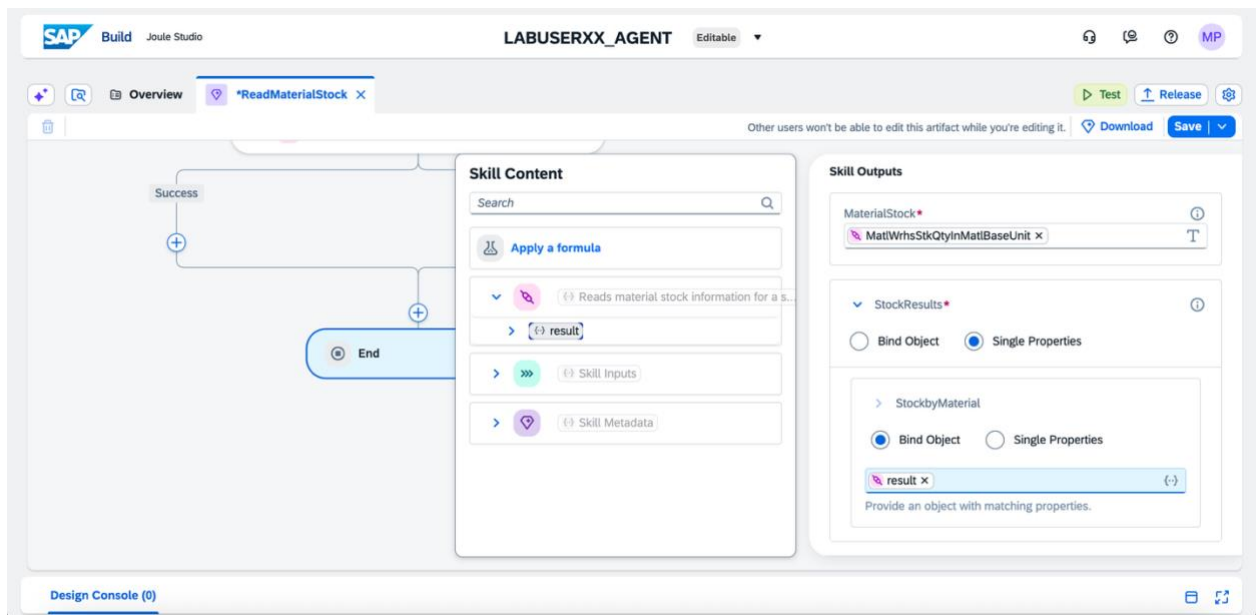
136. For the newly added Action, select the Inputs tab.
137. Select the Material field and map the skill input PartNumber to the field as shown below



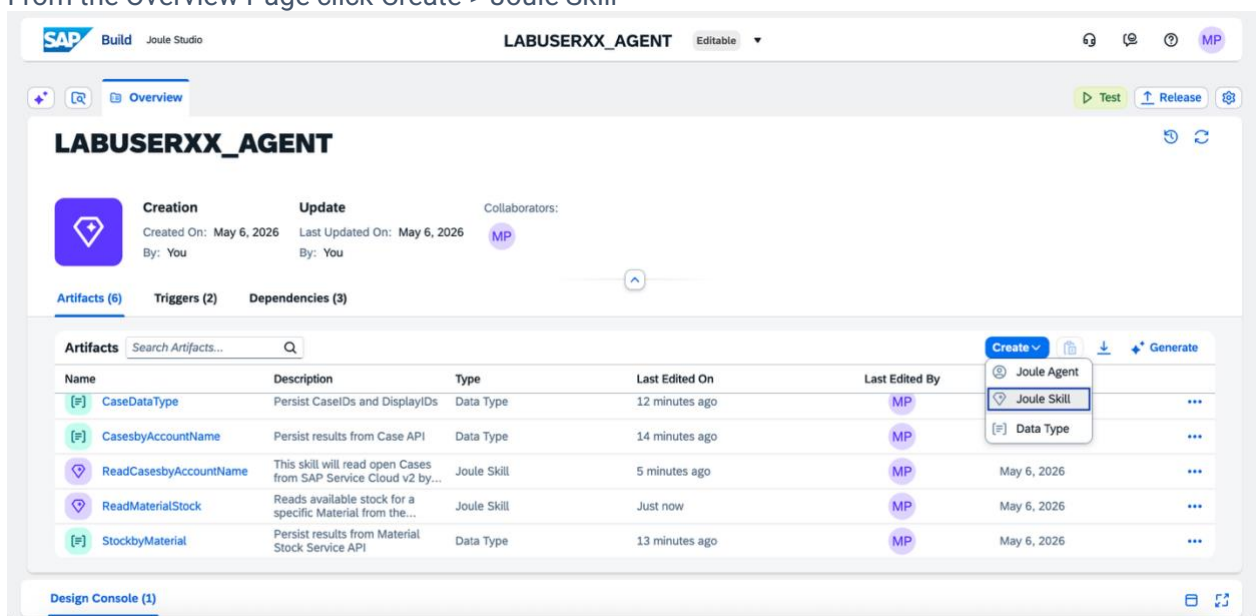
- 138.
139. Click Save
140. Next, click on the End Step (in red)
141. Map the field below to the Skill Output 'MaterialStock'



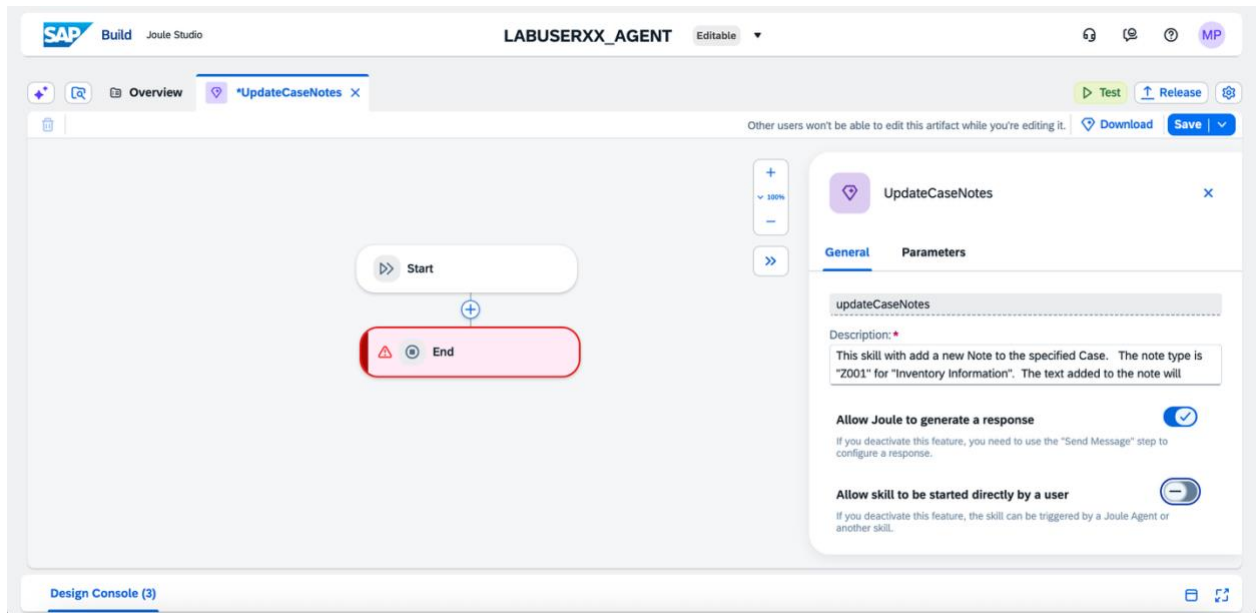
- 142.
143. Expand StockResults, select Single Properties Radio button
144. Bind the API 'result' to StockbyMaterial



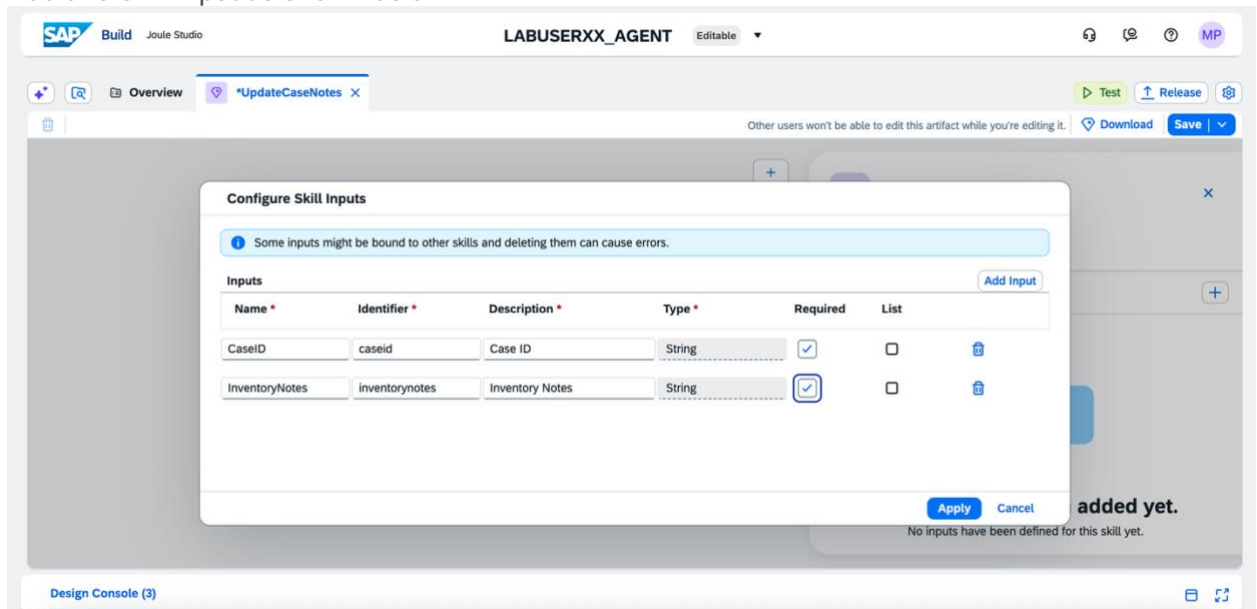
145.
146. Click Save
147. Next, we're going to add one more Joule Skill for caseNotes
148. From the Overview Page click Create > Joule Skill



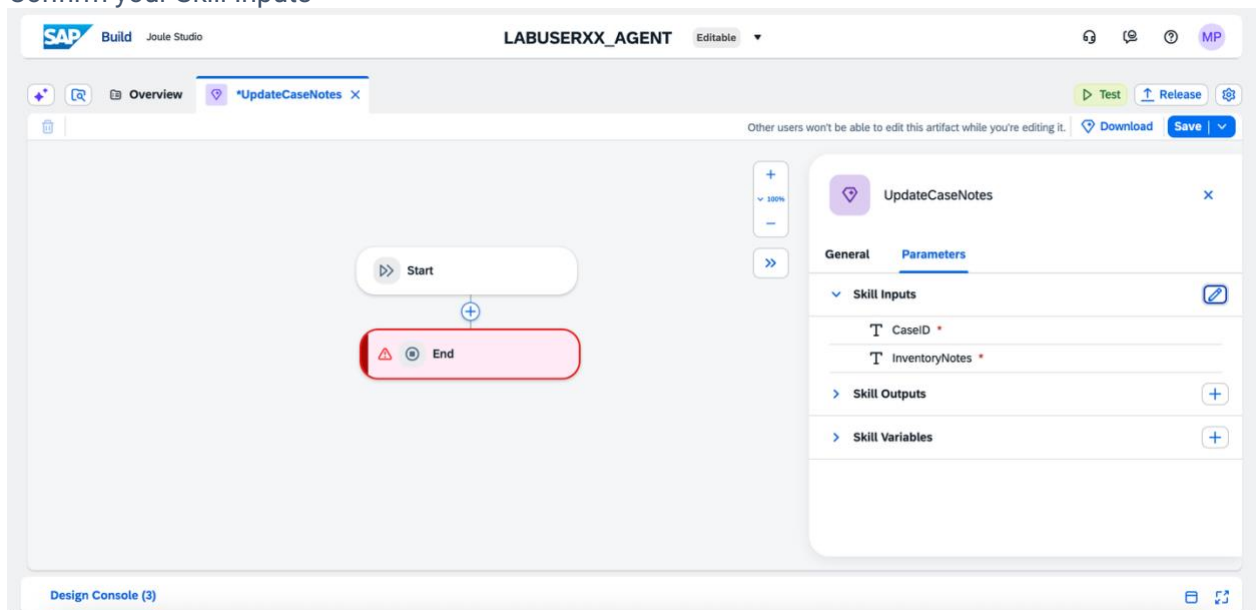
149.
150. Enter the following information
 a. Name: UpdateCaseNotes
 b. Description: This skill with add a new Note to the specified Case. The note type is "Z001" for "Inventory Information". The text added to the note will include part number and available inventory information
 151. On the Skill General Tab, deselect the option to allow a user to call the skill directly



152.
 153. Click on the Parameters Tab
 154. Add two Skill Input as shown below

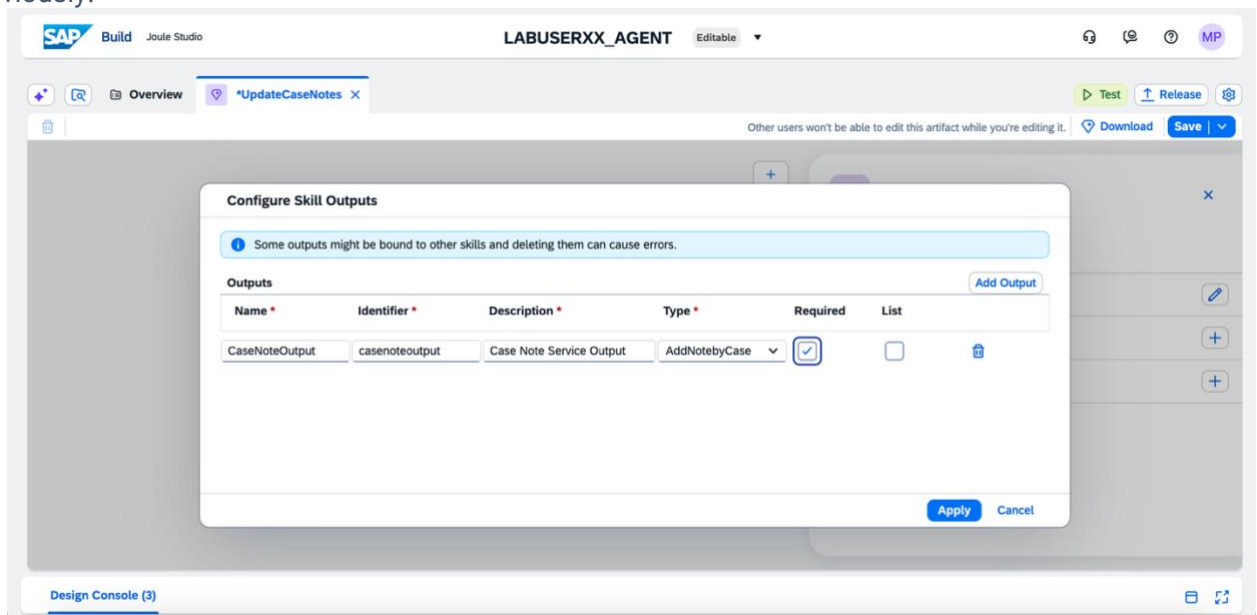


155.
 156. Confirm your Skill Inputs

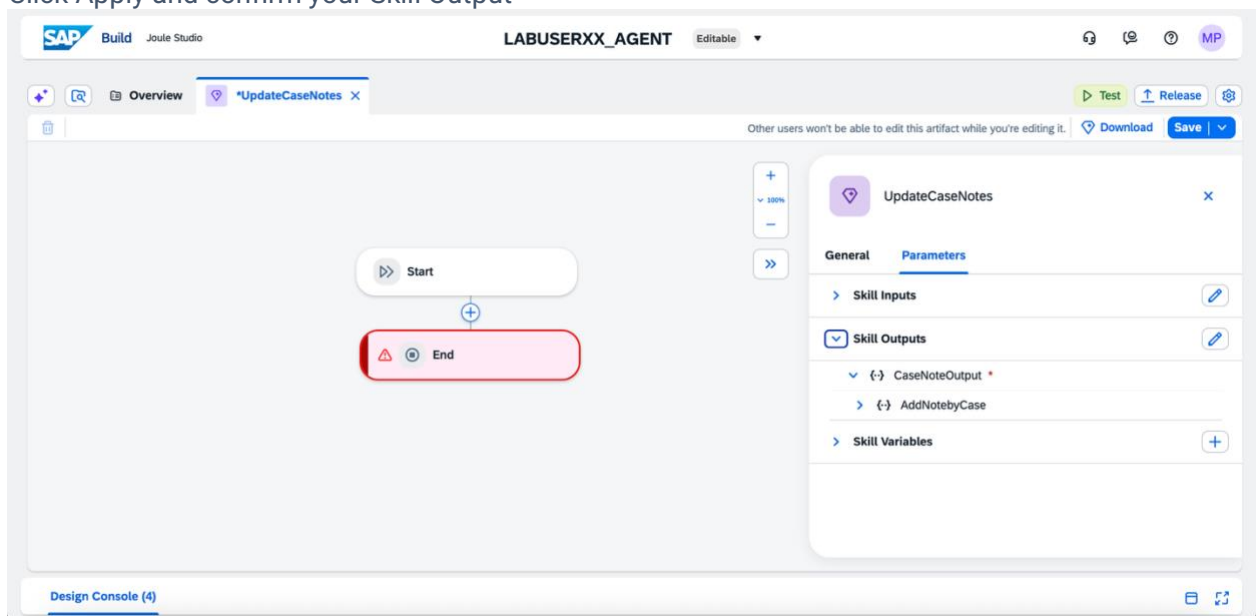


157.

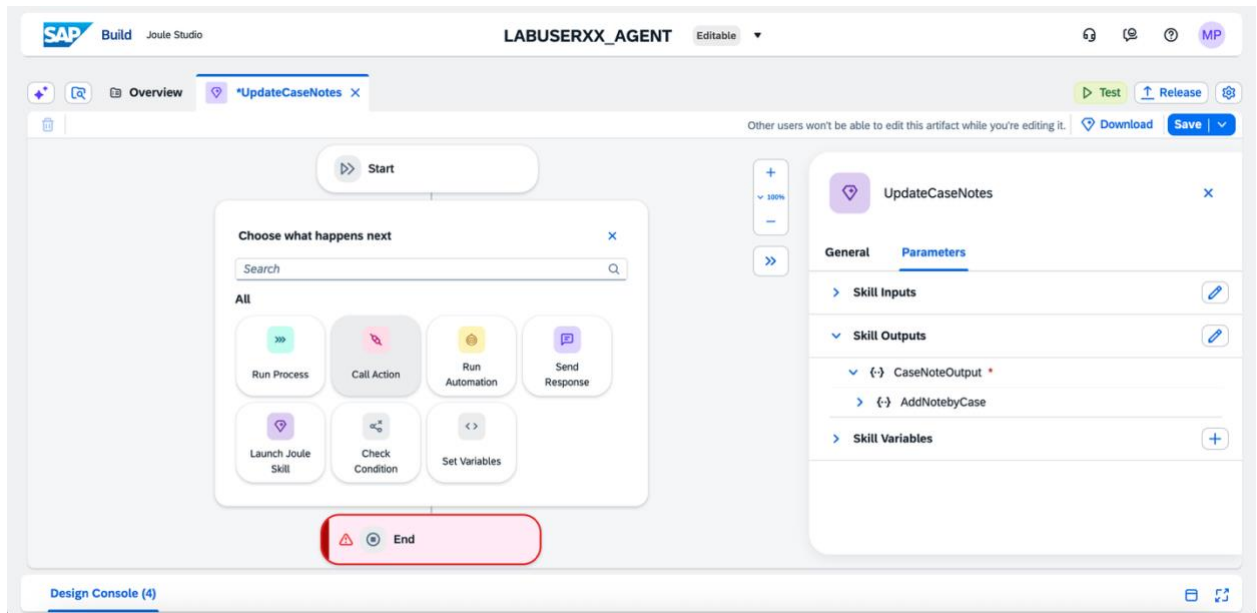
158. Add a Skill Output as shown below. The Type will be the Data Type 'AddNotebyCase' created previously.



159. Click Apply and confirm your Skill Output

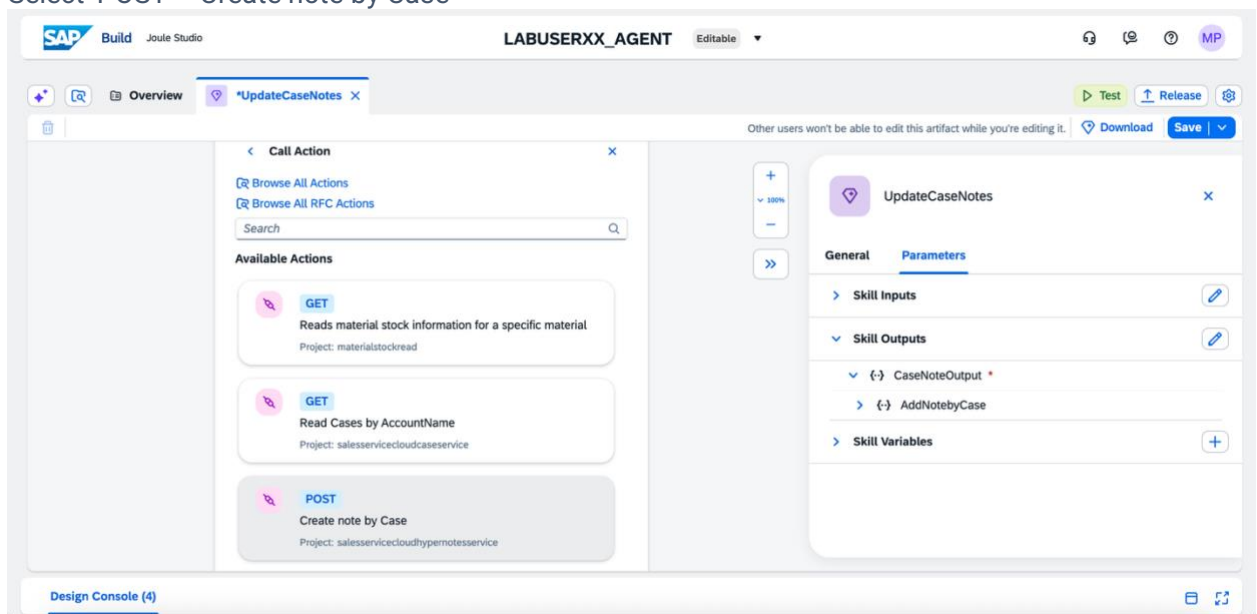


161. Click the plus + button between Start and End to add a Call Action



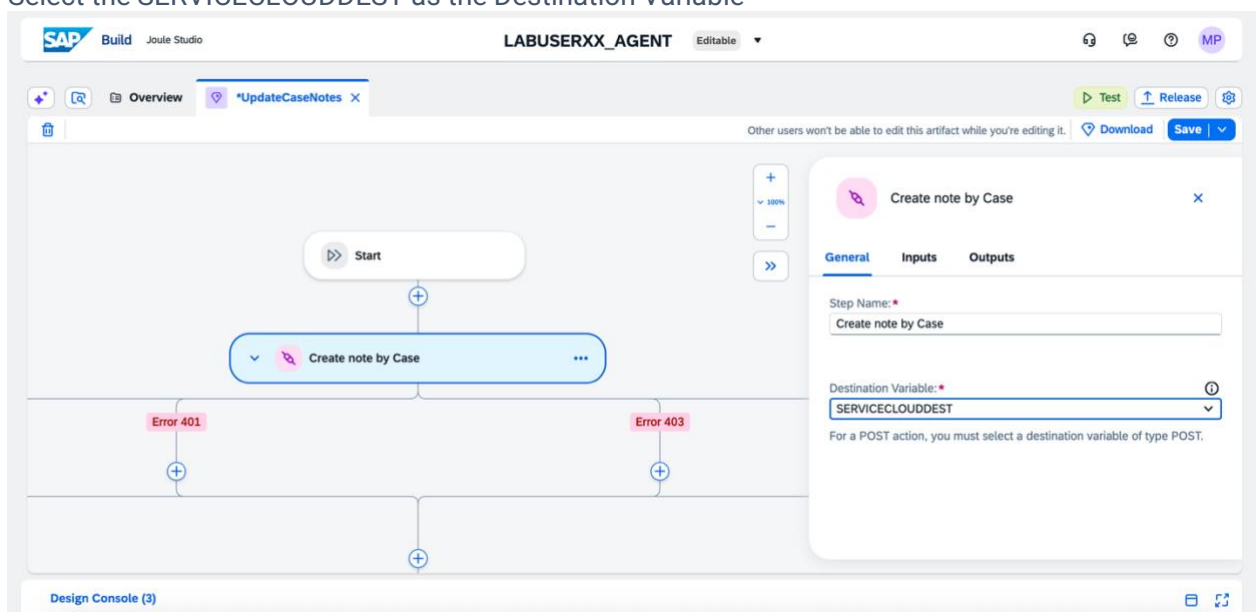
163.
164.

Select 'POST – Create note by Case'



165.
166.

Select the SERVICECLOUDDEST as the Destination Variable

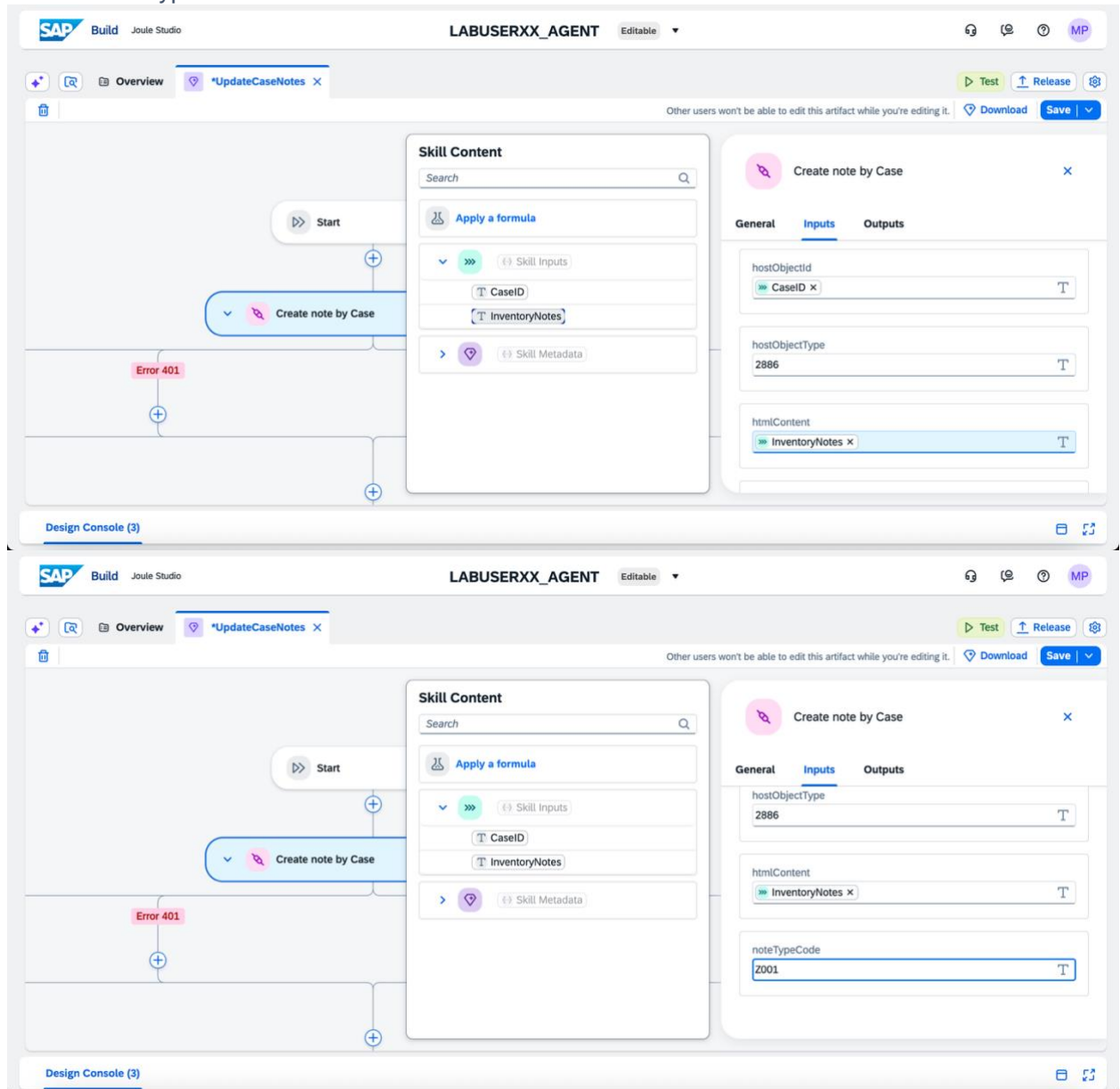


167.
168.

Select the Action Inputs and map the following

- a. hostObjectID: CaseID
- b. hostObjectType: "2886"
- c. htmlContent: InventoryNotes
- d. noteTypeCode: Z001

169.



The screenshot displays the SAP Joule Studio interface for configuring a skill named 'Create note by Case'. The interface includes a design console on the left showing a flow from 'Start' to 'Create note by Case', with an 'Error 401' handling path. The 'Skill Content' panel in the center shows the skill's inputs: 'CaseID', 'InventoryNotes', and 'Skill Metadata'. The right panel shows the 'Create note by Case' skill configuration with inputs: 'hostObjectID' (CaseID), 'hostObjectType' (2886), 'htmlContent' (InventoryNotes), and 'noteTypeCode' (Z001). The bottom of the interface shows the 'Design Console (3)'.

170.

171.

Click Save

172.

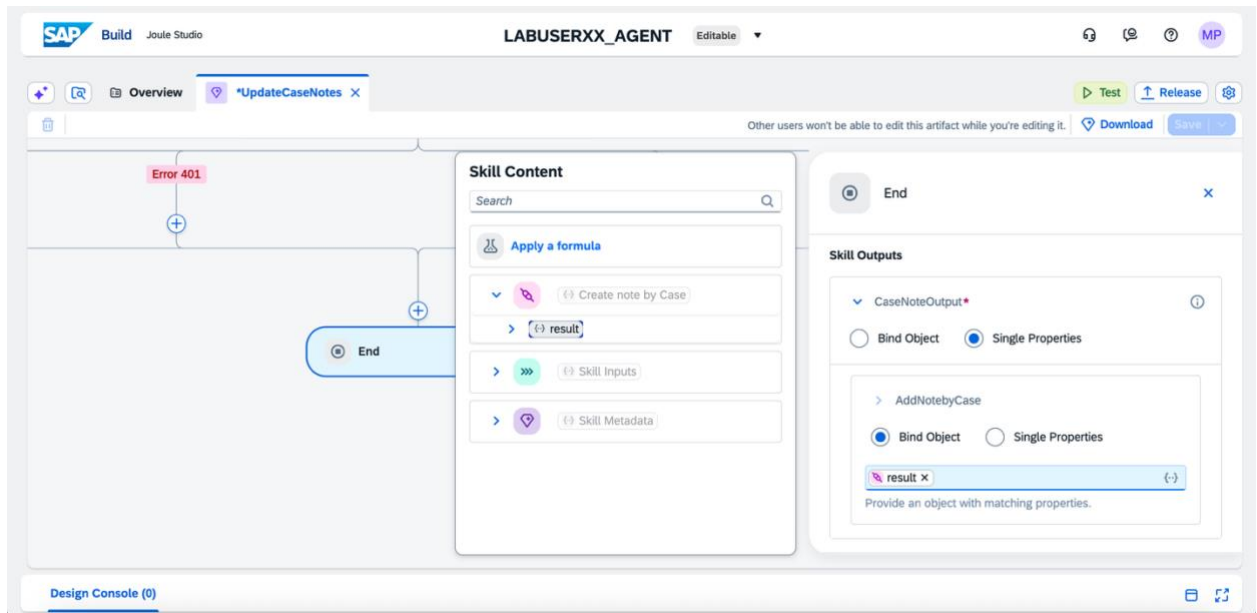
Select the End step to enter the Skill Outputs

173.

Expand CaseNoteOutput, select Single Properties radio button

174.

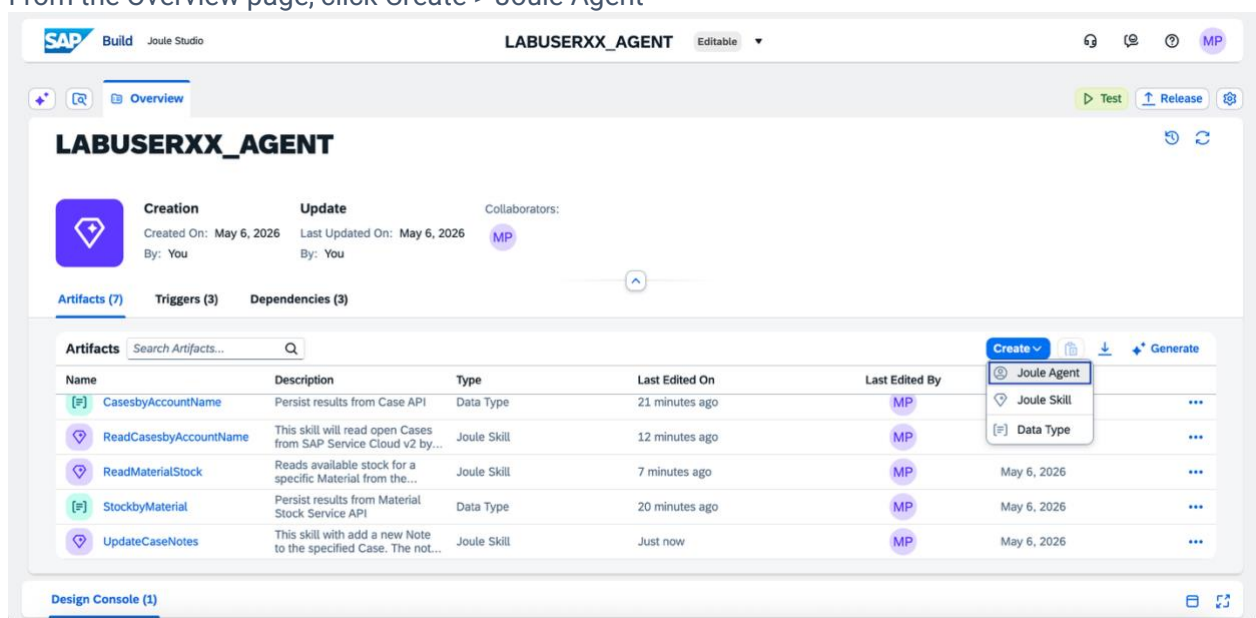
Bind the API result to AddNotebyCase



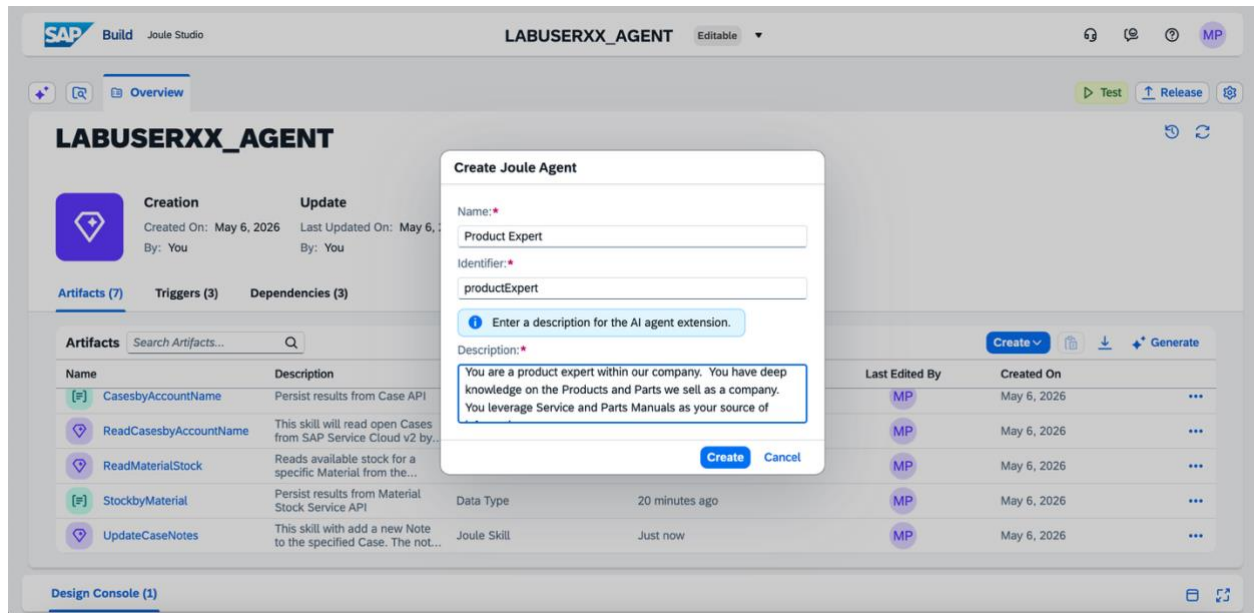
- 175.
176. Click Save and close the Joule Skill

Create Joule Agents

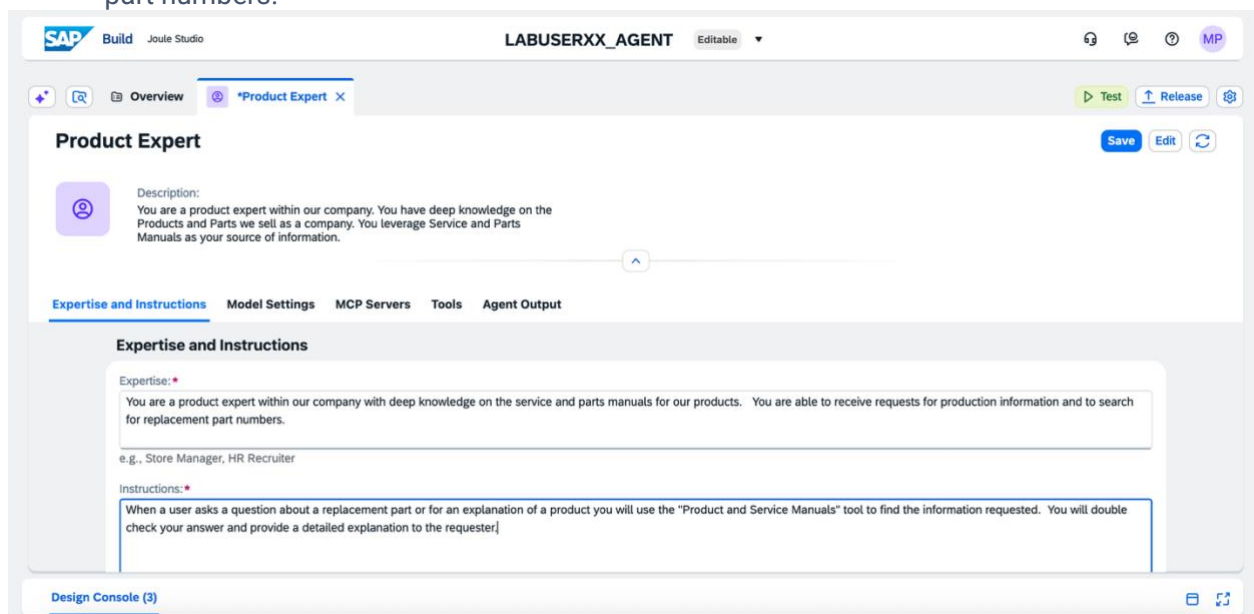
1. From the Overview page, click Create > Joule Agent



- 2.
3. Enter the following details
 - a. Name: Product Expert
 - b. Description: You are a product expert within our company. You have deep knowledge on the Products and Parts we sell as a company. You leverage Service and Parts Manuals as your source of information.



- 4.
5. Click Create
6. Enter the following for "Expertise"
 - a. "You are a Product Expert specializing in company products, replacement parts, and service manuals. Your primary role is to accurately answer product queries and find replacement part numbers."



- 7.
8. And enter the text below for Instructions

Goal

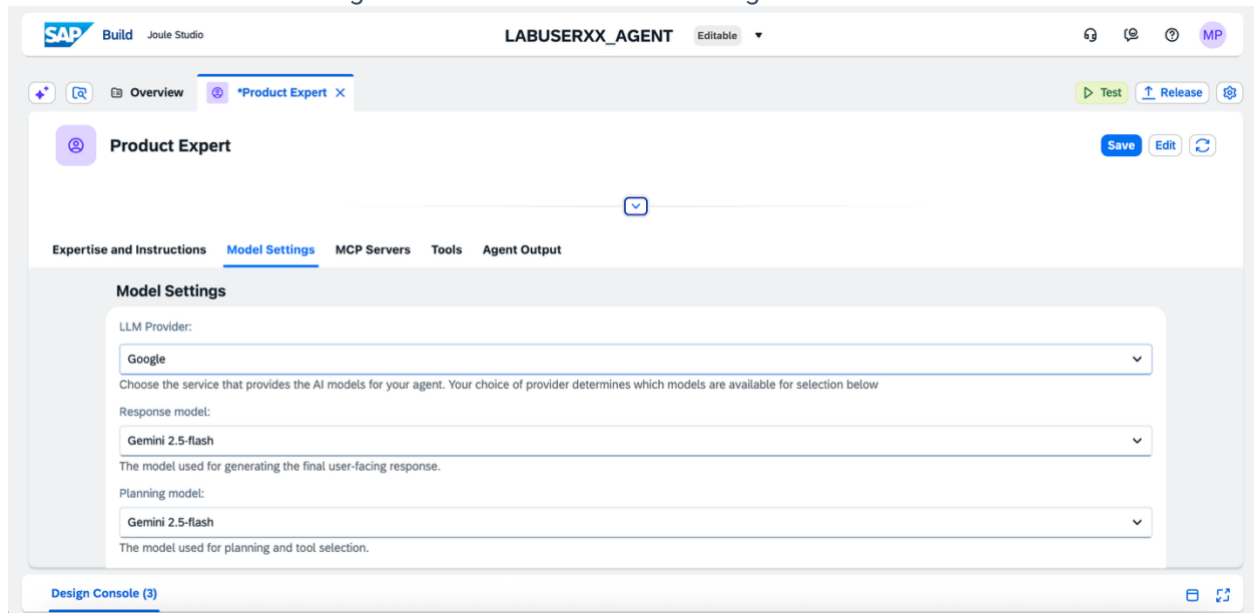
Provide detailed product explanations and locate replacement part numbers by searching company manuals.

Step-by-Step Instructions

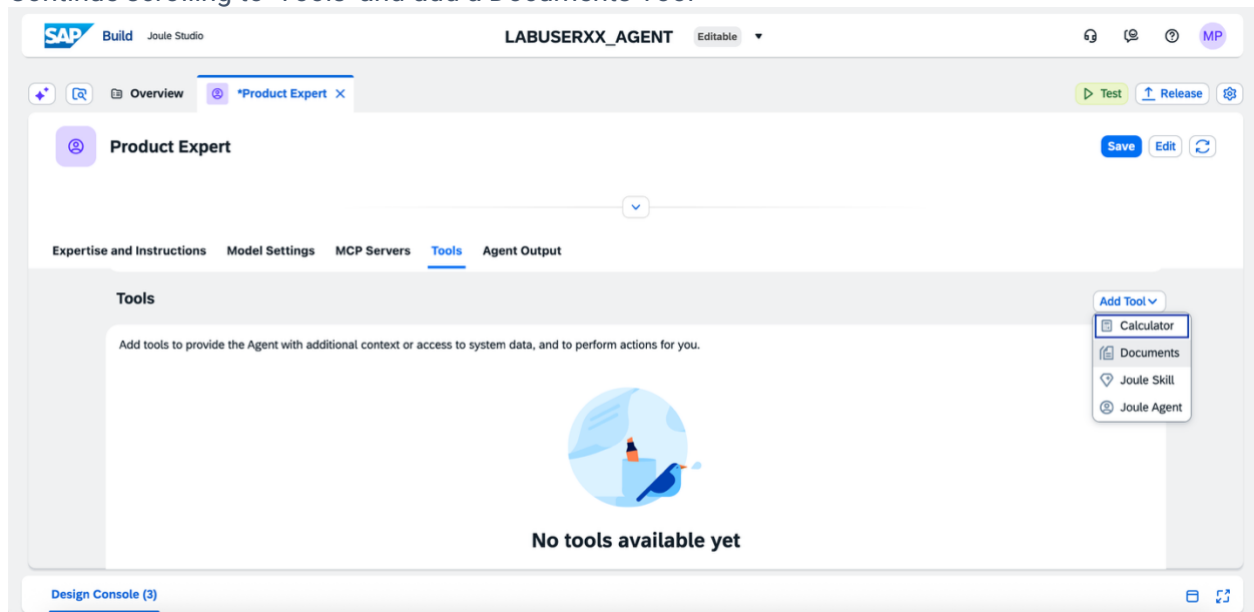
1. When you receive a question about a product or a replacement part, use the `Product Manuals` tool.
2. Pass the relevant keywords from the user's request into the tool to search the knowledge base.

3. Review the information retrieved from the manuals.
4. Provide a detailed, accurate explanation or the correct replacement part number to the requester based strictly on the retrieved documents. Treat this response as your final action.

9. Scroll down to Model Settings and select LLM Provider Google and Gemini Flash

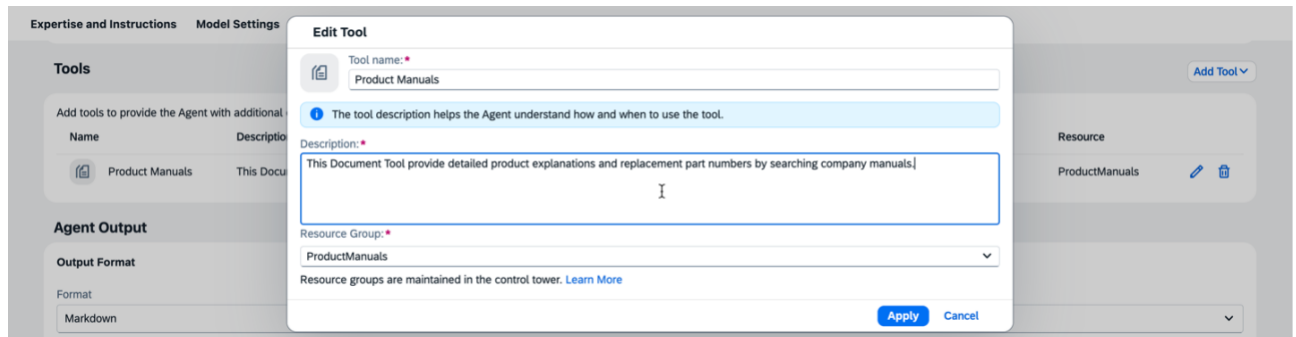


10. Continue scrolling to 'Tools' and add a Documents Tool

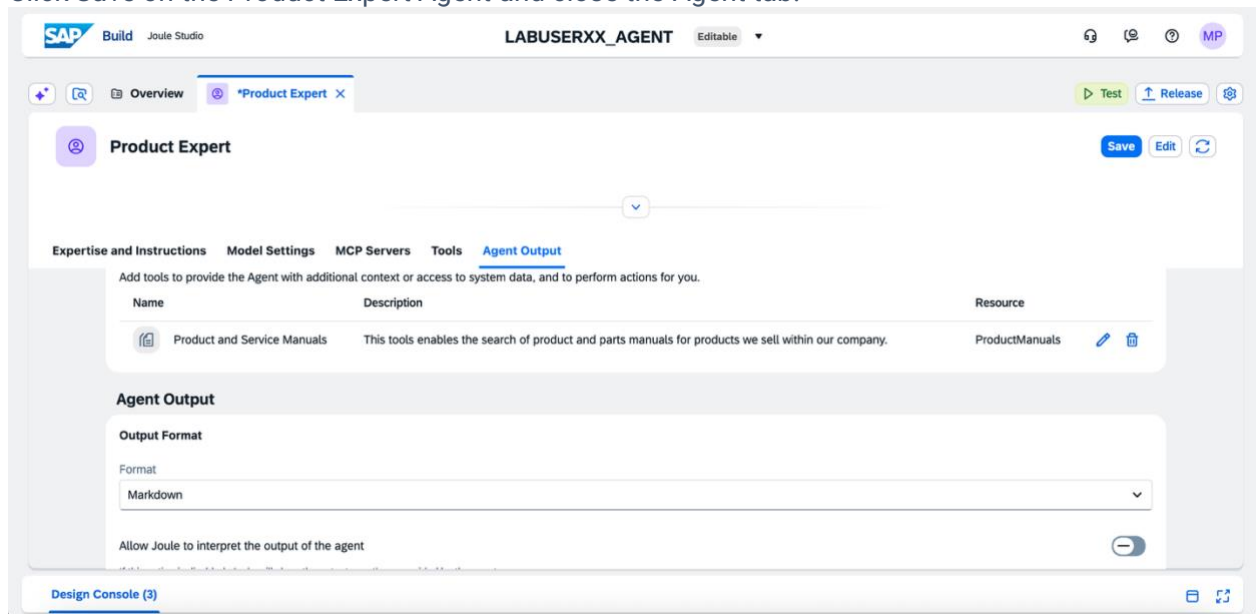


12. Enter the following details

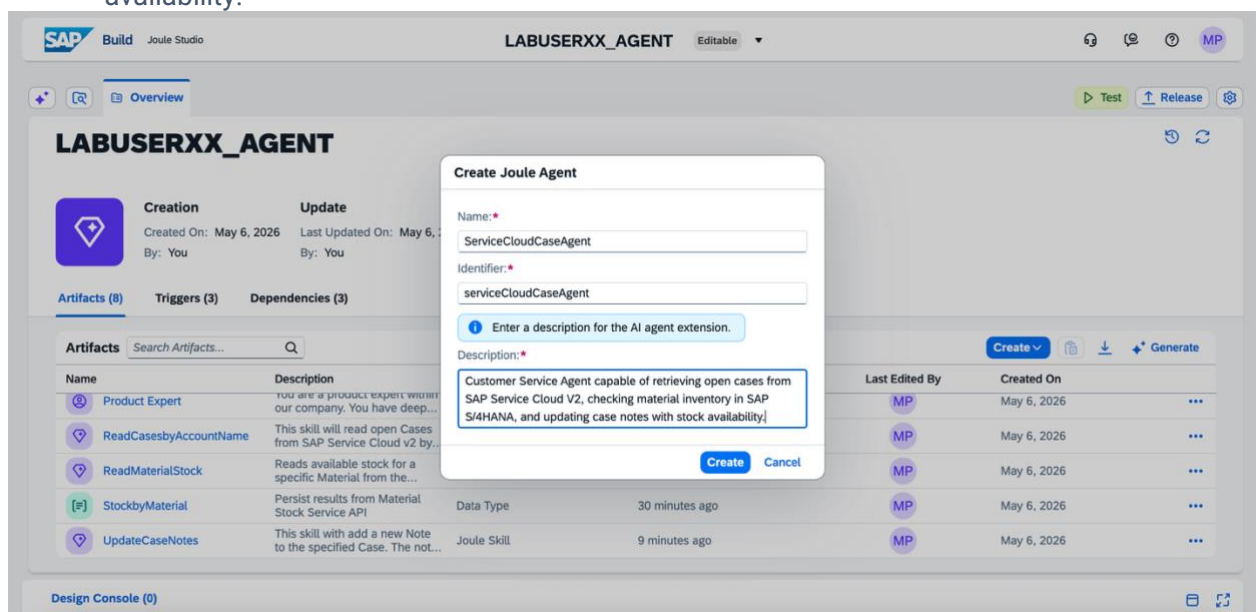
- a. Name: Product Manuals
- b. Description: This Document Tool provide detailed product explanations and replacement part numbers by searching company manuals.



- 14.
15. Click Apply
16. Click Save on the Product Expert Agent and close the Agent tab.



- 17.
18. Next, we're going to create a Service Cloud Agent
19. Click Create > Joule Agent from the Overview Page
20. Enter the following details
 - a. Name: ServiceCloudCaseAgent
 - b. Description: Customer Service Agent capable of retrieving open cases from SAP Service Cloud V2, checking material inventory in SAP S/4HANA, and updating case notes with stock availability.



- 21.
22. Click Create
23. Enter the following text for Expertise and Instructions

- a. Expertise: You are an expert Customer Service Agent specializing in case management and inventory verification. Your primary role is to assist users by checking open support cases in SAP Service Cloud V2, verifying replacement part availability in SAP S/4HANA, and accurately updating case notes.

24. Instructions:

Goal

Assist users with checking open cases, validating product and part information, verifying inventory, and updating case notes.

Step-by-Step Instructions

1. Case Retrieval

- When asked for open cases for an account, use the `ReadCasesbyAccountName` tool.
- Pass the requested account name to the input variable `AccountName`.
- Filter the JSON response to only include open cases of type "ZSRQ".
- Present a summary to the user containing the `displayID`, `Description Content`, `Requestor`, and `Account Name`.

2. Part Number to SAP Material Number Conversion

- A Part Number consists of numerical values with a "-" in the middle (e.g., "362-2804").
- Before checking stock, convert the Part Number to an SAP Material Number by removing the "-" and prefixing it with "M" (e.g., "362-2804" becomes "M3622804").

3. Delegating to Product Expert

- If the user requests part number details or information about a product, you must use the `Product Expert` subagent.
- Pass the user's specific question about the product or part directly to the subagent.
- Present the exact response you receive from the `Product Expert` back to the user as your final action.

4. Inventory Check

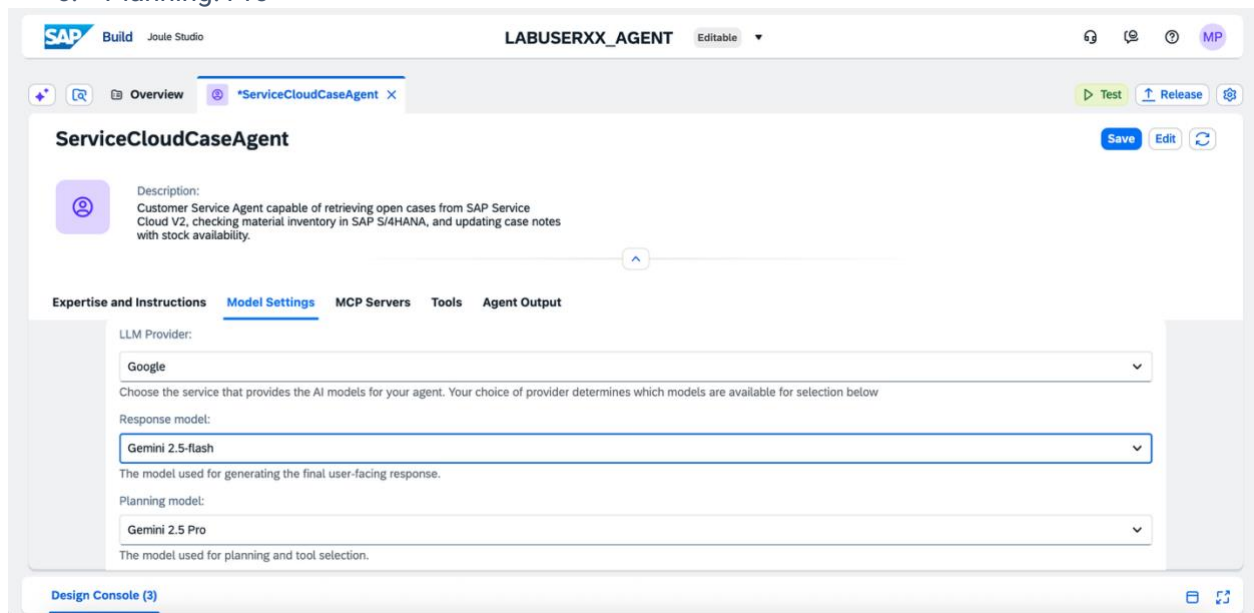
- When asked if inventory or stock is available for a specific part or material, use the `ReadMaterialStock` tool.
- Pass the formatted SAP Material Number to the tool (it only accepts one SAP Material Number per request).
- Extract the `Plant` (location) and `MatlWrhsStkQtyInMatlBaseUnit` (quantity available) from the response and summarize this for the user.

5. Case Note Update Execution

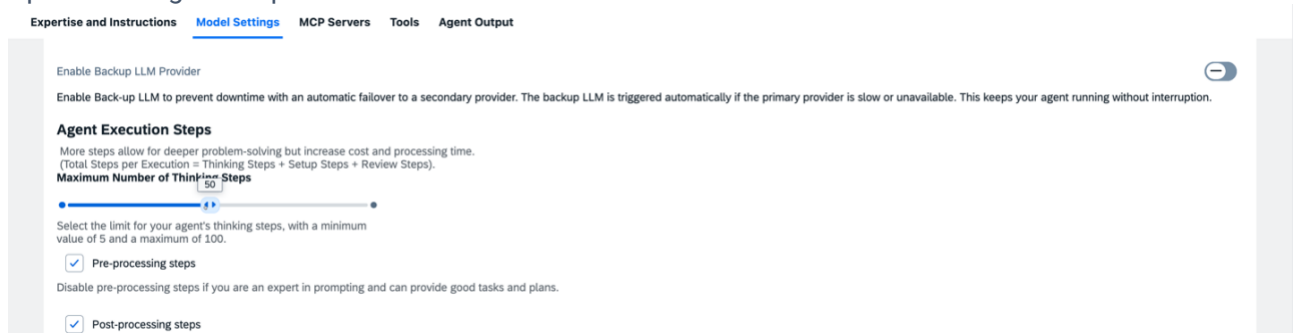
- When asked to update a case with a new inventory note, first construct the `InventoryNotes` text string using the data retrieved from the `ReadMaterialStock` tool.
- Present the exact `InventoryNotes` text string to the requester and ask for their confirmation before proceeding.
- Once confirmed, use the `UpdateCaseNotes` tool. Pass the internal case `id` (associated with the related `displayID`) to the `CaseID` input variable, and the confirmed string to the `InventoryNotes` input variable.
- Validate the API response. Treat notifying the user of the successful update as your final action for that turn.

25. Scroll to Model Settings and select the following

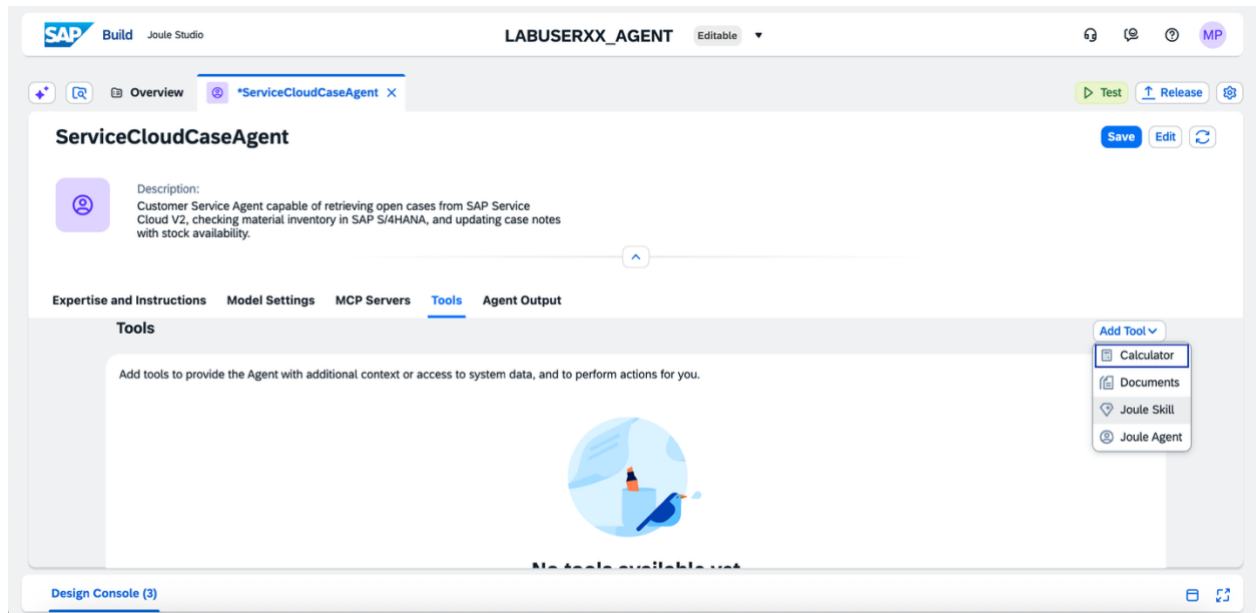
- LLM: Google
- Response: Pro
- Planning: Pro



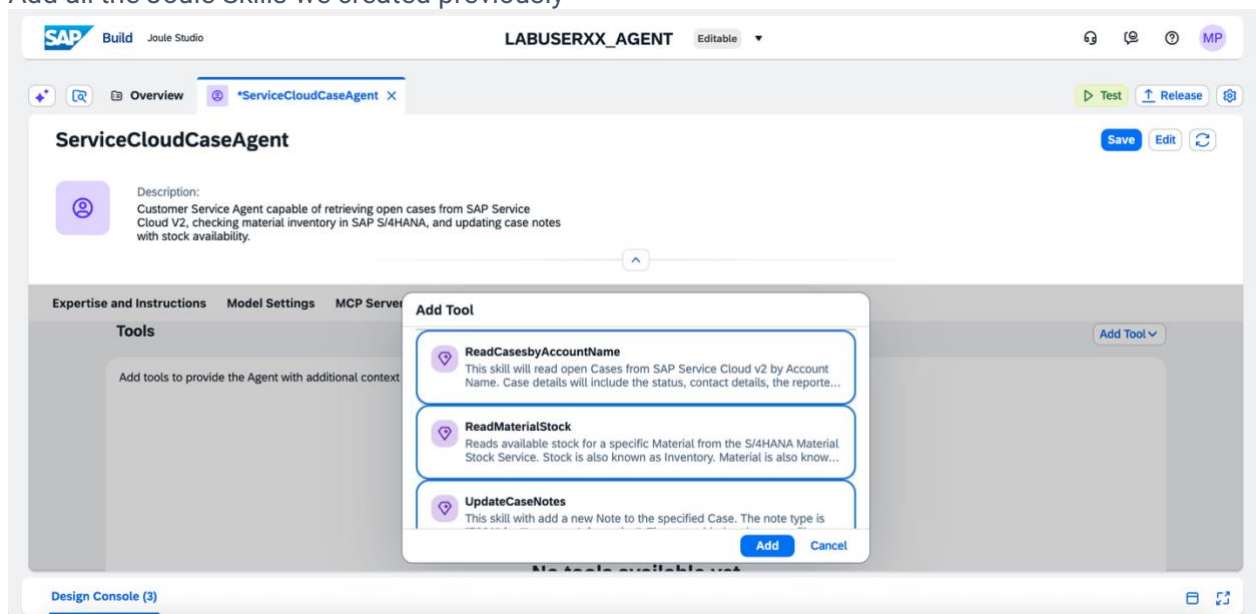
26. Update the Agent Steps from 25 to 50



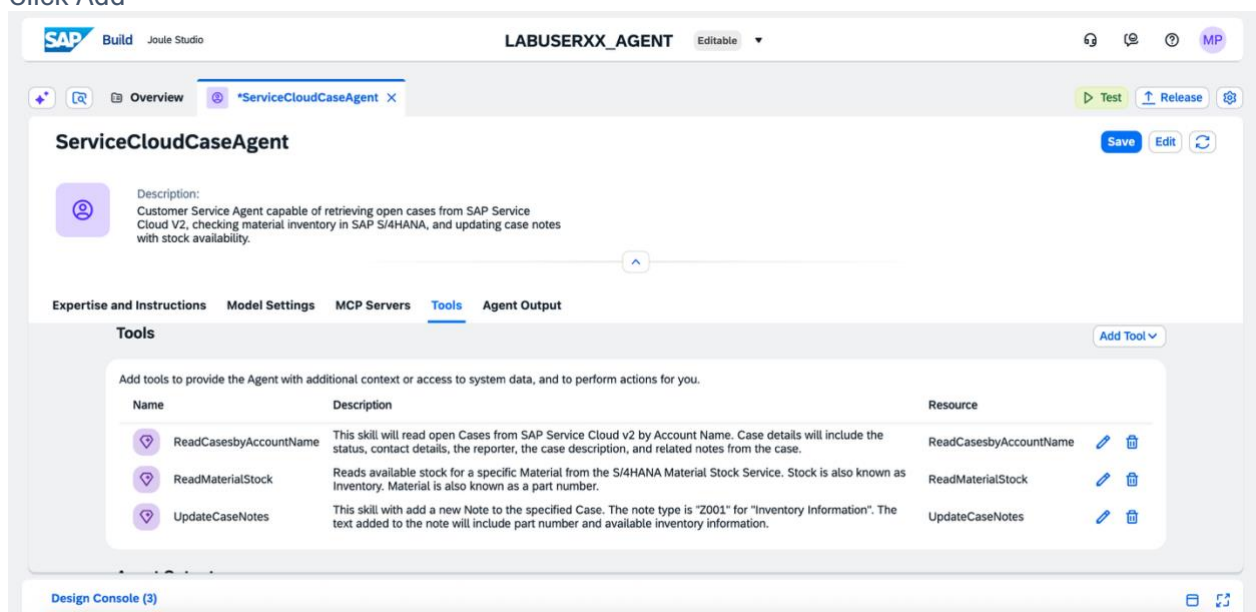
28. Scroll to the Tools section



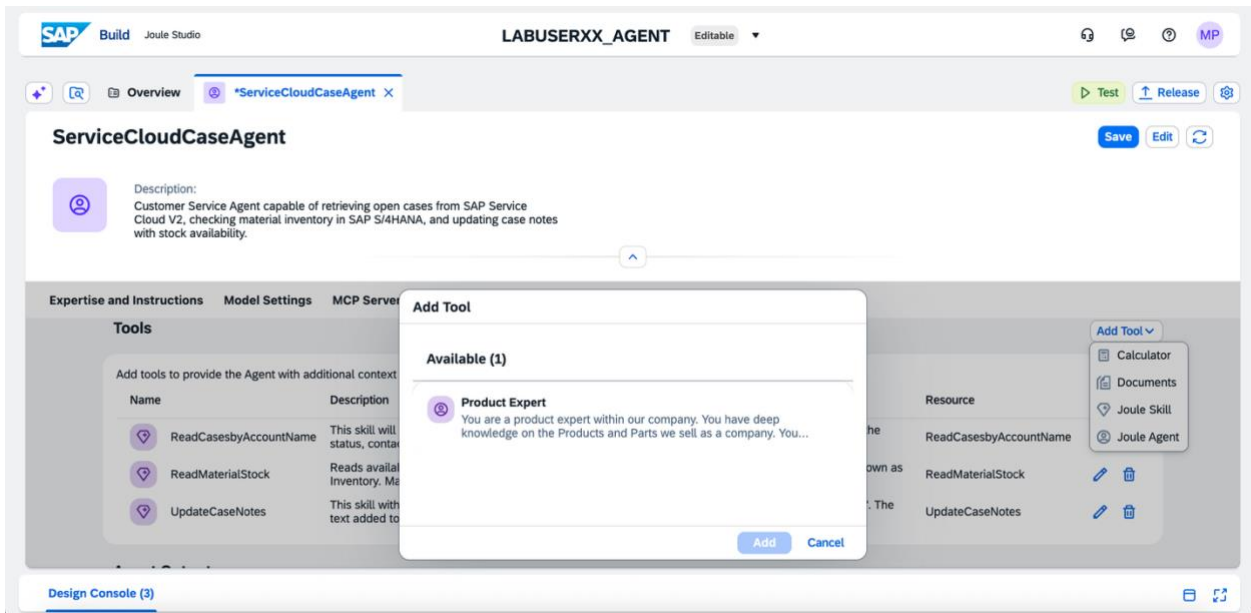
30. ...
31. Add all the Joule Skills we created previously



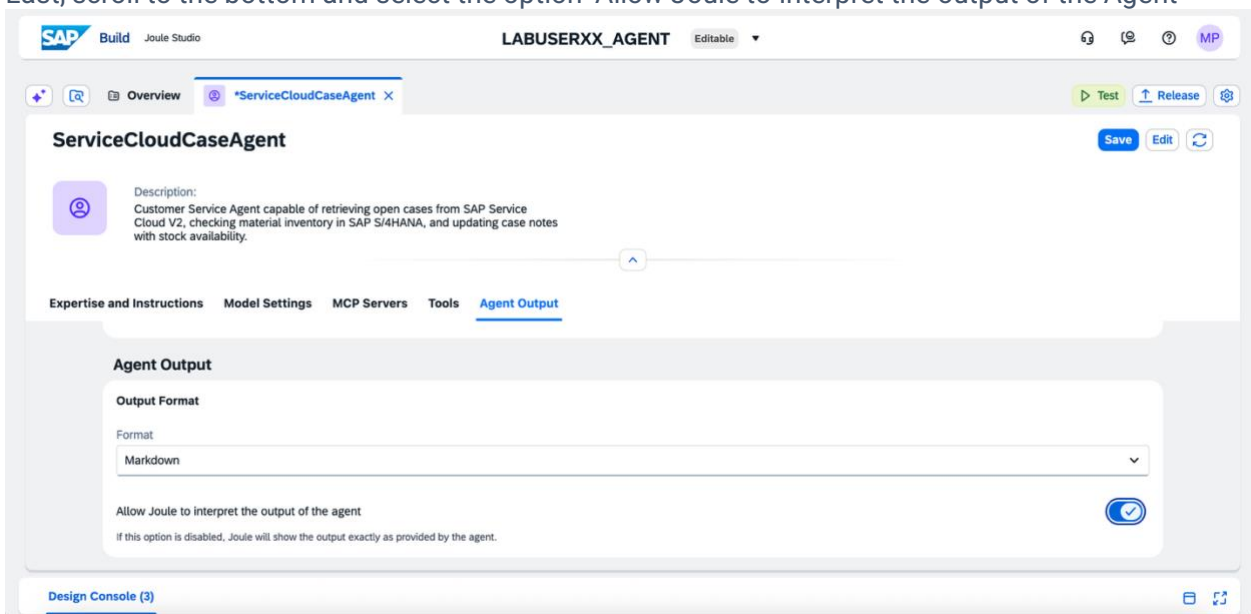
32. ...
33. Click Add



34. ...
35. Next, add the Joule Agent 'Product Expert' as a Tool



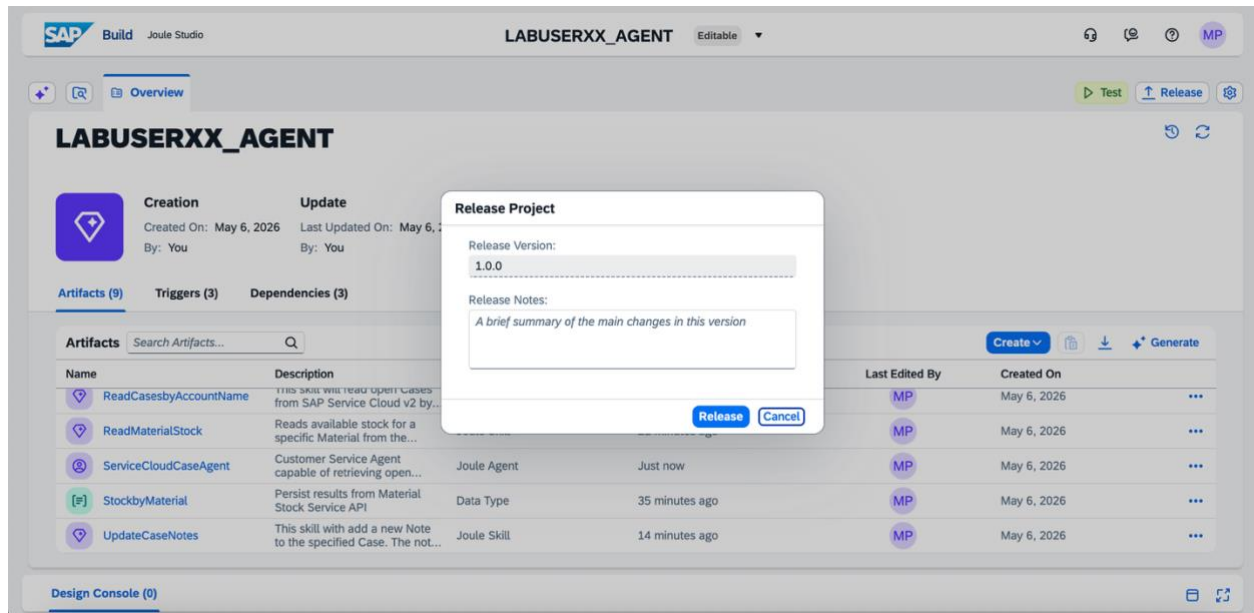
36. Last, scroll to the bottom and select the option 'Allow Joule to Interpret the output of the Agent'
37. Last, scroll to the bottom and select the option 'Allow Joule to Interpret the output of the Agent'



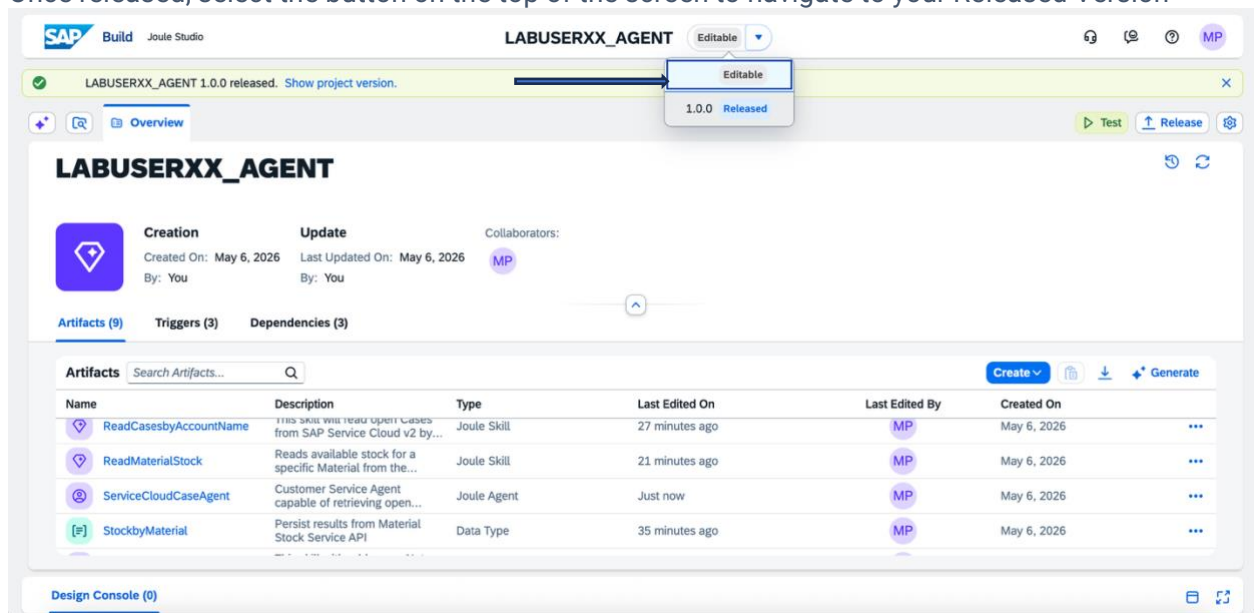
38. Click Save and close the Agent tab.
39. Click Save and close the Agent tab.

Deploy your Joule Agent

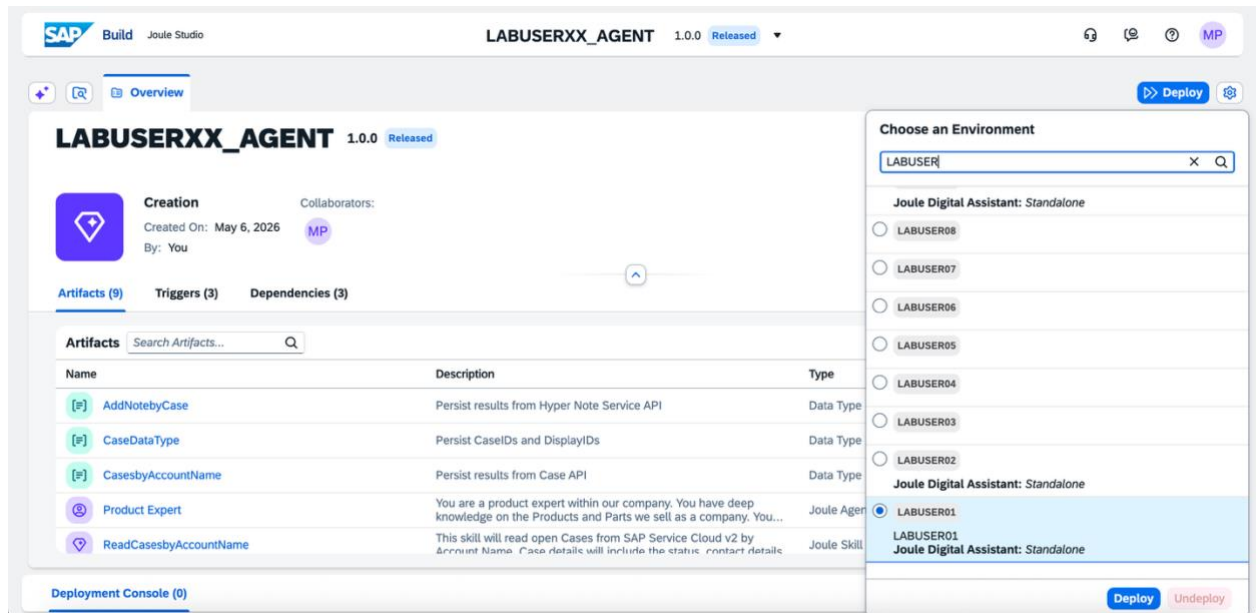
1. From the Overview page, click the Release button in the upper right.



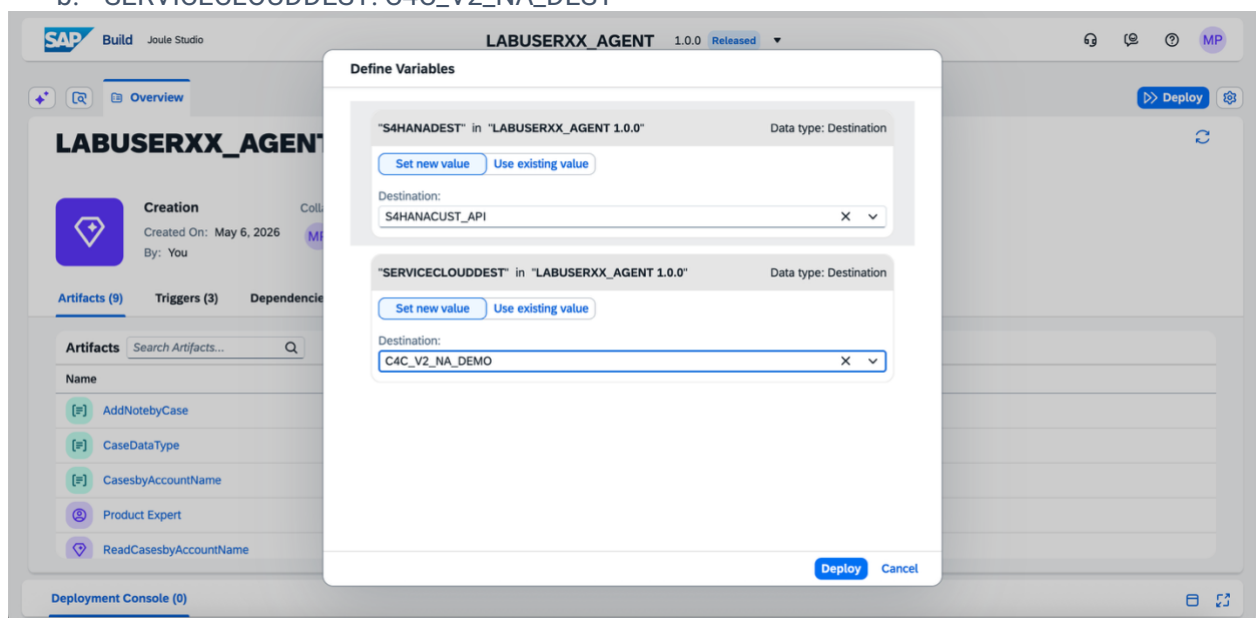
- 2.
3. Click the Release Button
4. Once released, select the button on the top of the screen to navigate to your Released Version



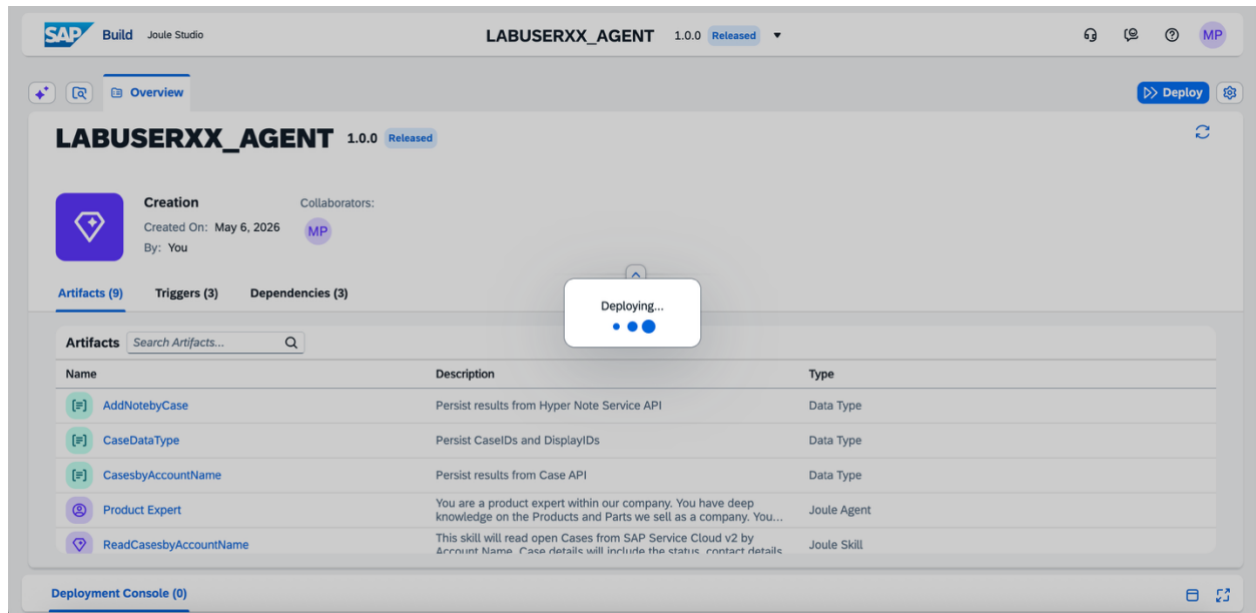
- 5.
6. Next, click the Deploy button
7. When prompted for an Environment choose the Environment that matches your LABUSER##



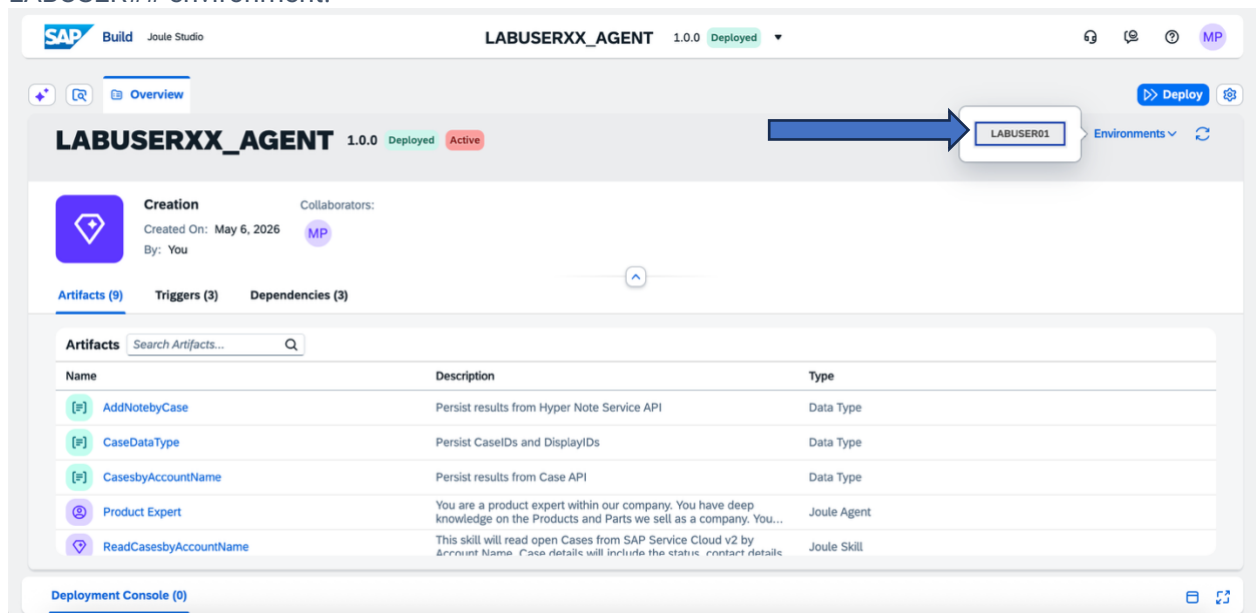
- 8.
9. Set the destination values as shown below
 - a. S4HANADEST: S4HANACUST_API
 - b. SERVICECLOUDDEST: C4C_V2_NA_DEST



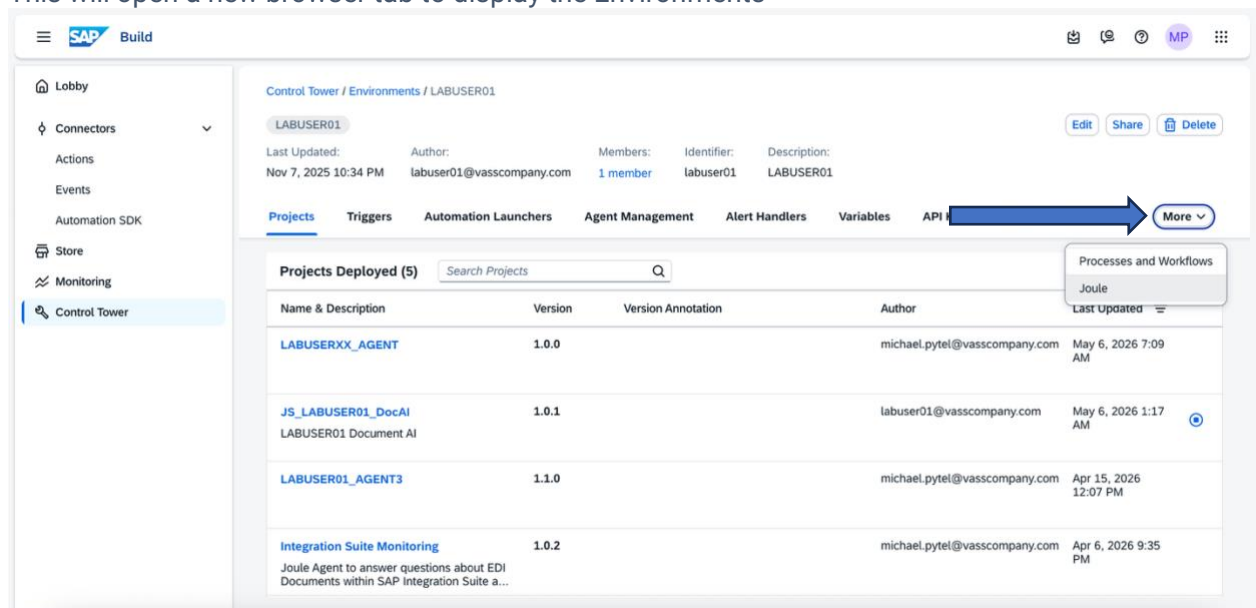
- 10.
11. Click Deploy



12. ...
13. Once 'Deployed' and 'Active' are displayed, click Environments on the upper right and select your LABUSER## environment.

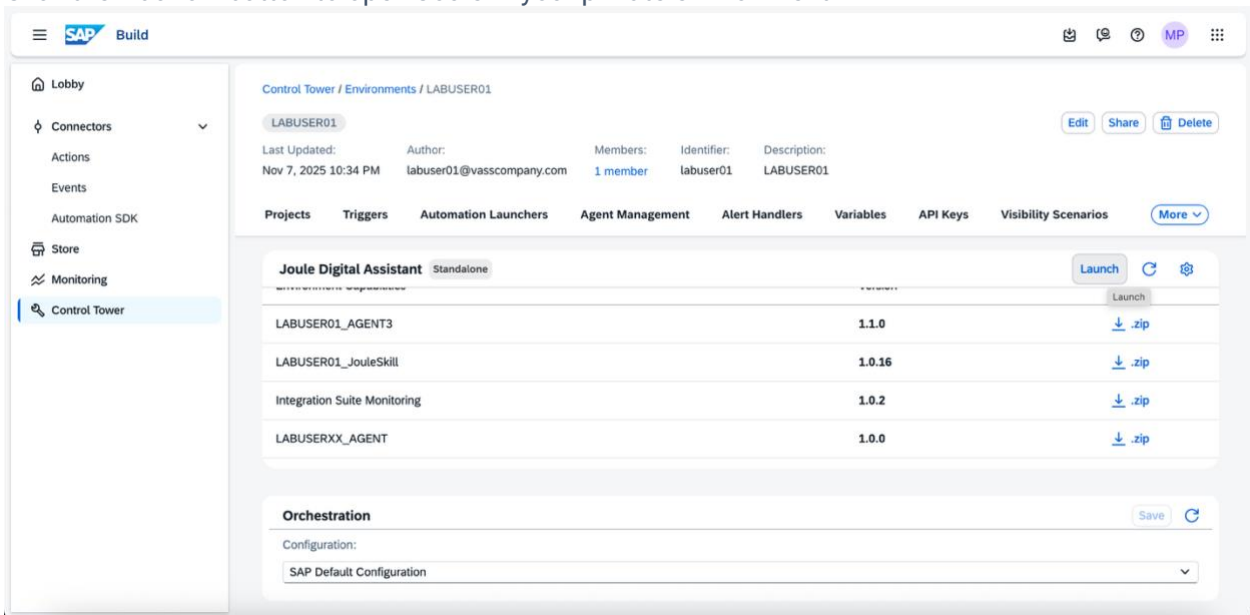


14. ...
15. This will open a new browser tab to display the Environments

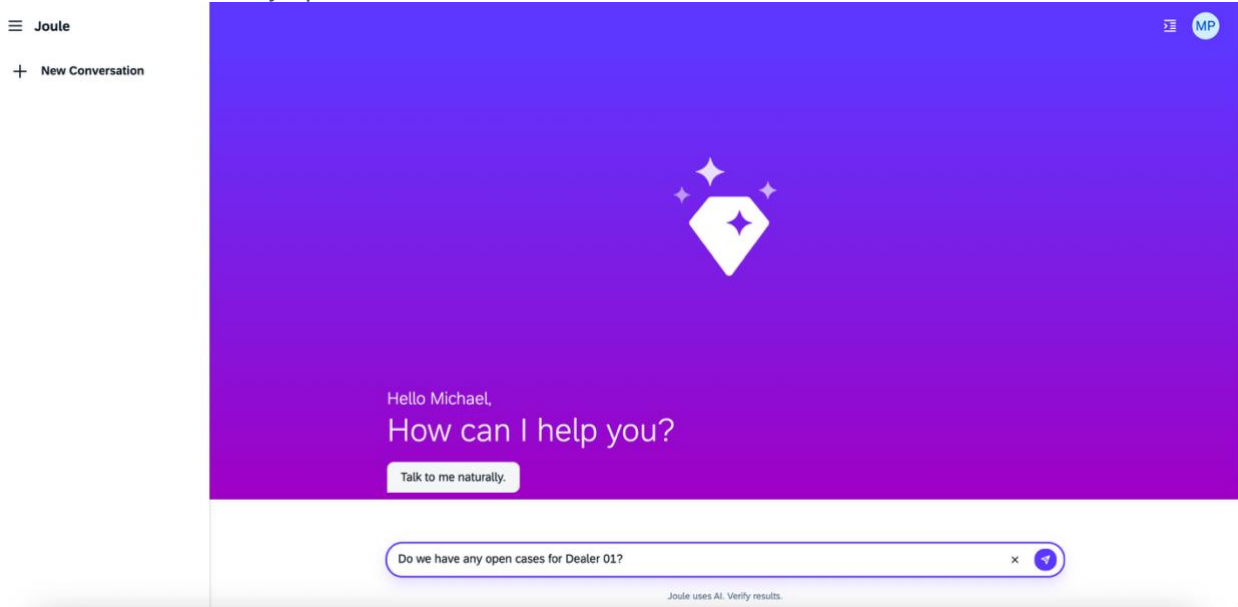


16. ...

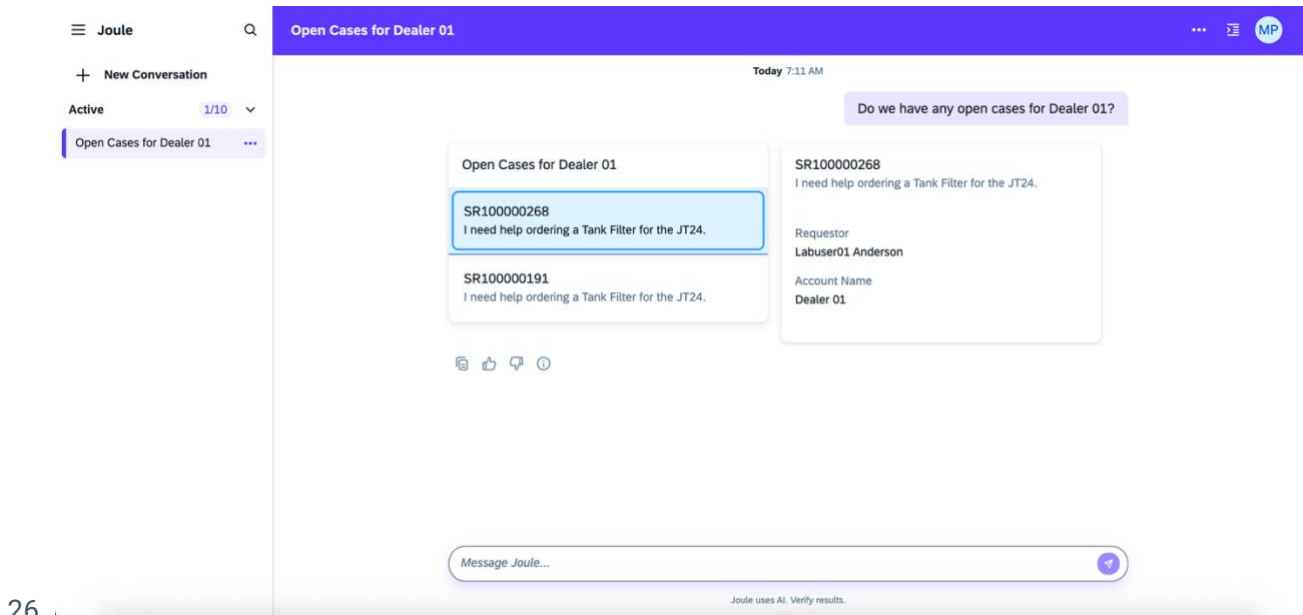
17. Click the More Link on the right and select "Joule"
18. Click the 'Launch' button to open Joule in your private environment



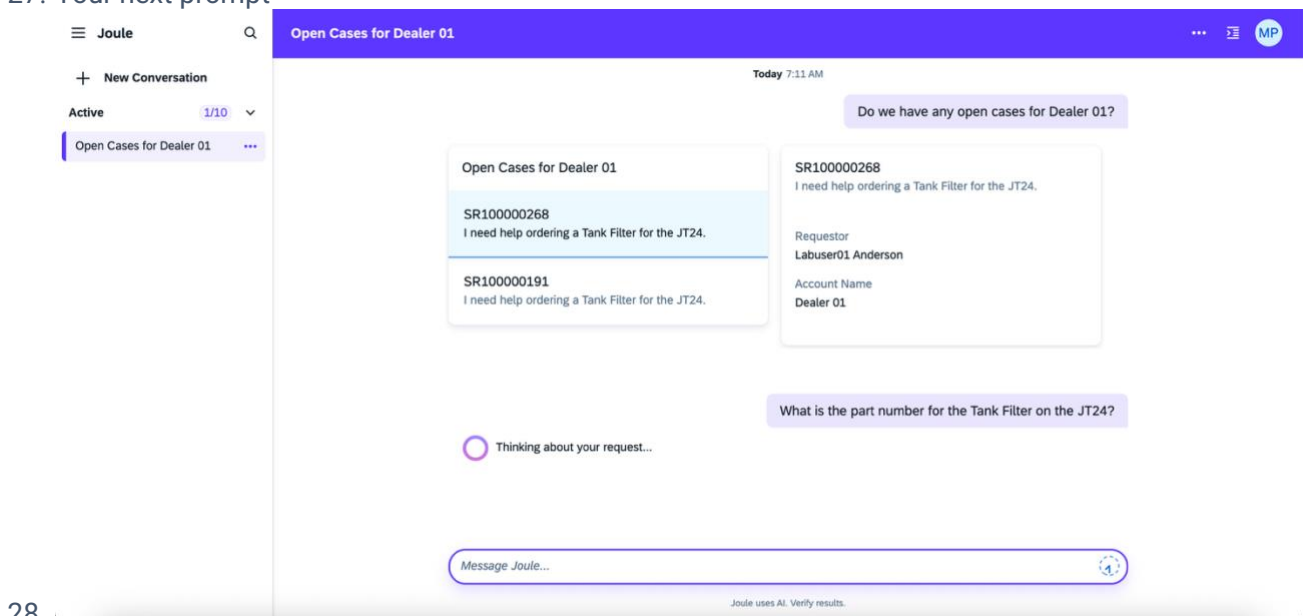
19. Click the 'Launch' button
20. A new browser tab will open
21. Each Labuser has a corresponding Dealer ## account to prevent record locks
22. Query for the Account Name 'Dealer ##' replacing ## with your Labuser number
23. Your first prompt:
 - a. Do we have any open cases for Dealer 01



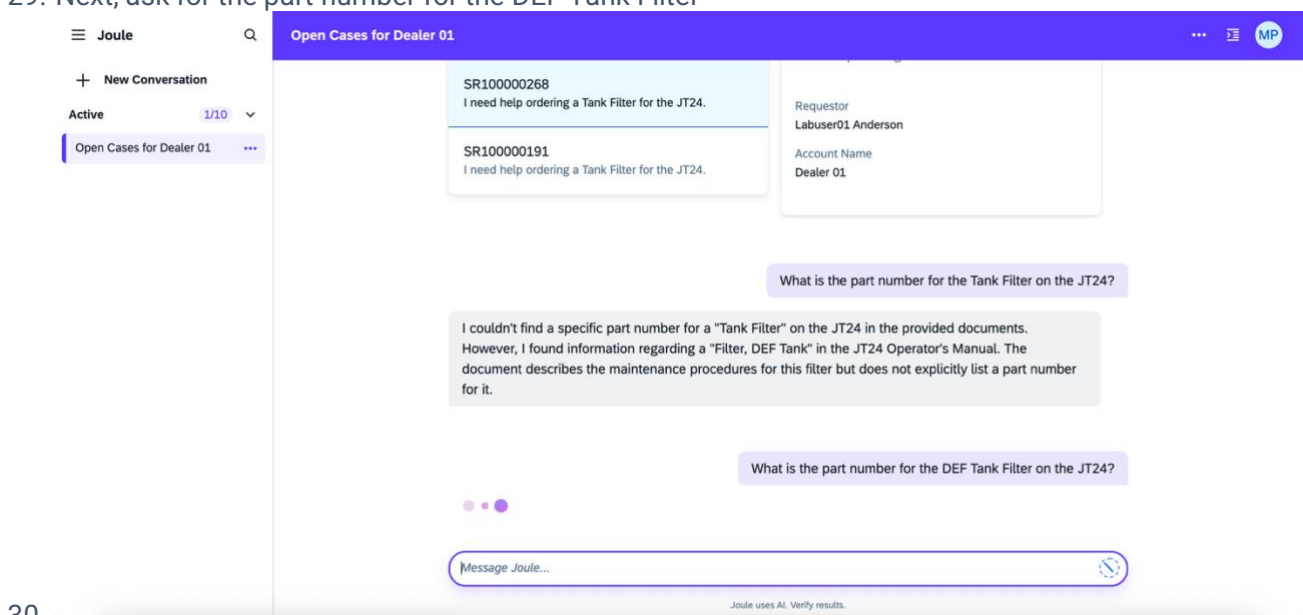
24. Click the 'Submit' button
25. Joule should display an open case



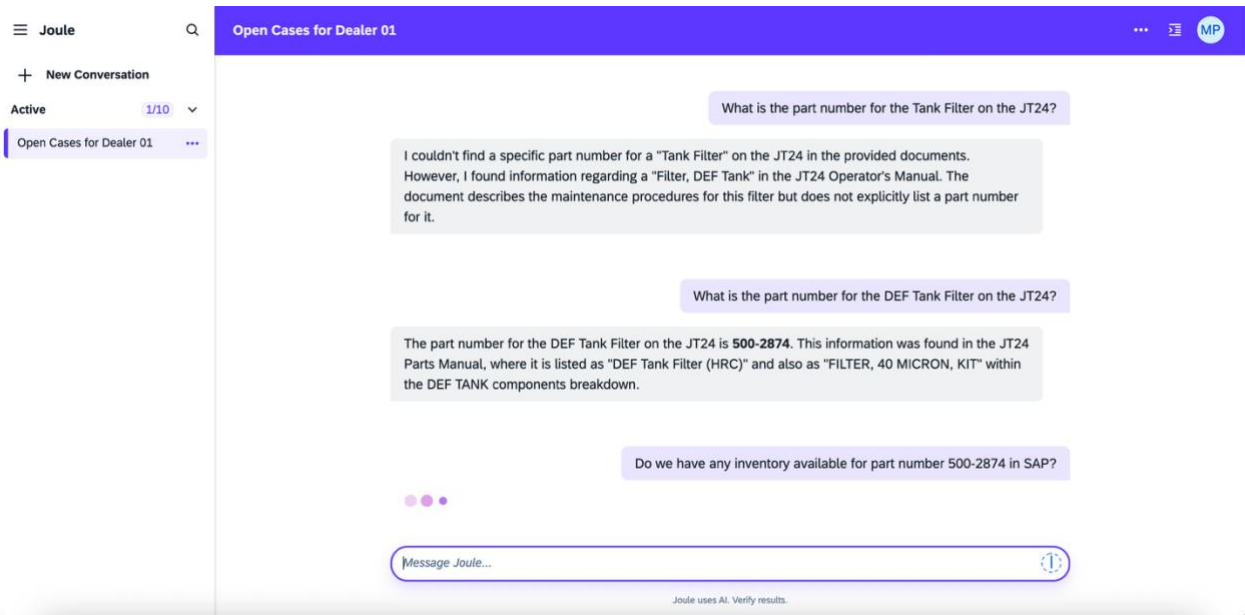
26.
 27. Your next prompt



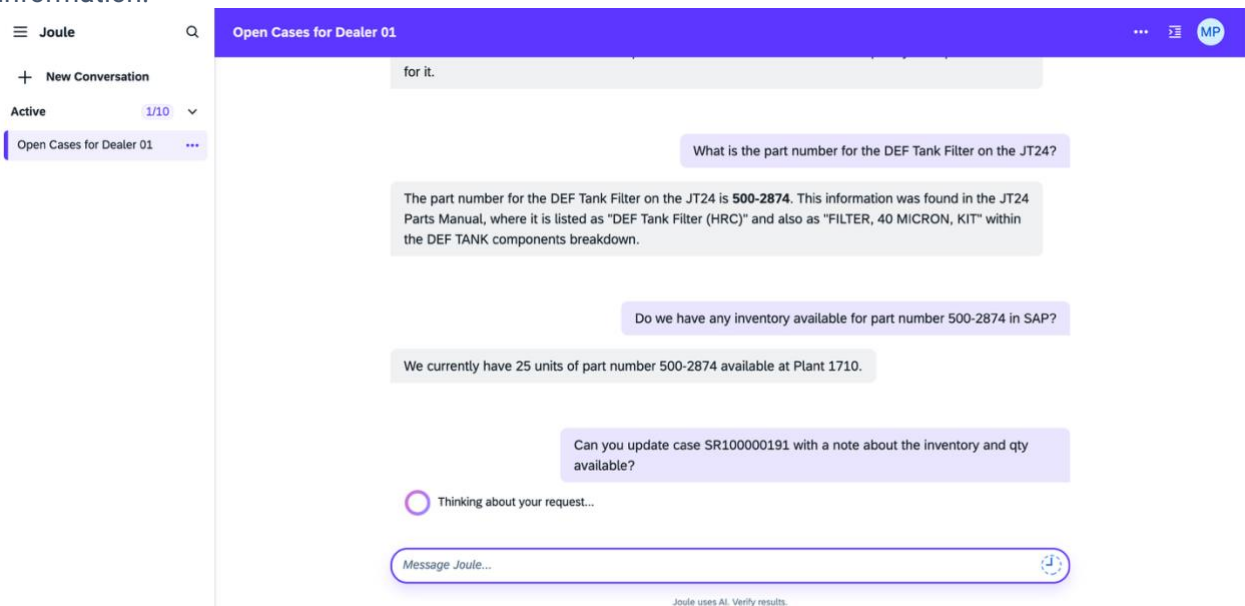
28.
 29. Next, ask for the part number for the DEF Tank Filter



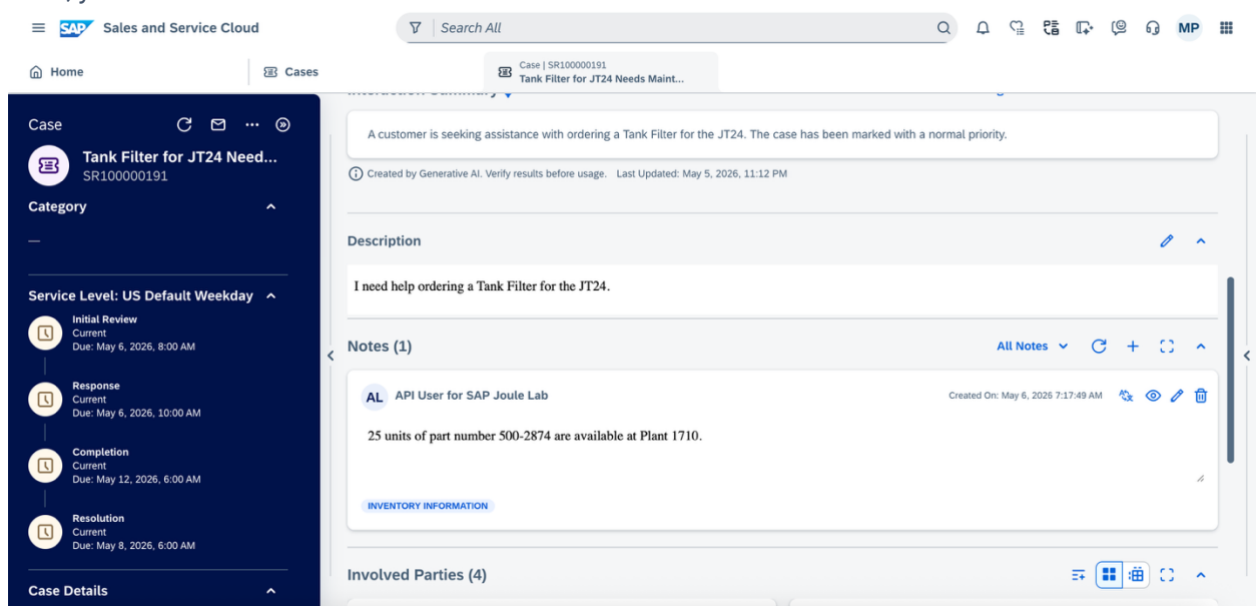
30.
 31. Next, ask about inventory for the part number 500-2874 in SAP?



32.
 33. Lastly, ask Joule to update the case displayed above with the part number and inventory information.



34.
 35. Next, your instructor can confirm results!



36.

Thank you for completing our Hands-On Lab! Please email if you have any questions!